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Metabolic risk factors of non-communicable diseases in Ethiopia: a systematic review and meta-analysis.

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Title: Metabolic risk factors of non-communicable diseases in Ethiopia: a systematic review and meta-analysis.

List of authors

Tilahun Tewabe Alamnia, [1,2*] Wubshet Tesfaye,[3] Solomon Abrha, [3,4] and Matthew Kelly.[2]

Authors' institutional affiliations

[1,2*]Tilahun Tewabe Alamnia (Corresponding author, MSc, [1]College of Medical and Health Sciences, Bahir Dar University, Ethiopia; [2]Department of Global Health, Research School of Population Health, Australian National University, Australia. Email: Tilahun.Alamnia@anu.edu.au, Phone number +61474173615)

[3]Wubshet Tesfaye (PhD, Institute of Health Research, University of Canberra, Australia. Email: Wubshet.Tesfaye@canberra.edu.au)

[3,4]Solomon Abrha (MSc, [3]University of Canberra, Australia. [4]College of Medicine and Health Sciences, Mekelle University, Ethiopia. Email: Solomon.Bezabh@canberra.edu.au)

[2]Matthew Kelly (PhD, Department of Global Health, Research School of Population Health, Australian National University, Australia. Email: Matthew.Kelly@anu.edu.au)

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Abstract

Objectives: Non-communicable diseases (NCDs) are causing a new and devastating health burden in low-income countries. In Ethiopia 39% of deaths are NCD related, but the health system is under-prepared. Evidence of risk factor distributions is needed to inform response. The research question for this review investigates the best estimates of prevalence and sex and age distributions of metabolic risk factors for NCDs among Ethiopian adults.

Research design and methods: This systematic review used the Preferred Reporting Items for Systematic Review and Meta-Analysis guidelines. Studies published until January 06, 2021 were searched from PubMed, Scopus, ProQuest, and Web of Science databases, reference lists and grey literature. Studies reporting prevalence of metabolic risk factors: overweight/obesity, hypertension, impaired glucose homeostasis, and metabolic syndrome among adults were eligible for this systematic review and meta-analysis. Two authors independently extracted data and used the Joanna Briggs Institute (JBI) tool for appraisal. Pooled prevalence's were based on random effect models using Stata version 16. Subgroup analyses examined differences by region, study year, sample size, and settings.

Results: From 6087 records, 74 studies including 104,382 participants were included. Most showed high prevalence of metabolic risk factors. Meta-analysis revealed pooled prevalence of metabolic risk factors from 12%-24% with the highest observed for overweight/obesity (23.9%, 95 % CI: 19.9-28.0) and hypertension (21.1%, 95% CI: 18.7-23.5) followed by metabolic syndrome (14.7%, 95 % CI: 9.8-19.6) and impaired glucose tolerance (12.4%, 95% CI: 8.7-16.1). The prevalence of overweight/obesity was higher in women. All metabolic risk factors were higher among people aged above 45 years.

Conclusions: Nearly 1 in 5 Ethiopian adults has at least one metabolic risk factor for NCDs, highlighting the need for a coordinated effort among policymakers, health care providers, non-governmental stakeholders, and the community to implement appropriate preventive measures for management of metabolic risk factors for NCDs.

Strengths and limitations of the study

- This is the first systematic review and meta-analysis combining studies with similar characteristics in the context of metabolic risk factors for NCDs in Ethiopia.
- Prevalence of metabolic risk factors was between 12 and 24% revealing the likelihood of continuing increases in non-communicable disease incidence in the future for this population.

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- Studies included had heterogeneity of outcomes and variations in sample sizes which hampered the ability to calculate combined pooled prevalence of multiple risk factors. We did not find studies conducted in some regions of the country like Afar, Benishangul-Gumuz, Gambella regions, so our prevalence estimates might not reflect the real situation in these regions.

Introduction

Non-communicable diseases (NCDs) are the leading cause of death worldwide, but they disproportionately affect people living in Low and Middle-Income Countries (LMICs).[1] In 2017, more than 73% of global deaths (41 million) were attributed to NCDs.[1] Of these, more than 80% were due to cardiovascular diseases, cancer, chronic respiratory diseases, and diabetes mellitus (DM).[1,2] About 75% (28 million) of all NCDs deaths and 82% (16 million) of the premature deaths occurred in LMICs. In the WHO African Region, the burden of NCDs is gradually increasing and is anticipated to overtake the burden of mortality and morbidity from communicable diseases by the year 2030.[3] The same holds true in Ethiopia where, NCDs were estimated to cause around 43% of deaths in 2019.[4]

Most NCD related deaths are attributed to major risk factors which can be broadly categorized into behavioral (tobacco smoking, alcohol drinking, sedentary lifestyle, and high dietary salt intakes) and metabolic (overweight/obesity, high fasting plasma glucose, high blood pressure, and high level of blood triglycerides) factors.[3] The continual increase in the prevalence of metabolic risk factors, often reinforced by behavioral factors, has been associated with a direct increase in prevalence of NCDs, especially among adults between the ages of 30 and 70 years.[2]

According to the 2017 Global Burden of Disease Study, metabolic risk factors contributed to a substantial proportion (26%) of the total Disability-Adjusted Life Years (DALYs). High blood pressure, high fasting plasma glucose, high body-mass index, and high cholesterol are responsible for 9%, 7%, 6% , and 4% of global DALYs, respectively.[5] Besides, the 2019 estimates on DM prevalence revealed that 463 million people (9%) have diabetes and this number is projected to rise to 700 million (11%) by 2045.[6] LMICs has been disproportionately affected by these problems and the burden of these diseases is likely to increase in developing countries in Africa.[5]

The 2030 United Nations agenda for sustainable development has set a target to reduce one-third of premature mortality form NCDs[7]. In line with this, the World Health Organization (WHO) has set out a milestone plan to prevent and control premature mortality from

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cardiovascular diseases, cancer, DM, or chronic respiratory diseases by 25% in 2025, which can be achieved by reducing the burden of metabolic risk factors and its contributing behavioral problems among adults using affordable and accessible intervention strategies.[8] However, to implement these interventions and achieve the intended goals, up-to-date, representative, and comprehensive evidence is always needed. The Ethiopian government prepared national guidelines for NCDs prevention and management in line with the global target in 2016,[9] but there is a lack of evidence on the successes of ongoing implementation and program evaluations of the proposed strategies. There is also a lack of comprehensive data on the prevalence of the immediate predictors of NCDs to implement focused interventions and make policy decisions.

There has been no systematic review of the prevalence of metabolic risk factors in Ethiopia. Although some studies have reviewed the prevalence of hypertension, their results are inconclusive due to lack of methodological rigor such as including participants with varying characteristics and/or were performed more than five years ago. There is no available comprehensive evidence for the remaining three risk factors: impaired glucose tolerance, overweight/obesity, and metabolic syndrome. Furthermore, previous individual studies have shown a conflicting association between metabolic risk factors and participants' characteristics, such as sex and age. For instance, some showed that hypertension was higher in women and elders, while others in men and in younger people. To this end, we aimed to conduct this systematic review and meta-analysis to synthesize nationwide data on the prevalence of metabolic risk factors for NCDs in Ethiopian adults. Also, we aimed to examine the distribution of the pooled prevalence of these risk factors across different regions, years, and settings as well as examine associations with the sex and age of participants.

We hypothesized that metabolic risk factors for NCDs are higher among adults due to the life-course cumulative increase in risk caused by ageing and behavioral factors in this age group. Estimating the prevalence of metabolic risk factors for NCDs in Ethiopia is essential to identify the priority areas for policy action, inform practices through various intersectoral collaborations, and decide future research directions. The result of this study can be used by the government, health authorities, non-governmental stakeholders, and health care providers to design and deliver health interventions to decrease the burden of NCDs.

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Methods

Study design

This systematic review and meta-analysis was conducted and reported in accordance with the Preferred Reporting Items for Systemic Reviews and Meta-analysis.[10] It includes key research questions, information sources, eligibility criteria, study selection, data extraction, risk of bias assessment, and data analysis, results and discussion. The process of screening and selection of studies, the results of the analysis and narrative descriptions were supported with figures and tables for reporting this study.

Patient and public involvement

No patients or public were involved in the design, conduct, interpretation, or dissemination of the results of this study.

Research questions

The key research questions for this study were: 1) What is the pooled prevalence of metabolic risk factors among adults in Ethiopia? and 2) what is the association between metabolic risk factors and sex and age of participants?

Based on these research questions, we identified Population, Exposure, and Outcome (PEO) for eligibility criteria. Since the review was based on prevalence studies, we did not define controls. The population in this study was any adult living in Ethiopia. The adult was defined as person between the ages of 18-64. However, in some studies, they referred to adults over the age of 15. To maximize the generalizability of our study findings, we used both definitions for this systematic review and meta-analysis. Exposures were any type of variables that might contribute to metabolic risk factors, including sociodemographic and economic variables. The outcome variables for this study were metabolic risk factors, such as overweight/obesity, hypertension, impaired glucose homeostasis, and metabolic syndrome.

Eligibility criteria

The eligibility criteria for this study were based on the research questions and the predefined PEO's for this review. These include studies conducted in Ethiopia and published in English language which reported a prevalence of at least one of the metabolic risk factors (overweight/obesity, impaired glucose tolerance, hypertension, and metabolic syndrome) in adults with a cross-sectional or cohort designs and published prior January 06, 2021. Whereas papers with only abstracts, qualitative studies, articles with a case control design, and research

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involving participants from vulnerable groups (children and pregnant women) or having pre-existing chronic medical conditions were excluded as these factors were considered potential confounders of the outcomes of interest. Studies conducted in other languages were not considered due to translation issues. However, there was no published article found other than English including in Amharic (Ethiopian official language). Eligible studies were independently screened and evaluated by two authors following the predefined data fields.

Information sources

First, we developed search strategies, identified key terms, and selected databases. These were developed in collaboration with all authors. We also searched PubMed Medical Subject Headings (MeSH) for additional keywords to expand our search scope to include all available evidence. Before we performed the main search, the search method was piloted in the selected databases. We did not use any restrictions, including language, to maximize our search results. Overall our key terms include: "metabolic risk factors", "cardiometabolic risk factors", "metabolic syndrome", "dyslipidemia*", "dyslipidaemia*", "hyperlipidemia*", "hyperlipidaemia*", "high cholesterol", "hypercholesterolemia", "hypercholesterolaemia", "triglycerides", "hypertriglyceridemia", "hypertriglyceridaemia", "overnutrition", "malnutrition", "overweight", "obesity", "obese", "blood glucose", "impaired glucose tolerance", "glucose intolerance", "high blood glucose", "hyperglycemia", "high fasting plasma glucose", "insulin resistance", "diabetes", "diabetes mellitus", "blood pressure", "high blood pressure", "hypertension", "systolic blood pressure", "diastolic blood pressure" and "Ethiopia". Then two reviewers (TA and WT) independently searched PubMed, Embase via Scopus, ProQuest, and Web of Science databases for relevant studies from their inception to January 06, 2021. The reference lists of the included articles in this systematic review and meta-analysis and grey literatures were reviewed and hand searched to identify additional relevant articles. The search results were then exported to Endnote X9 reference manager software and to Covidence for removing duplicates, screening, and study selection.

PubMed search strategy

(((“metabolic risk factors” OR “cardiometabolic risk factors” OR “metabolic syndrome” OR dyslipidemia* OR dyslipidaemia* OR Hyperlipidemia* OR hyperlipidaemia* OR “high blood cholesterol” OR hypercholesterolemia OR hypercholesterolaemia OR triglycerides OR hypertriglyceridemia OR hypertriglyceridaemia) OR (overnutrition OR malnutrition OR overweight OR obesity OR obese) OR (“blood glucose” OR “impaired glucose tolerance” OR

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"glucose intolerance" OR "high blood glucose" OR hyperglycemia OR "High fasting plasma glucose" OR "insulin resistance" OR diabetes OR "diabetes mellitus") OR ("blood pressure" OR "high blood pressure" OR hypertension OR "systolic blood pressure" OR "diastolic blood pressure")) AND (Ethiopia))

Study selection

To perform the screening, the records identified from the search results were exported to Endnote X9 software. After duplicate records were removed, two reviewers (TA and WT) independently screened the titles and abstracts of the retrieved articles guided by the predefined PEO and eligibility criteria. Following the selection of articles based on their titles and abstracts, a full text review was conducted for the selected studies for further relevance and inclusion. Any disagreements during the screening process were resolved via discussion.

Data collection process

Two authors (TA and WT) independently extracted key data from the included studies. The data extraction was conducted based on the modified version of the Joanna Briggs Institute (JBI) Meta-Analysis of Statistics and Review Instruments (JBI-MAStARI).[11] Data extracted included study characteristics (e.g., authors name, year of study, study setting, and study design, type of metabolic risk factors studied), participant characteristics (age, gender, disease condition) and study outcome measures (prevalence of metabolic risk factors). Any disagreements encountered during information extraction were resolved via discussion.

Outcome measures

The main outcome of interest for this study were the prevalence of metabolic risk factors among adults in Ethiopia reported in the original articles in percentage (number of cases/ total number of participants). We examined four metabolic risk factors: overweight/obesity, impaired glucose tolerance, hypertension, and metabolic syndrome. The outcome variables were categorized based on the following definitions: Overweight, defined as a body mass index (BMI) of 25-29.9 kg/m^2 and obesity as BMI ≥ 30 kg/m^2); [2,12] Impaired glucose tolerance (a fasting plasma glucose of 100 to 125 mg/dL or oral glucose tolerance test greater between 140 and 199 mg/dL); [13] High blood pressure (a blood pressure cut-off value of ≥140/90 mmHg on the average of blood pressure measurements using a sphygmomanometer); [2,14] and metabolic syndrome (three or more of the following based on adult treatment panel (ATP III) criteria: elevated triglyceride (≥150mg/dL), reduced high density lipoprotein(<40mg/dl for male, and <50 for female), high blood pressure (≥130/85 mg/dl, high fasting plasma glucose (≥110

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mg/dl), or increased waist circumference (≥102 cm for men and ≥88 cm for women); [15] or waist circumference of ≥94 cm for men and ≥80 cm for women, and two of elevated triglycerides (≥150 mg/dl) or treatment for this lipid abnormality; reduced HDL-C (<40 mg/dl in men and <50 mg/dL in women) or treatment for lipid abnormality; high blood pressure BP (≥130/85 mm Hg) or treatment for hypertension; impaired fasting serum glucose (≥100 mg/dl) or previously diagnosed with type 2 diabetes based on international diabetes federation criteria).[16] Additional outcomes include the variation of the metabolic risk factors by age and gender of participants.

Risk of bias assessment

Two reviewers (TA and WT) independently assessed the quality of the included studies using Joanna Briggs Institute (JBI) checklist[17] developed for methodological quality assessment of prevalence studies that includes criteria of inclusion, description of subjects and the setting, reliability and validity of exposure measurement, criteria of measurement, identification and strategies to deal confounding factors, validity and reliability of outcomes measurement, and statistical analysis used. The studies were scored on a scale of 0 to 1 for 8 items and studies with a total score of <4, 4-6 and 7-8 were considered to have low, moderate, and high methodological quality, respectively. Any disagreements in ratings the studies were resolved through discussion between the two reviewers.

Data analysis

Relevant information on the study area, study setting, study design, study sample, study response rate, and study outcome were extracted using Microsoft Excel and exported to Stata version 16 (StataCorp LLC, College Station, TX, USA) for the meta-analysis. The characteristics of included studies was summarized and presented using tables and figures (forest plots). The random effects restricted maximum likelihood (REML) meta-analysis model was used to estimate the pooled prevalence of studies with homogenous outcomes, given it is more robust in terms of variance estimation in small sample studies.[18] The Z-statistic was used for testing associations with $\alpha \leq 0.05$ as statistical significance level. Heterogeneity across studies was assessed using the I^2 statistic. Forest plots of the prevalence estimates along the respective 95% CIs were generated for individual studies and the pooled data to show graphic summaries of the analyses. After we identified studies that reported significant association of sex and age of participants with metabolic risk factors, log odds ratio was used to determine

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the pooled association between each metabolic risk factor and the sex and age of the participants.

Publication bias

The presence of significant publication bias was assessed using funnel plot and the Begg and Egger's test. The symmetry of the funnel plot was visually inspected to assess the normal distribution studies around the centre of funnel plot.[19] Then, the Begg's funnel plot and Egger test for small-study effects[20] were used to objectively assess the risk of bias across studies.

Subgroup analysis

Subgroup analyses were performed to examine the difference in prevalence of metabolic risk factors across the study regions: based on the nine regions and two city administration of the country; Years of the study: until 2000, 2000 to 2010, and 2011 to 2020; Study settings: community-based versus institution-based; And sample size: less than 400 versus 400. The sample size classification was based on the authors' discussion and consensus to compare pooled prevalence of studies with small sample size and large sample size. This. The results of subgroup analysis were presented in tables with prevalence's, 95% confidence intervals, and I^2 statistics test results. The statistical significance for subgroup analysis results was also determined at a p-value less than 0.05.

Trim-and- fill analysis

Trim-and-fill analysis were used to assess the effect of remaining studies that might have been included and fill imputed studies based on bias corrected pooled prevalence's. Statistical significance was declared if the p-value is <0.05.

Results

Study selection

A total of 6087 records were identified from the search result using databases and hand search of reference lists of the included studies. After removal of duplicates (n=2900), articles were screened based on their titles and abstracts. Then by using the inclusion and exclusion criteria ineligible articles (n=3045) were removed. A total 142 records were eligible for full-text screening. Of these, 14 articles were excluded due to their participants including children and adolescents, 27 studies were excluded due to participants with confounding chronic diseases like diabetes, 2 articles were excluded due to previous similar reports, 18 studies were excluded because they did not explicitly record, measure, or report the outcomes, and 7 articles were

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excluded due to their study design (case control). No unpublished relevant study was found, and no records were excluded due to a low-quality assessment. Finally, 74 articles involving a total of 104,382 participants were included in this systematic review and meta-analysis (figure 1).

Study characteristics

Out of 74 studies, 36 of them reported a point prevalence on hypertension,[21-56] 14 on impaired glucose tolerance,[57-70] 18 on overweight or obesity, [71-86] and the remaining eight studies on metabolic syndrome.[87-94] The studies were conducted in — Amhara (n=22), [22,25,26,29,31,35,38,42,47,50,58-60,62,67,69,72,75-78,86] Oromia (n=9), [32,33,43,49,66,68,84,88,94] Southern Ethiopia (n=15), [28,34,37,41,48,52,54-56,61,64,71,82,83,91] Tigray (n=5), [23,39,40,70,80] Somali (n=1), [27] Addis Ababa city administration (n=14), [21,24,30,44,45,51,53,57,63,65,74,85,87,89,90] Dire Dawa city administration (n=2) [36,93] regions, and five [46,73,79,81,92] of them were nationwide studies. All studies were cross-sectional in nature and most of them (n=52) were community-based studies. The sample sizes for the studies ranged from 68 to 10,938 participants (supplementary table 1).

Risk of bias

Amongst the included studies in this systematic review and meta-analysis, 66 (89%) had low risk of bias, while the remaining 8 (11%) of studies had moderate level of risk of bias. None of the included studies had high risk of bias (supplementary table 1).

Prevalence of hypertension

A total of 36 studies involving 47,204 adult participants reported on prevalence of hypertension were included in this analysis. Of these studies, most were conducted in Amhara region (n=11),[22,23,25,26,29,31,35,38,42,47,50] Addis Ababa (n=7),[21,24,30,44,45,51,53] Southern Ethiopia (n=9)[28,34,37,41,48,52,54-56] and there was one nationwide study. [46] The prevalence of hypertension among individual studies ranged between 3% in Amhara region [50] to 34.7% in Addis Ababa.[30] Overall, the pooled prevalence of hypertension was 21.1 (95% CI: 18.7- 23.4) but with a significant level of heterogeneity among the included studies (I^2= 97.6%; p≤ 0.001) (figure 2).

Prevalence of overweight/obesity

With regard to overweight/obesity, 16 studies involving 28,822 adult participants reported on prevalence of overweight/obesity were included in this analysis [71-86]. Of these, six studies were from the Amhara region.[72,75-78,86] Looking at the distribution of overweight or obesity, the prevalence span from 9.7% from a nationwide study[79] to 56.5% in a study conducted in Addis

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Ababa. [84] The pooled prevalence of overweight/ obesity was estimated at 23.9% (95% CI: 19.9-28.0) but the included studies had significant level of heterogeneity (I^2= 98.8%; p≤ 0.001) (figure 3).

Prevalence of impaired glucose tolerance

Fourteen studies covering 12,604 adult participants and reporting prevalence of impaired glucose tolerance were included in this meta-analysis. [57-70] Six of these studies were conducted in Amhara region[58-60,62,67,69] and three in Addis Ababa.[57,63,65] The prevalence of impaired glucose tolerance ranged from 2.6% in Southern Ethiopia[61] to 21.7% in Addis Ababa.[65] Overall, the pooled prevalence of impaired glucose tolerance was 12.4% (95% CI: 8.7-16.1) but there was heterogeneity among the included studies in this meta-analysis (I^2= 98.3%; p≤ 0.001) (figure 4).

Prevalence of metabolic syndrome

Besides metabolic syndrome, eight studies targeting 15,932 adult participants which reported on prevalence of metabolic syndrome were included in this analysis.[87-94] Three of the these studies [87,89,90] were from Addis Ababa, while the remaining were from Southern Ethiopia,[91] Oromia,[88,94] Dire Dawa, [93] and nationwide.[92] Metabolic syndrome prevalence varied from 4.8% (95% CI: 1.73%,7.87%) based on national sample [92] to 26% (95% CI: 19.61%,32.39%) in Oromia region.[88] In this meta-analysis, the pooled prevalence of metabolic syndrome among Ethiopian adults was 14.7% (95% CI: 9.8%, 19.6%) with a high level of heterogeneity among the included studies (I^2= 98.6%; p≤ 0.001) (figure 5).

Subgroup analysis

Subgroup analysis was performed to investigate sources of heterogeneity among the studies included in the pooled prevalence of metabolic risk factors using regions of study, years of study, settings of the study, and sample size. The subgroup analysis showed that the pooled prevalence of hypertension was 20.9% in studies done after 2010, 21.4% in research conducted with study participants above 400, 20.9% in studies performed at community level, 27.0% in studies done in Addis Ababa. The result of subgroup analysis also revealed the pooled prevalence of impaired glucose tolerance was 11.5% in studies done later 2010, and 12.9% in research done with sample size less than or equal to 400, 15.6% in studies conducted in institution, and 18.6% in studies done in Addis Ababa city administration. With regard to the pooled prevalence of overweight/obesity, it was 23.9% in research performed since 2010, 24.6% in research done in Amhara region, 22.3% in studies done at community level, and

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38.6% in original studies conducted with study participants up to 400. Similarly, the pooled prevalence of metabolic syndrome was 23.0% in research performed with study participants less than or equal to 400, 14.7% in studies done in the last ten years, and 18.9% in studies performed in institutions (table 1).

Table 1. Prevalence of metabolic risk factors among adults in Ethiopia grouped by characteristics of included studies.

Type of metabolic risk factor	Subgroup		No of studies	Prevalence with 95% CL	I^2 and p value
Hypertension	Year of study	Before 2000	1	3.00 (0.84, 5.15)	-
		2000 to 2010	3	25.12 (20.10, 30.15)	47.4%, p= 0.14
		After 2010	32	20.90 (17.78, 24.01)	86.2%, p≤ 0.001
	Region of the study	Addis Ababa city administration	7	27.05 (22.63 to 33.47)	86.1%, p= 0.00
		Amhara	10	19.89 (12.54 to 27.25)	95.7%, p≤ 0.001
		Oromia	4	19.18 (11.08 to 27.30)	88.1%, p=0.61
		SNNP	9	19.25 (15.22, 23.18)	77.5%, p≤ 0.001
		Tigray	3	16.83 (13.64, 20.02)	0.0%, p= 0.86
		Somali	1	28.30 (21.74, 34.85)	-
		Dire Dawa city administration	1	24.40 (18.13, 30.66)	-
		Ethiopia	1	15.60 (10.21, 20.98)	-
	Setting	Community based	26	20.97 (17.23, 24.71)	92.0%, p≤ 0.001
		Institution based	10	21.12 (14.80, 27.44)	92.3%, p≤ 0.001
	Sample size	≤ 400	5	17.98 (6.90, 29.07)	85.9%, p≤ 0.001
		> 400	31	21.43 (18.67, 24.19)	96.4%, p≤ 0.001
Impaired glucose tolerance	Year of study	Before 2010	1	21.60 (15.578, 27.622)	-
		After 2010	13	11.49 (8.18, 14.81)	88.6%, p≤ 0.001
	Region of the study	Amhara	6	9.70 (5.73, 13.68)	82.4%, p≤ 0.001
		Addis Ababa city administration	3	18.56(12.59, 24.52)	69.8%, p= 0.037
		SNNP	2	8.99 (-4.028, 22.02)	95.2%, p≤ 0.001
		Oromia	2	13.80 (10.16, 17.43)	0.0%, p=0.609
		Tigray	1	12.50 (7.55, 17.45)	-
	Setting	Community based	10	10.87(7.21, 14.53)	88.6%, p≤ 0.001
		Institution based	4	15.58 (8.20, 22.96)	87.9%, p≤ 0.001
	Sample size	≤ 400	5	12.91 (8.49, 17.32)	75.7%, p=0.02
		> 400	9	11.76 (7.36, 16.17)	91.4%, p≤ 0.001
Overweight/obesity	Year of study	Before 2010	1	27.20 (20.72, 33.67)	-
		After 2010	15	23.90 (18.14, 29.67)	93.3%, p≤ 0.001
	Region of the study	Amhara	6	24.66 (15.97, 33.35)	92.1%, p≤ 0.001
		Ethiopia	3	12.90(9.33, 16.47)	34.8%, p=0.216
		Addis Ababa city administration	2	25.56 (21.07, 30.05)	0.0%, p=0.601
		SNNP	3	20.94 (9.79, 32.09)	91.0%, p≤ 0.001
		Tigray	1	30.10 (23.42, 36.77)	-
	Setting	Community based	13	22.31 (17.27, 27.36)	90.0%, p≤ 0.001
		Institution based	3	31.97 (8.37, 55.56)	97.6%, p≤ 0.001
	Sample size	≤ 400	3	38.57 (21.83, 55.29)	94.1%, p≤ 0.001
		> 400	13	20.78 (16.04, 25.51)	89.1%, p≤ 0.001
Metabolic	Year of study	Before 2010	2	14.56 (10.17, 18.96)	18.9%, p=0.236

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syndrome		After 2010	6	14.72 (8.23, 21.21)	91.1%, p≤ 0.001
	Region of the study	Addis Ababa	3	16.37 (11.87, 20.87)	51.0%, p=0.13
		Oromia	2	22.91 (17.13, 28.68)	43.6%, p=0.18
		SNNP	1	9.60 (5.167, 14.03)	-
		Ethiopia	1	4.80 (1.72, 7.87)	-
		Dire Dawa city administration	1	9.50 (5.08, 13.91)	
	Setting	Community based	3	7.63 (4.22, 11.04)	55.6%, p= 0.105
		Institution based	5	18.93 (14.53, 23.32)	66.3%, p= 0.018
	Sample size	≤400	2	23.01 (17.43, 28.59)	39.4%, p=0.19
		>400	6	11.91 (7.39, 16.42)	83.0%, p≤0.001

Publication bias

The presence of publication bias was first assessed subjectively by observing the symmetrical distribution of the included studies in the funnel plot. The funnel plot test result showed that all the included studies in the pooled prevalence of metabolic risk factors were symmetrically distributed around the center within the 95% confidence limits (figure 6-9). Then, Eggers test for funnel plot was conducted. The result of Egger's test for the funnel plot revealed that there was no publication bias in the included studies for the pooled prevalence of hypertension (p = 0.411). But there was publication bias for the pooled prevalence of overweight/obesity, impaired glucose tolerance, and metabolic syndrome in the included studies ($p \leq 0.001$ for overweight/obesity, $p \leq 0.001$ for impaired glucose tolerance, and $p = 0.005$ for metabolic syndrome) (supplementary figure 1- 4)

Trim-and-fill analysis for the prevalence of metabolic risk factors

Trim-and -fill analysis was conducted to adjust publication bias for pooled prevalence of metabolic risk factors with the estimation of number of missing studies that might exist. The result showed that there were no studies imputed for missing studies for the pooled prevalence of overweight/obesity. Whereas the trim and fill analysis revealed three studies were imputed for the pooled prevalence of impaired glucose tolerance and after adjustment the estimated pooled prevalence of impaired glucose tolerance was 10.5 (7.2,13.7). While for the pooled prevalence of metabolic syndrome one studies was imputed for missing analysis and the estimated prevalence after adjustment was 13.3 (8.7, 17.9). Note that trim-and -fill analysis was not performed for hypertension since there was no publication bias among the included studies (supplementary figure 5 - 7).

The association of the sex of participants and prevalence of metabolic risk factors

To determine the association between metabolic risk factors and sex of participants log odds ratio was used after identifying studies reported significant association between sex of

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participants and the prevalence metabolic risk factors. Overall, 18 hypertension studies,[25-30,32,37,40-42,45,46,48,52,56,95] 3 impaired glucose tolerance studies,[61,65,67] 7 overweight/obesity studies,[71,75,80,82,83,85,86] and 7 metabolic syndrome studies [87-89,91-94] reported a significant link between prevalence and the sex of participants. The result of log odds ratio's analysis revealed that the prevalence of hypertension, impaired glucose tolerance, and metabolic syndrome did not show significant difference between men and women. However, a significant difference was observed in the prevalence of overweight/obesity between men and women, with women having 60% higher chance of being overweight/obesity than men [AOR = 0.41 (95% CL: 0.26-0.65)] (figure 10).

The association of the age of participants and prevalence of metabolic risk factors

Similarly, to determine the association of age of participants and the prevalence of metabolic risk factors log odds ratio were calculated after identifying studies reporting significant associations between the age of participants and prevalence metabolic risk factors. Out of 36 studies on hypertension, 26 studies [22,24,26-32,34,35,37-45,47,48,51,54-56] reported significant association between the age of the participants and the prevalence of hypertension. Similarly, nine studies [71,73-75,77,79,81,82,84] on overweight/obesity, six studies [57,58,60,61,63,70] on impaired glucose tolerance, and six studies [87,88,91-94] on metabolic syndrome reported significant associations between the age of participants and the respective prevalence. The result of log odds ratio showed that the prevalence of metabolic risk factors was significantly associated with the age of participants. The prevalence of hypertension was 3.4 times higher in people aged 45 years and older than people below 45 years old [AOR =3.4 (95% CI: 2.6-4.2)]. The prevalence of overweight/obesity was 1.6 times higher among people above 45 years old than participants below 45 years old [AOR = 1.6 (95% CI: 1.1-2.4)]. Similarly, impaired glucose tolerance [AOR =2.6 (95% CI: 1.4-4.8)] and metabolic syndrome [AOR =3.3 (95% CI: 2.2-4.7)] were significantly higher among people above 45 years old *(*figure 11)

Discussion

To the best of our knowledge, this is the first systematic review and meta-analysis to comprehensively examine the prevalence estimates of metabolic risk factors for NCDs in Ethiopia. There was one review[95] reporting the pooled prevalence of hypertension, but this was conducted more than five years ago and considered all adults irrespective of co-existing conditions. Given the lack of comprehensive evidence on risk factors of NCDs at national level,

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this review targeted metabolic risk factors for NCDs in Ethiopia. The findings of this review are likely to inform future policy decisions.

The findings revealed the high prevalence of metabolic risk factors for NCDs in Ethiopia — nearly one in five participants suffer from at least one of the studied metabolic risk factors (hypertension, overweight/obesity, impaired glucose tolerance, and metabolic syndrome). This aligns with previous reports on the prevalence of NCDs in Ethiopia, where more than a quarter of deaths were attributed to NCDs,[96] and is also similar to the 2014 report by the WHO.[2] Metabolic risk factors are immediate indicators for most NCDs, especially for cardiovascular diseases and DM. They are highly associated with, and reinforced by, behavioral risk factors such as alcohol consumption, tobacco smoking, poor dietary habits, and physical inactivity.[2] The high level of poverty, and unemployment could also contribute to the metabolic risk factors.[97-99] Therefore, the high prevalence of the metabolic predictors for NCDs in this study potentially indicate the presence of various behavioral and environmental stressors underlying these conditions that need to be examined in future studies.

In this study, we found a 21.1% pooled prevalence of hypertension, which is slightly higher than the 19% reported in a previous review in Ethiopia,[95] but slightly lower than the 24% prevalence reported by the WHO.[100] Also, the pooled prevalence is lower than the prevalence (57%) in older adults found in a multiple country African review,[101] or the prevalence (32.3%) of hypertension reported in LMICs collectively.[102] The change in hypertension prevalence observed over the past five years could be due to the change in lifestyle (behavioral) or other factors linked with hypertension, although it is also important to note a smaller number of studies were included in the previous review.[95] Regarding the study in African countries,[101] the higher prevalence could be due to the inclusion of older cohort, a population group already at risk of high blood pressure. Also, the prevalence of smoking[103] and alcohol consumption[104] is lower in Ethiopia compared to the rest of Africa. Although the exact causes of hypertension in Ethiopia are not entirely understood, several factors such as smoking, overweight or obesity, physical inactivity, high salt intake, and alcohol consumption habits would have undeniable role for the reported high prevalence. Also, other factors like stress, increased age, urbanization, and genetics could contribute to the reported prevalence of high blood pressure.

Our findings also indicated that the pooled prevalence of overweight / obesity was 23.9%, which is lower than the global prevalence (39%)[105] but higher than the prevalence of overweight (15.9%) reported in demographic health survey analysis of 32 sub-Saharan African

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countries[106] and among urban women reported by 2011 Ethiopian demographic health survey (12%).[73] Poor diet intake including high levels of simple carbohydrates, increased caloric consumption, use of medications, reduced physical activity, and other psychosocial factors are among the predictors of overweight/obesity that could also be applicable in the Ethiopian context. In addition, although not considered in this study, early age malnutrition (including undernutrition) causes metabolic abnormalities like hypothyroidism and insulin resistance later in life and thus could contribute to the increasing overweight and obesity in the country.[107,108] Therefore, due attention should be given to the potentially modifiable determinants of overweight or obesity to mitigate the subsequent risk of NCDs earlier in life.

This study also showed that the pooled prevalence of impaired glucose tolerance in Ethiopia was 12.4%. This finding is higher than the prevalence (9.1%) reported in 2015 national NCDs survey,[92] the pooled prevalence (6.0%) among childbearing age women in sub-Saharan Africa;[109] comparable to the prevalence (12.0%) of impaired glucose homeostasis in sub-Saharan Africa;[110]and lower than the prevalence (22.1%) reported in patients with aldosteronism.[111] The differences may have occurred due to methodological variations, participants characteristics and the available evidence in respective settings. Nevertheless, our finding is a stark reminder on the burden of impaired glucose tolerance putting significant proportion of the adult population at greater risk of DM and other complications. This underlines the need to adequately and timely manage factors that contribute to impaired glucose tolerance, including less physical inactivity (sedentary lifestyle), overweight or obesity, family history of DM, history of high blood pressure, and high cholesterol.

We found a 15% pooled prevalence for metabolic syndrome in this study. This is much higher than a national NCDs survey that reported 4.8% prevalence [92] and comparable to a 16% pooled prevalence among HIV patients.[112] However, higher prevalence were reported in people with psychiatric disorders (19%),[113] arthritis (30.5%),[114] and sedentary lifestyle (73%).[115] The differences in the prevalence among the studies could be explained by the fact that we only considered healthy adult populations without any underlining health conditions in this study, which could have underestimated the magnitude of the burden in the community.

High levels of triglycerides and metabolic syndromes cause significant proportion of heart diseases thereby contributing to the ever-increasing prevalence of NCDs in LMICs and high-income countries alike.[2] This is because of the thickening and hardening of the walls of blood vessels (arteries), which are most of the time associated with poorly controlled obesity/diabetes

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and improper feeding of saturated fat, sugar, salty foods and limited intake of fruits and vegetables.[116] Poor diet with high-fat content and highly refined, processed foods like sugars, white bread, and dry cereals are associated with greater risks of developing insulin resistance and all other aspects of the metabolic syndrome.[117] Thus, it is important to educate people on the value of healthy diets (which are low in cholesterol, sodium, and sugar contents), staying physically active, quitting smoking, and decreasing alcohol intake to reduce metabolic syndrome and its long-term consequences.

Finally, metabolic risk factors showed variation based on gender — overall, risks appeared to be higher in women. This may seem counterintuitive given some behavioral risk factors like smoking and alcohol consumption, which are often associated with metabolic risk factors like high blood pressure, are more common in men.[118] In contrast, there are studies showing the greater tendency of physical inactivity or sedentary lifestyle among women. [119] Nevertheless, a similar trend was observed in Tanzania, where women had a higher prevalence of obesity, and metabolic syndrome.[120] Consistent with our findings, a previous study also indicated that obesity is more common among women,[118] whereas, another study showed men in some LMICs tended to be physically more active.[119] These findings, collectively, suggest that the metabolic risk factors, particularly overweight/obesity, are more common in women, highlighting their increased risk for NCDs. Irrespective of the causes, the findings highlight the need to address this problem through addressing behavioral risk factors that cause metabolic disorders and promoting community screening for early detection of the conditions. Given the link between socioeconomic status and NCDs,[121,122]strategies aimed at addressing socioeconomic disparities could help mitigate factors contributing to stressful life events thereby reducing high blood pressure and subsequent NCDs.

Implications for practice

This review highlighted the high prevalence of metabolic risk factors among adults in Ethiopia. The findings will alert health care workers, health care institutions and the government to focus on prevention activities to reduce risk factors for NCDs, with special attention to the adult population.

This study informs primary health care providers and public health workers who are working in the community to consider preventions to reduce metabolic risk factors for NCDs in adults. The current primary health care programs in the country hugely focus on maternal and child health programs. However, over the past three decades, NCDs and risk factors have been

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increasing rapidly, warranting further research into the causes and strategies to rectify them. Lifestyle education and health promotion strategies mainly aiming at modifying behavioral risk factors for NCDs can be instrumental in realizing the national and global targets. Therefore, primary health care providers need to be involved in awareness creation programs through health education, community-based screening, identification, and referral of cases that need health care interventions like hypertension and diabetes. Working with adult and youth centers, campaigning and awareness creation can also help to reduce these risk factors in the community. This review showed metabolic risk factors were more common in women and older adults. Thus, considering targeted approaches tailored for these segments of the community during prevention activities will potentially result in positive outcomes.

Health care workers in hospitals and health centers should consider screening for metabolic risk factors for adults at least by measuring height, weight, blood pressure, and blood glucose for all adults along with screening for behavioral risk factors such as physical inactivity, dietary habits, tobacco usage and alcohol consumption. It will be important to include topics related to metabolic risk factors prevention for all people who come to the health facilities in the morning health education programs. Follow up and counseling should be also considered for peoples who are found to have metabolic risk factors. In addition, proper training for health care providers increases their ability to diagnose and manage metabolic risk factors and prevent NCDs.

This study will inform the regional government to prepare management guidelines to revert the rise of metabolic risk factors to decrease the burden of non-communicable diseases. Regional and national health programs should consider non-communicable diseases preventions as one of their majors by designing appropriate interventions such as by preparing protocols and guidelines, equipping primary health care setups with human power, resource, and making accessible for the community. The government of Ethiopia should also consider strategies working on social determinants of health, such as urban planning, education, employment, and public safety which help to reduce risk behaviors to reduce burden metabolic risk factors. Further, it is important to have proper program evaluation in place to oversee the implementation of the strategies.

Implications for research

Based on the result of this review, the following recommendation are drawn for future research and researchers.

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This review showed that most studies used relatively small sample sizes to represent the community, most of which were conducted by less than a thousand participants. Although they were statistically representative of the community, we recommend large scale surveys to guide public health interventions and inform decision making. Nearly all studies were cross sectional in nature that showed the problem at a specific point of time. Thus, long term studies, such as prospective cohorts can help to identify the causes of metabolic risk factors in the community.

No intervention has been tested to reduce the metabolic risk factors in Ethiopia. Thus, to identify feasible and effective interventions researchers should consider randomized control trails. Community based interventional studies help to inform the governments interventions that are efficient to reduce metabolic risk factors. In addition, studies additional studies are needed to understand behavioral factors that contribute metabolic risk factors to design and implement appropriate interventions.

Social determinants of health such as gender and age were found to be associated with metabolic risk factors. They are currently the main research and public health agenda in many countries. We also recommend Ethiopian researchers to assess the social gradients of health such as gender, age, education, occupation, income, housing, physical and social environment, public safety, transportation, and access to health care.

Strengths and limitations

This review is the first of its nature combining studies with similar characteristics in the context of metabolic risk factors for NCDs in Ethiopia in a single study, producing strong, evidence to inform decision making. This study strictly adhered to the Preferred Reporting Items for Systemic Reviews and Meta-analysis (PRISMA) guidelines during the design, conducting and reporting of our findings to increase robustness of the methods. The review uses prevalence studies to estimate the level of metabolic risk factors in adults, and most of the included studies had low risk of bias. The number of participants incorporated in this review were more than hundred thousand. Thus, our prevalence estimate will fairly represent the adult population in Ethiopia. Furthermore, the study showed age and sex differences of metabolic risk factors in adults which make our result easy to understand, interpret, put in action.

However, the review has no a priori registered protocol. Although most of the included studies in this review had low risk of bias to represent prevalence's of metabolic risk factors among adults in Ethiopia, we did not find studies conducted in Afar, Benishangul-Gumuz, Gambella regions, so our prevalence estimates might not precisely reflect the real situations in these

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regions. In addition, the higher level of prevalence for some studies and the low sample size could have somehow hampered the pooled prevalence and thereby limiting to highlight the true nature of the problem. The heterogeneity among the included studies precluded us from performing further analyses to identify individual study and clinical factors contributing to metabolic risk factors.

Conclusion

This review showed a high level of metabolic risk factors for NCD in Ethiopia — one in five persons has at least one of the metabolic risk factors. Since metabolic risk factors are immediate precursors for NCDs, they need to be managed and, when possible, reversed to reduce their long-term consequences on health-related quality of life and other health outcomes. These results are important reminders of the need to effectively implement the multilayered strategies proposed by the Ethiopian government, including strengthening national health policy on NCDs and their risk factors, strengthening national and regional NCDs units, and raising awareness on the burden of these diseases among health workers and the wider community. Further, it is important to have proper program evaluation in place to oversee the implementation of the strategies. Lifestyle education and health promotion strategies mainly aiming at modifying behavioral risk factors for NCDs can be instrumental in realizing the national and global targets. This study will inform the regional government to prepare management guidelines to revert the rise of metabolic risk factors to decrease the burden of non-communicable diseases, and for researchers to consider community-based interventional studies.

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Figure legends

Figure 1 Study selection flow diagram.

Figure 2 A forest plot of prevalence (%) of hypertension among adults in Ethiopia.

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1 Figure 3 A forest plot of prevalence (%) of overweight/obesity among adults in Ethiopia.

2 Figure 4 A forest plot of prevalence (%) of impaired glucose tolerance among adults in Ethiopia.

3 Figure 5 A forest plot of prevalence (%) of metabolic syndrome among adults in Ethiopia.

4 Figure 6 Distribution of studies included in hypertension analysis in the funnel graph.

5 Figure 7 Distribution of studies included in overweight/obesity analysis in the funnel graph.

6 Figure 8 Distribution of studies included in impaired glucose tolerance in the funnel graph.

7 Figure 9 Distribution of studies included in metabolic syndrome analysis in the funnel graph.

8 Figure 10 A Forest plot of the association gender of participants with the prevalence of
9 metabolic risk factors in Ethiopia (*female as reference*).

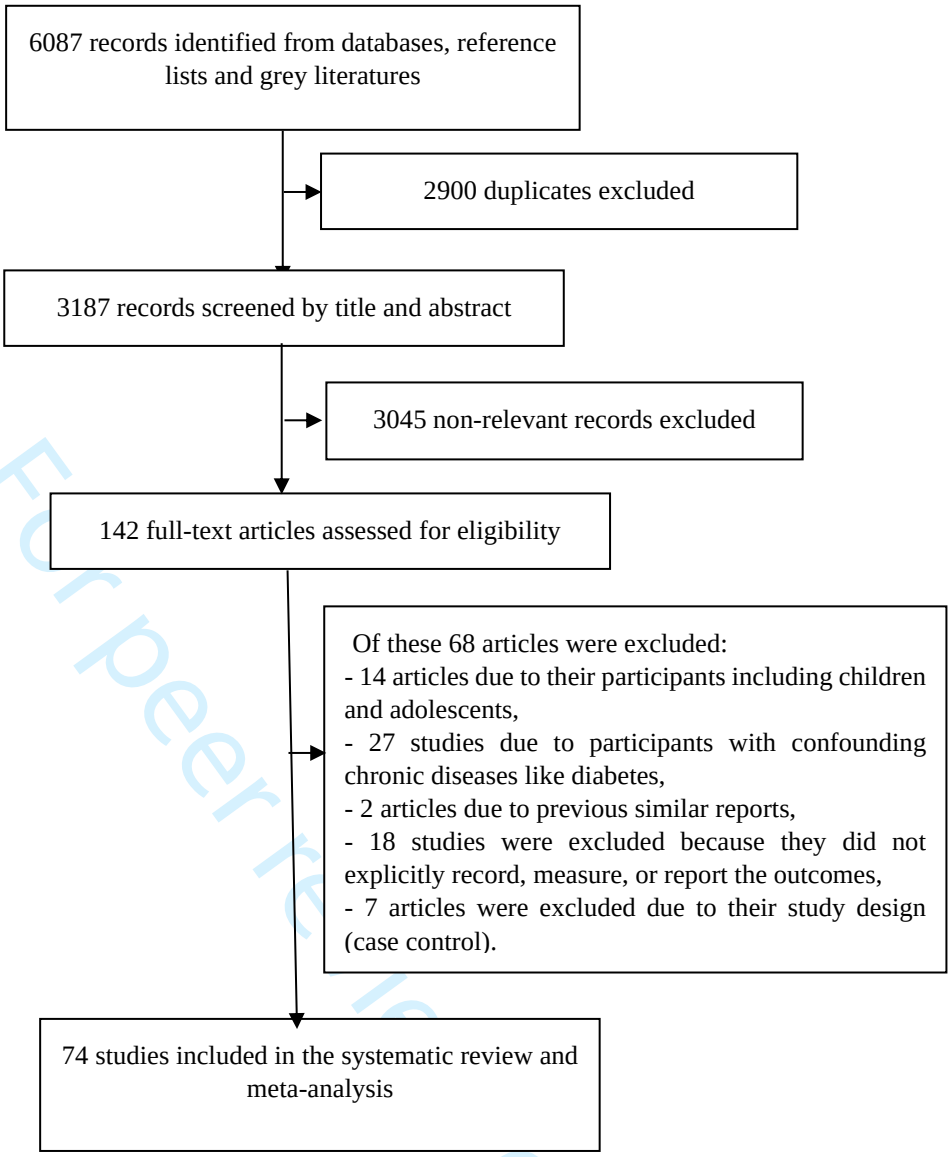
10 Figure 11 A forest plot of the association of age of participant with the prevalence of metabolic
11 risk factors in Ethiopia *(people under the age of 45 years as a reference)*.

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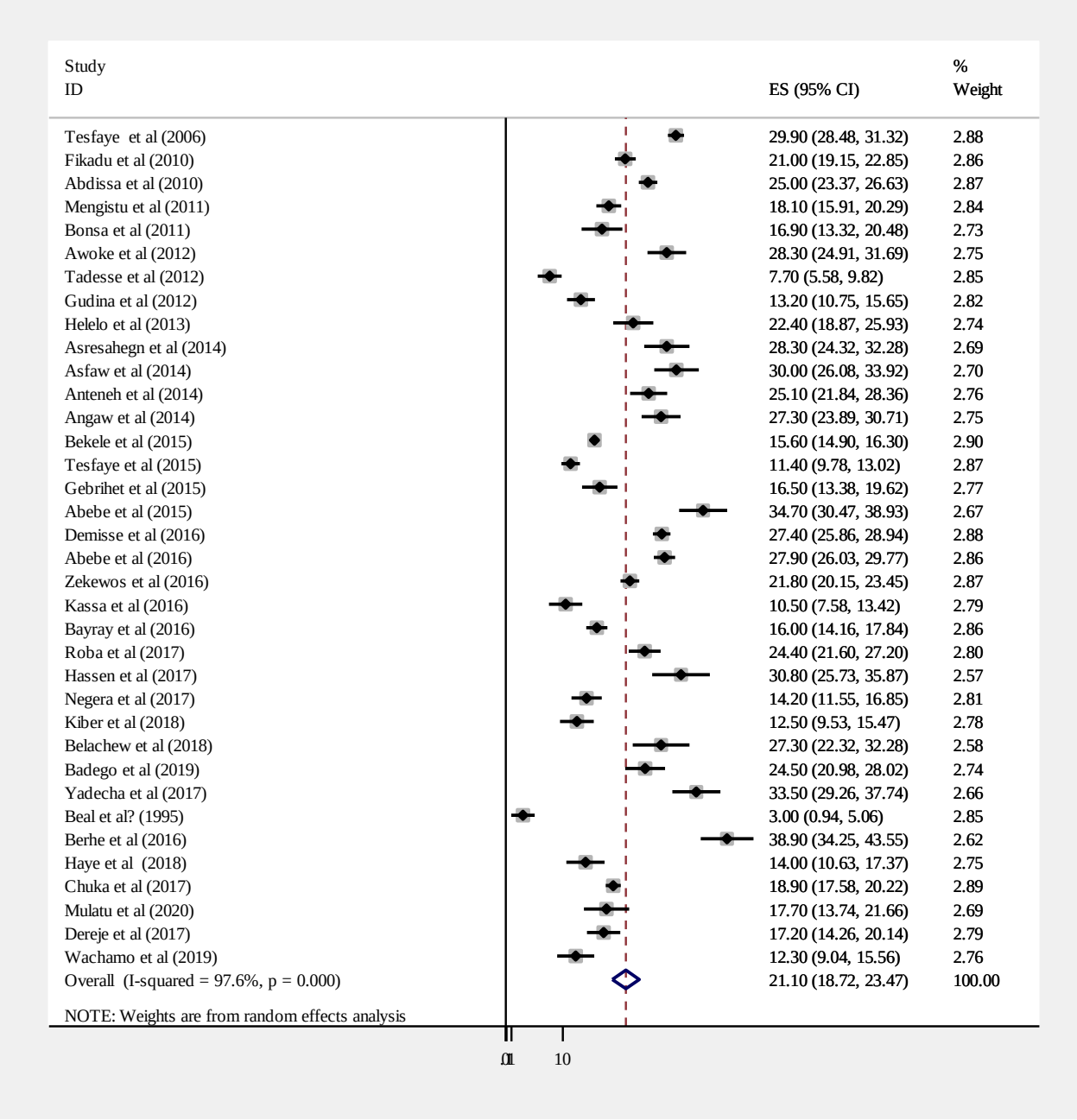
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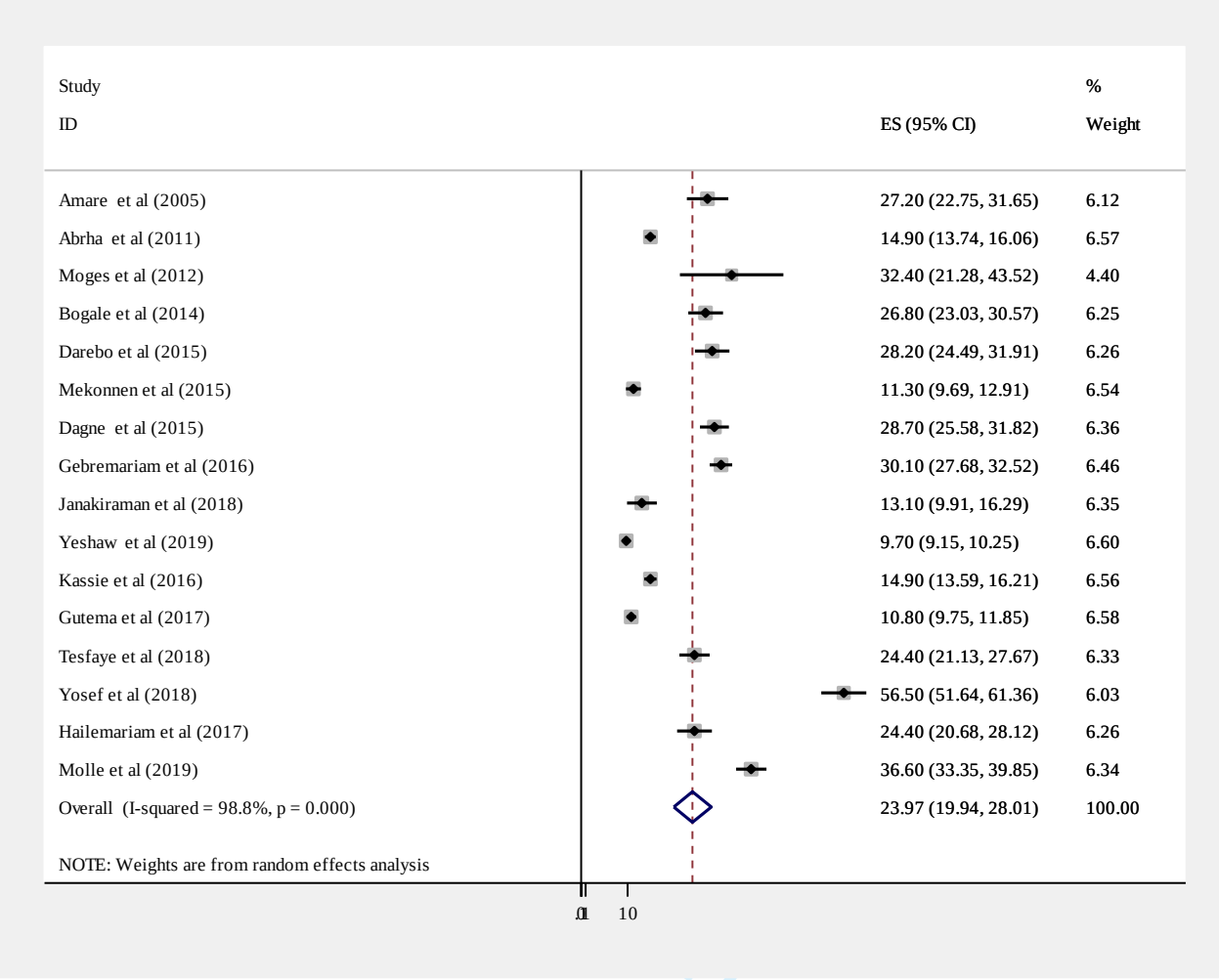
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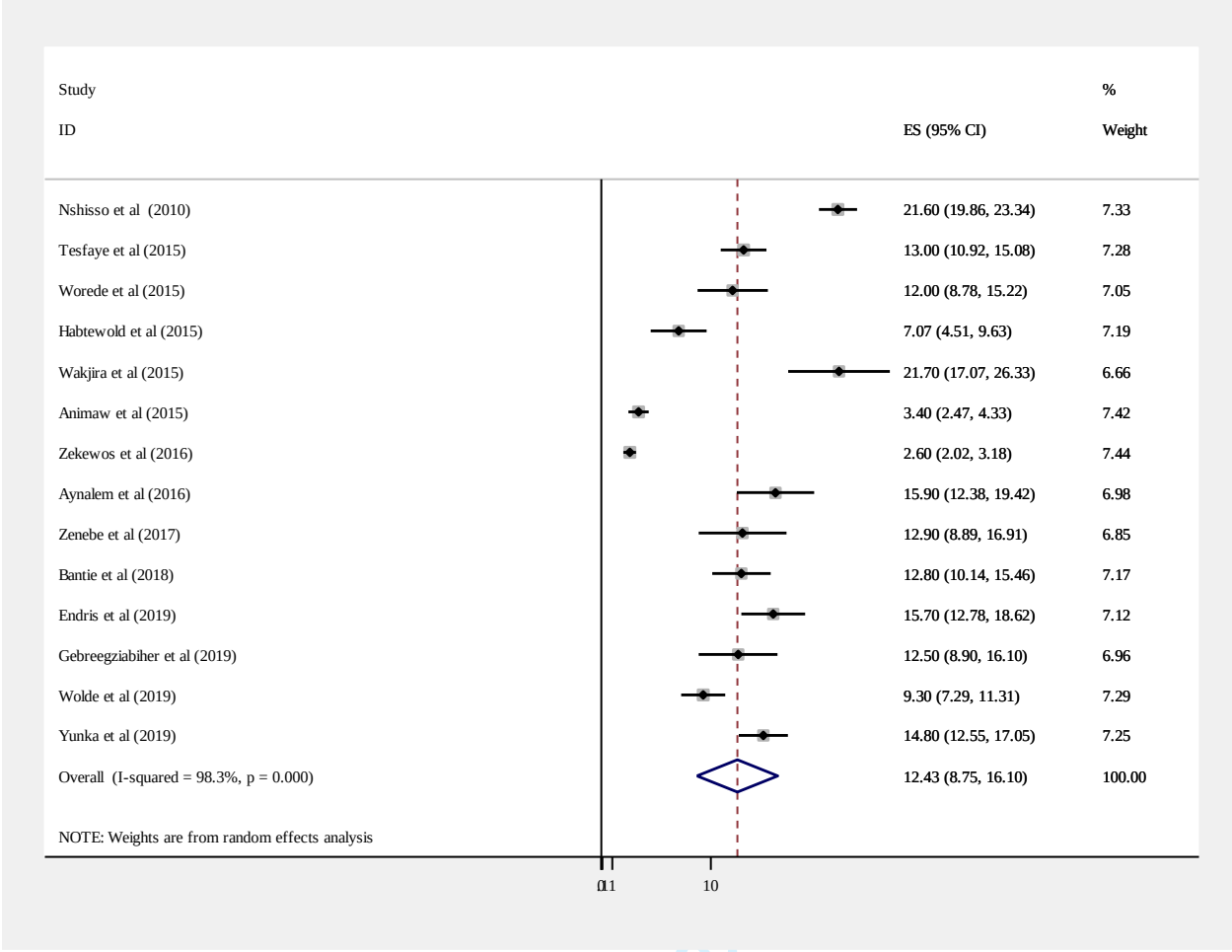
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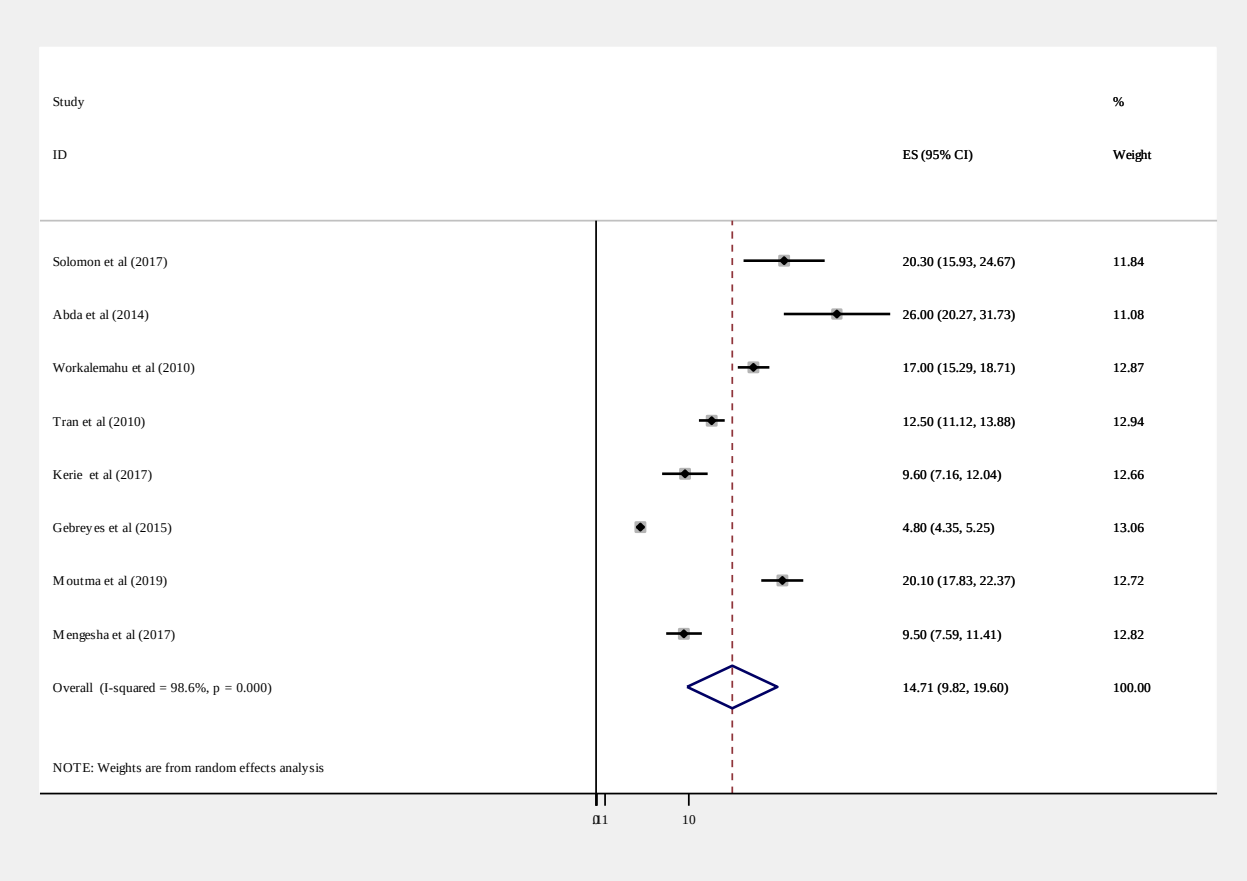

Study
ID
ES (95% CI)
%
Weight
Tesfaye et al (2006) 29.90 (28.48, 31.32) 2.88
Fikadu et al (2010) 21.00 (19.15, 22.85) 2.86
Abdissa et al (2010) 25.00 (23.37, 26.63) 2.87
Mengistu et al (2011) 18.10 (15.91, 20.29) 2.84
Bonsa et al (2011) 16.90 (13.32, 20.48) 2.73
Awoke et al (2012) 28.30 (24.91, 31.69) 2.75
Tadesse et al (2012) 7.70 (5.58, 9.82) 2.85
Gudina et al (2012) 13.20 (10.75, 15.65) 2.82
Helelo et al (2013) 22.40 (18.87, 25.93) 2.74
Asresahegn et al (2014) 28.30 (24.32, 32.28) 2.69
Asfaw et al (2014) 30.00 (26.08, 33.92) 2.70
Anteneh et al (2014) 25.10 (21.84, 28.36) 2.76
Angaw et al (2014) 27.30 (23.89, 30.71) 2.75
Bekele et al (2015) 15.60 (14.90, 16.30) 2.90
Tesfaye et al (2015) 11.40 (9.78, 13.02) 2.87
Gebrihet et al (2015) 16.50 (13.38, 19.62) 2.77
Abebe et al (2015) 34.70 (30.47, 38.93) 2.67
Demisse et al (2016) 27.40 (25.86, 28.94) 2.88
Abebe et al (2016) 27.90 (26.03, 29.77) 2.86
Zekewos et al (2016) 21.80 (20.15, 23.45) 2.87
Kassa et al (2016) 10.50 (7.58, 13.42) 2.79
Bayray et al (2016) 16.00 (14.16, 17.84) 2.86
Roba et al (2017) 24.40 (21.60, 27.20) 2.80
Hassen et al (2017) 30.80 (25.73, 35.87) 2.57
Negera et al (2017) 14.20 (11.55, 16.85) 2.81
Kiber et al (2018) 12.50 (9.53, 15.47) 2.78
Belachew et al (2018) 27.30 (22.32, 32.28) 2.58
Badego et al (2019) 24.50 (20.98, 28.02) 2.74
Yadecha et al (2017) 33.50 (29.26, 37.74) 2.66
Beal et al? (1995) 3.00 (0.94, 5.06) 2.85
Berhe et al (2016) 38.90 (34.25, 43.55) 2.62
Haye et al (2018) 14.00 (10.63, 17.37) 2.75
Chuka et al (2017) 18.90 (17.58, 20.22) 2.89
Mulatu et al (2020) 17.70 (13.74, 21.66) 2.69
Dereje et al (2017) 17.20 (14.26, 20.14) 2.79
Wachamo et al (2019) 12.30 (9.04, 15.56) 2.76
Overall (I-squared = 97.6%, p = 0.000) 21.10 (18.72, 23.47) 100.00
NOTE: Weights are from random effects analysis
10



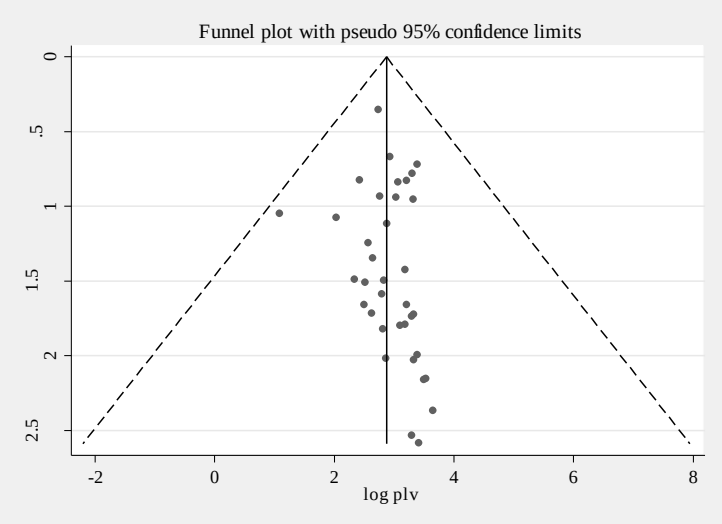
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Study
ID
ES (95% CI)
%
Weight
Nshisso et al (2010)
21.60 (19.86, 23.34)
7.33
Tesfaye et al (2015)
13.00 (10.92, 15.08)
7.28
Worede et al (2015)
12.00 (8.78, 15.22)
7.05
Habtewold et al (2015)
7.07 (4.51, 9.63)
7.19
Wakjira et al (2015)
21.70 (17.07, 26.33)
6.66
Animaw et al (2015)
3.40 (2.47, 4.33)
7.42
Zekewos et al (2016)
2.60 (2.02, 3.18)
7.44
Aynalem et al (2016)
15.90 (12.38, 19.42)
6.98
Zenebe et al (2017)
12.90 (8.89, 16.91)
6.85
Bantie et al (2018)
12.80 (10.14, 15.46)
7.17
Endris et al (2019)
15.70 (12.78, 18.62)
7.12
Gebreegziabiher et al (2019)
12.50 (8.90, 16.10)
6.96
Wolde et al (2019)
9.30 (7.29, 11.31)
7.29
Yunka et al (2019)
14.80 (12.55, 17.05)
7.25
Overall (I-squared = 98.3%, p = 0.000)
12.43 (8.75, 16.10)
100.00
NOTE: Weights are from random effects analysis
01
1
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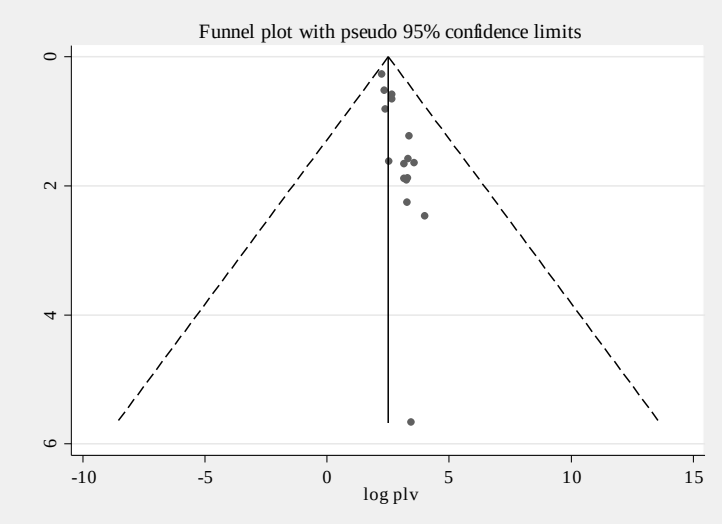
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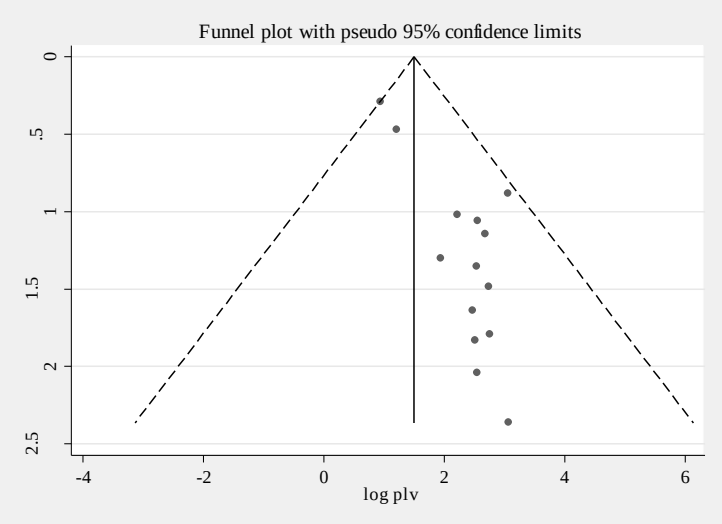
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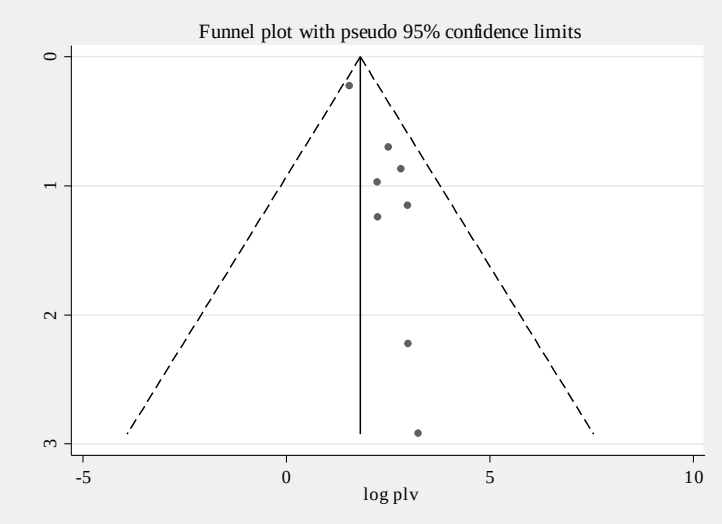
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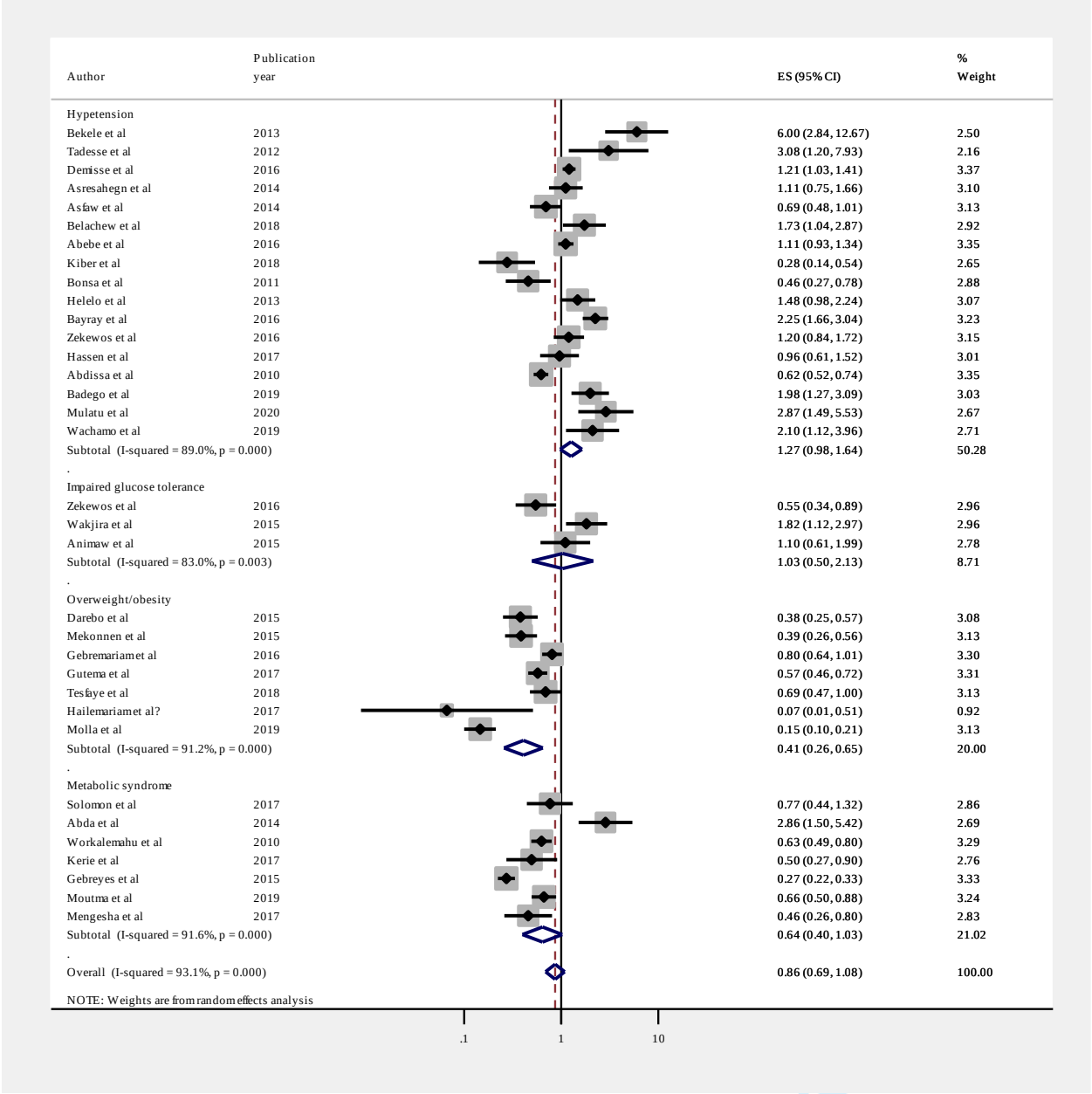
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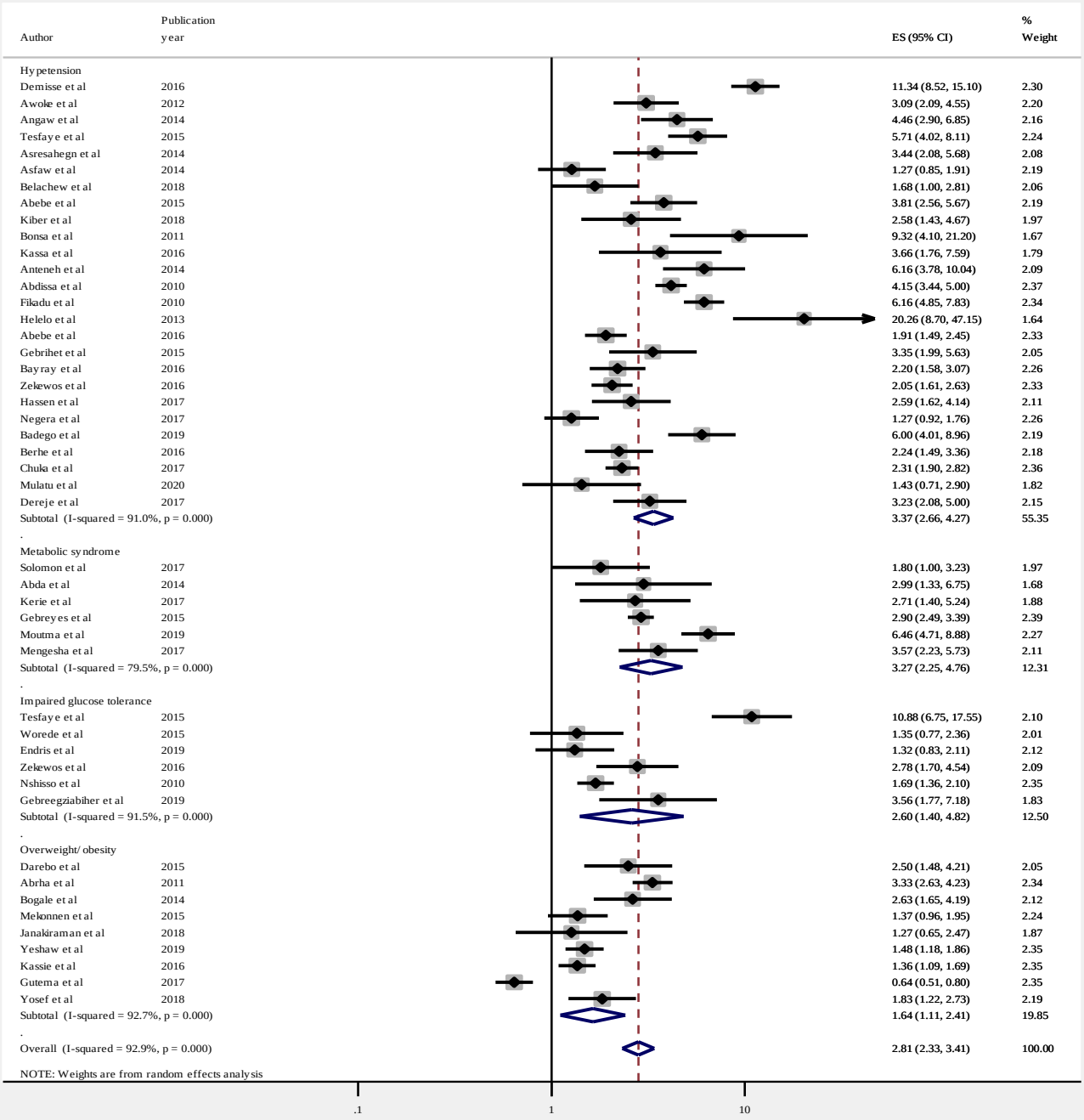
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Author
Publication year
ES (95% CI)
% Weight
Hypetension
Bekele et al 2013 6.00 (2.84, 12.67) 2.50
Tadesse et al 2012 3.08 (1.20, 7.93) 2.16
Demisse et al 2016 1.21 (1.03, 1.41) 3.37
Asresahegn et al 2014 1.11 (0.75, 1.66) 3.10
Asfaw et al 2014 0.69 (0.48, 1.01) 3.13
Belachew et al 2018 1.73 (1.04, 2.87) 2.92
Abebe et al 2016 1.11 (0.93, 1.34) 3.35
Kiber et al 2018 0.28 (0.14, 0.54) 2.65
Bonsa et al 2011 0.46 (0.27, 0.78) 2.88
Helelo et al 2013 1.48 (0.98, 2.24) 3.07
Bayray et al 2016 2.25 (1.66, 3.04) 3.23
Zekewos et al 2016 1.20 (0.84, 1.72) 3.15
Hassen et al 2017 0.96 (0.61, 1.52) 3.01
Abdissa et al 2010 0.62 (0.52, 0.74) 3.35
Badego et al 2019 1.98 (1.27, 3.09) 3.03
Mulatu et al 2020 2.87 (1.49, 5.53) 2.67
Wachamo et al 2019 2.10 (1.12, 3.96) 2.71
Subtotal (I-squared = 89.0%, p = 0.000) 1.27 (0.98, 1.64) 50.28
Impaired glucose tolerance
Zekewos et al 2016 0.55 (0.34, 0.89) 2.96
Wakjira et al 2015 1.82 (1.12, 2.97) 2.96
Animaw et al 2015 1.10 (0.61, 1.99) 2.78
Subtotal (I-squared = 83.0%, p = 0.003) 1.03 (0.50, 2.13) 8.71
Overweight/obesity
Darebo et al 2015 0.38 (0.25, 0.57) 3.08
Mekonnen et al 2015 0.39 (0.26, 0.56) 3.13
Gebremariamet al 2016 0.80 (0.64, 1.01) 3.30
Gutema et al 2017 0.57 (0.46, 0.72) 3.31
Tesfaye et al 2018 0.69 (0.47, 1.00) 3.13
Hailemariamet al? 2017 0.07 (0.01, 0.51) 0.92
Molla et al 2019 0.15 (0.10, 0.21) 3.13
Subtotal (I-squared = 91.2%, p = 0.000) 0.41 (0.26, 0.65) 20.00
Metabolic syndrome
Solomon et al 2017 0.77 (0.44, 1.32) 2.86
Abda et al 2014 2.86 (1.50, 5.42) 2.69
Workalemahu et al 2010 0.63 (0.49, 0.80) 3.29
Kerie et al 2017 0.50 (0.27, 0.90) 2.76
Gebreyes et al 2015 0.27 (0.22, 0.33) 3.33
Moutma et al 2019 0.66 (0.50, 0.88) 3.24
Mengesha et al 2017 0.46 (0.26, 0.80) 2.83
Subtotal (I-squared = 91.6%, p = 0.000) 0.64 (0.40, 1.03) 21.02
Overall (I-squared = 93.1%, p = 0.000) 0.86 (0.69, 1.08) 100.00
NOTE: Weights are from random effects analysis
.1
1
10

sex


Author
Publication year
ES (95% CI)
% Weight
Hypetension
Demisse et al 2016 11.34 (8.52, 15.10) 2.30
Awoke et al 2012 3.09 (2.09, 4.55) 2.20
Angaw et al 2014 4.46 (2.90, 6.85) 2.16
Tesfaye et al 2015 5.71 (4.02, 8.11) 2.24
Asresahegn et al 2014 3.44 (2.08, 5.68) 2.08
Asfaw et al 2014 1.27 (0.85, 1.91) 2.19
Belachew et al 2018 1.68 (1.00, 2.81) 2.06
Abebe et al 2015 3.81 (2.56, 5.67) 2.19
Kiber et al 2018 2.58 (1.43, 4.67) 1.97
Bonsa et al 2011 9.32 (4.10, 21.20) 1.67
Kassa et al 2016 3.66 (1.76, 7.59) 1.79
Anteneh et al 2014 6.16 (3.78, 10.04) 2.09
Abdissa et al 2010 4.15 (3.44, 5.00) 2.37
Fikadu et al 2010 6.16 (4.85, 7.83) 2.34
Helelo et al 2013 20.26 (8.70, 47.15) 1.64
Abebe et al 2016 1.91 (1.49, 2.45) 2.33
Gebrihet et al 2015 3.35 (1.99, 5.63) 2.05
Bayray et al 2016 2.20 (1.58, 3.07) 2.26
Zekewos et al 2016 2.05 (1.61, 2.63) 2.33
Hassen et al 2017 2.59 (1.62, 4.14) 2.11
Negera et al 2017 1.27 (0.92, 1.76) 2.26
Badego et al 2019 6.00 (4.01, 8.96) 2.19
Berhe et al 2016 2.24 (1.49, 3.36) 2.18
Chuka et al 2017 2.31 (1.90, 2.82) 2.36
Mulatu et al 2020 1.43 (0.71, 2.90) 1.82
Dereje et al 2017 3.23 (2.08, 5.00) 2.15
Subtotal (I-squared = 91.0%, p = 0.000) 3.37 (2.66, 4.27) 55.35
Metabolic syndrome
Solomon et al 2017 1.80 (1.00, 3.23) 1.97
Abda et al 2014 2.99 (1.33, 6.75) 1.68
Kerie et al 2017 2.71 (1.40, 5.24) 1.88
Gebreyes et al 2015 2.90 (2.49, 3.39) 2.39
Moutma et al 2019 6.46 (4.71, 8.88) 2.27
Mengesha et al 2017 3.57 (2.23, 5.73) 2.11
Subtotal (I-squared = 79.5%, p = 0.000) 3.27 (2.25, 4.76) 12.31
Impaired glucose tolerance
Tesfaye et al 2015 10.88 (6.75, 17.55) 2.10
Worede et al 2015 1.35 (0.77, 2.36) 2.01
Endris et al 2019 1.32 (0.83, 2.11) 2.12
Zekewos et al 2016 2.78 (1.70, 4.54) 2.09
Nshisso et al 2010 1.69 (1.36, 2.10) 2.35
Gebreegziabiher et al 2019 3.56 (1.77, 7.18) 1.83
Subtotal (I-squared = 91.5%, p = 0.000) 2.60 (1.40, 4.82) 12.50
Overweight/ obesity
Darebo et al 2015 2.50 (1.48, 4.21) 2.05
Abrha et al 2011 3.33 (2.63, 4.23) 2.34
Bogale et al 2014 2.63 (1.65, 4.19) 2.12
Mekonnen et al 2015 1.37 (0.96, 1.95) 2.24
Janakiraman et al 2018 1.27 (0.65, 2.47) 1.87
Yeshaw et al 2019 1.48 (1.18, 1.86) 2.35
Kassie et al 2016 1.36 (1.09, 1.69) 2.35
Gutema et al 2017 0.64 (0.51, 0.80) 2.35
Yosef et al 2018 1.83 (1.22, 2.73) 2.19
Subtotal (I-squared = 92.7%, p = 0.000) 1.64 (1.11, 2.41) 19.85
Overall (I-squared = 92.9%, p = 0.000) 2.81 (2.33, 3.41) 100.00
NOTE: Weights are from random effects analysis
.1
1
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Supplementary table 1: Characteristics of included studies (n=74) on metabolic risk factors for NCDs in Ethiopia.

Author	Year of study	Study area	Region of the study	Study setting	Age of participant	Types of metabolic risk factor studied	Response rate (%)	Sample size	Quality score of studies	Number of participants with Outcome	Prevalence (%)
Tesfaye et al [1]	2006	Addis Ababa	Addis Ababa City administration	Community	25–64	Hypertension	88.1	4001	7	981	29.9
Awoke et al [2]	2012	Gondar	Amhara	Community	35+	Hypertension	97.6	679	8	192	28.3
Mengistu et al[3]	2011	Humera	Tigray	Community	18+	Hypertension	100	1183	8	214	18.1
Angaw et al[4]	2014	Addis Ababa	Addis Ababa City administration	Institution	18+	Hypertension	96	655	8	172	27.3
Tadesse et al[5]	2012	Gondar	Amhara	Institution	18+	Hypertension	100	610	8	47	7.7
Demisse et a[6]	2016	Gondar	Amhara	Community	18+	Hypertension	94.8	3227	8	837	27.4
Asresahegn et al[7]	2014	Jigjiga	Somali	Community	25–65	Hypertension	98.9	492	8	138	28.3
Asfaw et al[8]	2014	Hossana	South/SNNPR	Community	25–65	Hypertension	99.8	525	7	157	30
Belachew et al[9]	2018	Bahir Dar	Amhara	Institution	18+	Hypertension	100	308	6	84	27.3
Abebe et al[10]	2015	Addis Ababa	Addis Ababa City administratio	Institution	18+	Hypertension	100	487	7	169	34.7

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Kiber et al[11]	2018	Debre Markos	Amhara	Community	18+	Hypertension	95.6	477	8	57	12.5
Bonsa et al[12]	2011	Bedele	Oromia	Community	15+	Hypertension	93.8	422	7	67	16.9
Gudina et al[13]	2012	Jimma	Oromia	Institution	15+	Hypertension	100	734	7	97	13.2
Kassa et al[14]	2016	Hawassa	South/SNNPR	Institution	18+	Hypertension	97.25	422	8	43	10.5
Anteneh et al[15]	2014	Bahir Dar	Amhara	Community	30+	Hypertension	99.6	681	8	149	25.1
Roba et al[16]	2017	Dire Dawa	Dire Dawa City council	Community	25-64	Hypertension	96.5	903	8	213	24.4
Helelo et al[17]	2013	Durame	South/SNNPR	Community	31+	Hypertension	96.6	536	7	116	22.4
Abebe et al[18]	2013	Dabat	Amhara	Community	18+	Hypertension	97.3	2200	8	598	27.9
Gebrihet et al[19]	2015	Aksum	Tigray	Community	18+	Hypertension	96	544	8	86	16.5
Bayray et al[20]	2016	Mekelle	Tigray	Institution	18+	Hypertension	99.7	1525	8	243	16
Zekewos et al[21]	2016	Bona District	South/SNNPR	Community	25+	Hypertension	82.1	2401	8	196	21.8
Hassen et al[22]	2017	Kombolcha	Amhara	Community	18+	Hypertension	100	318	8	96	30.8
Negera et al[23]	2017	Illubabor Zone	Oromia	Community	18+	Hypertension	97	668	8	92	14.2
Fikadu et al [24]	2010	Addis Ababa	Addis Ababa City administration	Community	18+	Hypertension	100	1866	8	392	21
Abdissa et al[25]	2010	Addis Ababa	Addis Ababa City	Community	18+	Hypertension	100	2716	8	677	25

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Bekele et al[26]	2015	Ethiopia	Ethiopia	Community	15-69	Hypertension	95.5	10,260	6	1527	15.6
Tesfaye et al[27]	2015	Bahir Dar	Amhara	Community	18+	Hypertension	95.5	1472	8	160	11.4
Badego et al[28]	2019	Sidama	South/SNNPR	Institution	18+	Hypertension	94.9	575	8	134	24.5
Yadecha et al[29]	2017	Gimbi	Oromia	Community	18+	Hypertension	98.9	476	7	157	33.5
Beall et al[30]	1995	Ambaras	Amhara	Community	18+	Hypertension	100	263	6	8	3
Berhe et al[31]	2016	Yekatit12	Addis Ababa City administration	Institution	25-64	Hypertension	97.9	423	8	106	38.9
Haye et al[32]	2018	Akaki	Addis Ababa City administration	Institution	18+	Hypertension	98.5	407	7	54	14
Chuka et al[33]	2017	Arba Minch	South/SNNPR	Community	25-64	Hypertension	99.4	3368	8	633	18.9
Mulatu et al[34]	2020	Mizan Aman	South/SNNPR	Community	18+	Hypertension	98.6	356	7	62	17.7
Dereje et al[35]	2017	Hossana	South/SNNPR	Community	18+	Hypertension	98.9	634	8	108	17.2
Wachamo et al[36]	2019	Hawela	South/SNNPR	Community	18+	Hypertension	98.2	390	7	47	12.3

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Tesfaye et al[37]	2015	Addis Ababa	Addis Ababa City administration	Institution	18+	Impaired glucose tolerance	93.3	1003	8	120	13
Worede et al[38]	2015	Koladiba	Amhara	Community	≥20	Impaired glucose tolerance	100	392	8	47	12
Bantie et al[39]	2018	Bahir Dar	Amhara	Community	18+	Impaired glucose tolerance	100	607	7	70	12.8
Endris et al[40]	2019	Dessie	Amhara	Community	18+	Impaired glucose tolerance	98.2	598	8	92	15.7
Zekewos et al[41]	2016	Bona District	South/SNNPR	Community	15+	Impaired glucose tolerance	91.4	2922	8	69	2.6
Habtewold et al[42]	2015	Debre Birhan	Amhara	Institution	15+	Impaired glucose tolerance	95.3	385	7	26	7.1
Nshisso et al[43]	2010	Addis Ababa	Addis Ababa City administration	Institution	18+	Impaired glucose tolerance	100	2153	7	408	21.6
Aynalem et al[44]	2016	Mizan	South/SNNPR	Community	15+	Impaired glucose tolerance	97.1	414	7	64	15.9
Wakjira et al[45]	2015	Addis Ababa	Addis Ababa City administration	Facility	20+	Impaired glucose tolerance	100	304	7	66	21.7
Zenebe et al[46]	2017	Jimma	Oromia	Community	40+	Impaired glucose tolerance	98.1	269	8	34	12.9
Animaw et al[47]	2015	Bahir Dar	Amhara	Community	18+	Impaired glucose tolerance	95.5	1472	7	48	3.4

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Gebreegziabiher et al[48]	2019	Mekelle	Tigray	Community	20+	Impaired glucose tolerance	98.8	325	6	40	12.5
Wolde et al[49]	2019	Gonder	Amhara	Community	18+	Impaired glucose tolerance	96	805	7	72	9.3
Yunka et al[50]	2019	Jimma	Oromia	Community	18+	Impaired glucose tolerance	95.8	955	7	135	14.8
Darebo et al[51]	2015	Hawassa	South/SNNPR	Community	18+	Overweight/ obesity	92.7	565	8	148	28.2
Moges et al[52]	2012	Gondar	Amhara	Community	18+	Overweight/ obesity	100	68	8	13	32.4
Abrha et al[53]	2011	Ethiopia	Ethiopia	Community	15-49	Overweight/ obesity	100	3602	8	534	14.9
Bogale et al[54]	2014	Addis Ababa	Addis Ababa City administration	Institution	18+	Overweight/ obesity	98.87	531	8	141	26.8
Mekonnen et al[55]	2015	Bahir Dar	Amhara	Community	18+	Overweight/ obesity	94.7	1484	8	159	11.3
Dagne et al[56]	2015	Dessie	Amhara	Community	18–64	Overweight/ obesity	93.29	805	8	216	28.7
Janakiraman et al[57]	2018	Gondar	Amhara	Institution	25+	Overweight/ obesity	89	429	8	50	13.1
Amare et al[58]	2005	Gondar	Amhara	Community	18 +	Overweight/ obesity	93	384	6	97	27.2
Yeshaw et al[59]	2019	Ethiopia	Ethiopia	Community	15-49	Overweight/ obesity	100	10938	8	1058	9.7
Kassie et al[60]	2016	Ethiopia	Ethiopia	Community	15-49	Overweight/ obesity	100	2848	8	425	14.9

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Gutema et al[61]	2017	Arba Minch	South/SNNPR	Community	25-64	Overweight/ obesity	99.4	3368	8	361	10.8
Tesfaye et al[62]	2018	Dilla	South/SNNPR	Community	18-64	Overweight/ obesity	95.6	663	8	155	24.4
Yosef et al[63]	2018	Modjo	Oromia	Institution	18-64	Overweight/ obesity	100	400	6	126	56.5
Hailemariam et al[64]	2017	Addis Ababa	Addis Ababa City administration	Community	18-64	Overweight/ obesity	95	512	7	113	24.4
Molla et al[65]	2019	Gonder	Amhara	Community	18+	Overweight/ obesity	91.5	845	7	174	36.6
Gebremariam et al[66]	2016	Mekelle	Tigray	Community	18+	Overweight/ obesity	90.4	1380	6	415	30.1
Solomon et al[67]	2017	Addis Ababa	Addis Ababa City administration	Institution	18+	Metabolic syndrome	100	325	8	66	20.3
Abda et al[68]	2014	Jimma	Oromia	Institution	20+	Metabolic syndrome	100	225	8	59	26
Workalemahu et al[69]	2010	Addis Ababa	Addis Ababa City administration	Institution	18+	Metabolic syndrome	83.24	1843	7	313	17
Tran et al[70]	2010	Addis Ababa	Addis Ababa City	Institution	18+	Metabolic syndrome	88.78	2205	6	346	12.5

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Kerie et al[71]	2017	Mizan	South/SNNPR	Community	18+	Metabolic syndrome	95.7	558	8	51	9.6
Gebreyes et al[72]	2015	Ethiopia	Ethiopia	Community	15-69	Metabolic syndrome	84.5	8673	8	757	4.8
Moutma et al[73]	2019	Eastern Ethiopia	Oromia	Institution	19-64	Metabolic syndrome	97	1200	8	234	20.1
Mengesha et al[74]	2017	Dire Dawa	Dire Dawa city council	Community	25-64	Metabolic syndrome	96.6	903	7	83	9.5

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Egger's test

Std_Eff	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
slope	2.754683	.1588469	17.34	0.000	2.431867	3.077498
bias	.1234977	.1484605	0.83	0.411	-.1782103	.4252056

Supplementary figure 1: Eggers's test for detection of publication bias for studies included in hypertension.

Egger's test

Std_Eff	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
slope	2.136301	.075485	28.30	0.000	1.974402	2.2982
bias	.6232349	.0960689	6.49	0.000	.4171876	.8292822

Supplementary figure 2: Eggers's test for detection of publication bias for studies included in overweight/obesity.

Egger's test

Std_Eff	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
slope	.6438857	.1736983	3.71	0.003	.2654296	1.022342
bias	1.377838	.2219967	6.21	0.000	.8941491	1.861528

Supplementary figure 3: Eggers's test for detection of publication bias for studies included in impaired glucose tolerance.

Egger's test

Std_Eff	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
slope	1.37992	.1365355	10.11	0.000	1.045829	1.71401
bias	1.060526	.2416536	4.39	0.005	.4692211	1.651831

Supplementary figure 4: Eggers's test for detection of publication bias for studies included in metabolic syndrome.

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. metatrim pre sep, reffect graph funnel

Note: default data input format (theta, se_theta) assumed.

Meta-analysis

       |  Pooled      95% CI          Asymptotic      No. of
Method |     Est   Lower   Upper  z_value  p_value   studies
-------+-------------------------------------------------------
Fixed  |  13.384  12.999  13.769   68.133    0.000      16
Random |  23.975  19.938  28.012   11.641    0.000

Test for heterogeneity: Q= 1220.870 on 15 degrees of freedom (p= 0.000)
Moment-based estimate of between studies variance = 64.187

Trimming estimator: Linear
Meta-analysis type: Random-effects model

iteration |  estimate     Tn     # to trim     diff
----------+-----------------------------------------
    1     |   23.975      64          0         136
    2     |   23.975      64          0           0

Note: no trimming performed; data unchanged

Filled
Meta-analysis

       |  Pooled      95% CI          Asymptotic      No. of
Method |     Est   Lower   Upper  z_value  p_value   studies
-------+-------------------------------------------------------
Fixed  |  13.384  12.999  13.769   68.133    0.000      16
Random |  23.975  19.938  28.012   11.641    0.000

Test for heterogeneity: Q= 1220.870 on 15 degrees of freedom (p= 0.000)
Moment-based estimate of between studies variance = 64.187
```

Supplementary figure 5: Trim-and-fill analysis for the prevalence of overweight/obesity among adults in Ethiopia.

```
. metatrim pre sep, reffect graph funnel

Note: default data input format (theta, se_theta) assumed.

Meta-analysis

       |  Pooled      95% CI          Asymptotic      No. of
Method |     Est   Lower   Upper  z_value  p_value   studies
-------+-------------------------------------------------------
Fixed  |   6.235   5.825   6.644   29.814    0.000      14
Random |  12.426   8.753  16.100    6.630    0.000

Test for heterogeneity: Q= 764.415 on 13 degrees of freedom (p= 0.000)
Moment-based estimate of between studies variance = 47.131

Trimming estimator: Linear
Meta-analysis type: Random-effects model

iteration |  estimate     Tn     # to trim     diff
----------+-----------------------------------------
    1     |   12.426      60          1         105
    2     |   11.762      66          2          12
    3     |   10.877      73          3          14
    4     |   10.435      74          3           2
    5     |   10.435      74          3           0

Filled
Meta-analysis

       |  Pooled      95% CI          Asymptotic      No. of
Method |     Est   Lower   Upper  z_value  p_value   studies
-------+-------------------------------------------------------
Fixed  |   5.808   5.413   6.203   28.822    0.000      17
Random |  10.467   7.201  13.733    6.281    0.000

Test for heterogeneity: Q= 831.028 on 16 degrees of freedom (p= 0.000)
Moment-based estimate of between studies variance = 44.962
```

Supplementary figure 6: Trim-and-fill analysis for the prevalence of impaired glucose tolerance among adults in Ethiopia.

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```
. metatrim pre sep, reffect graph funnel

Note: default data input format (theta, se_theta) assumed.

Meta-analysis

       |  Pooled      95% CI          Asymptotic      No. of
Method |     Est   Lower   Upper  z_value  p_value   studies
-------+-------------------------------------------------------
Fixed  |   7.055   6.663   7.446   35.317    0.000       8
Random |  14.708   9.820  19.596    5.897    0.000

Test for heterogeneity: Q= 500.276 on 7 degrees of freedom (p= 0.000)
Moment-based estimate of between studies variance = 47.566

Trimming estimator: Linear
Meta-analysis type: Random-effects model

iteration |  estimate    Tn    # to trim     diff
----------+-----------------------------------
    1     |   14.708     21         1          36
    2     |   13.294     22         1           2
    3     |   13.294     22         1           0

Filled
Meta-analysis

       |  Pooled      95% CI          Asymptotic      No. of
Method |     Est   Lower   Upper  z_value  p_value   studies
-------+-------------------------------------------------------
Fixed  |   7.025   6.634   7.415   35.248    0.000       9
Random |  13.298   8.682  17.915    5.646    0.000

Test for heterogeneity: Q= 505.145 on 8 degrees of freedom (p= 0.000)
Moment-based estimate of between studies variance = 47.100
```

Supplementary figure 7: Trim-and-fill analysis for the prevalence of metabolic syndrome among adults in Ethiopia.

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PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) Checklist

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You must report the page number in your manuscript where you consider each of the items listed in this checklist. If you have not included this information, either revise your manuscript accordingly before submitting or note N/A.

Section/Topic	Item No.	Checklist item	Reported on Page No.
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3-4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	N/A
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	6
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	7
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	8

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Section/Topic	Item No.	Checklist item	Reported on Page No.
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	8
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	8
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	9
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	9
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	9
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	10
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	10
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	10-12
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	10-12
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	13
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	13-14
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	14-19
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	19
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	20
FUNDING			

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Section/Topic	Item No.	Checklist item	Reported on Page No.
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	20

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(6): e1000097. doi:10.1371/journal.pmed1000097

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MOOSE Checklist for Meta-analyses of Observational Studies

Title: Metabolic risk factors of non-communicable diseases in Ethiopia: a systematic review and meta-analysis.

Item No	Recommendation	Reported on Page No
Reporting of background should include		
1	Problem definition	2-4
2	Hypothesis statement	4
3	Description of study outcome(s)	4,7
4	Type of exposure or intervention used	NA
5	Type of study designs used	4,5
6	Study population	4
Reporting of search strategy should include		
7	Qualifications of searchers (eg, librarians and investigators)	1
8	Search strategy, including time period included in the synthesis and key words	6
9	Effort to include all available studies, including contact with authors	6
10	Databases and registries searched	6
11	Search software used, name and version, including special features used (eg, explosion)	NA
12	Use of hand searching (eg, reference lists of obtained articles)	6
13	List of citations located and those excluded, including justification	9, supplementary Table 1, Figure 1
14	Method of addressing articles published in languages other than English	6
15	Method of handling abstracts and unpublished studies	5, 6
16	Description of any contact with authors	NA
Reporting of methods should include		
17	Description of relevance or appropriateness of studies assembled for assessing the hypothesis to be tested	5
18	Rationale for the selection and coding of data (eg, sound clinical principles or convenience)	7
19	Documentation of how data were classified and coded (eg, multiple raters, blinding and interrater reliability)	8
20	Assessment of confounding (eg, comparability of cases and controls in studies where appropriate)	NA
21	Assessment of study quality, including blinding of quality assessors, stratification, or regression on possible predictors of study results	8
22	Assessment of heterogeneity	8
23	Description of statistical methods (eg, complete description of fixed or random effects models, justification of whether the chosen models account for predictors of study results, dose-response models, or cumulative meta-analysis) in sufficient detail to be replicated	8
24	Provision of appropriate tables and graphics	Table 1, Figures 1-11, supplementary table 1, supplementary figures 1-7
Reporting of results should include		

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25	Graphic summarizing individual study estimates and overall estimate	Figures 2-5
26	Table giving descriptive information for each study included	Supplementary Table 1
27	Results of sensitivity testing (eg, subgroup analysis)	11, Table 1
28	Indication of statistical uncertainty of findings	10-14

Item No	Recommendation	Reported on Page No
Reporting of discussion should include		
29	Quantitative assessment of bias (eg, publication bias)	10
30	Justification for exclusion (eg, exclusion of non-English language citations)	5
31	Assessment of quality of included studies	10, and supplementary Table 1
Reporting of conclusions should include		
32	Consideration of alternative explanations for observed results	14-18
33	Generalization of the conclusions (ie, appropriate for the data presented and within the domain of the literature review)	20
34	Guidelines for future research	19
35	Disclosure of funding source	20

From: Stroup DF, Berlin JA, Morton SC, et al, for the Meta-analysis Of Observational Studies in Epidemiology (MOOSE) Group. Meta-analysis of Observational Studies in Epidemiology. A Proposal for Reporting. *JAMA*. 2000;283(15):2008-2012. doi: 10.1001/jama.283.15.2008.

NA= Not Applicable

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Title: Metabolic risk factors of non-communicable diseases in Ethiopia: a systematic review and meta-analysis.

List of authors

Tilahun Tewabe Alamnia, [1,2*] Wubshet Tesfaye,[3] Solomon Abrha, [3,4] and Matthew Kelly.[2]

Authors' institutional affiliations

[1,2*]Tilahun Tewabe Alamnia (Corresponding author, MSc, [1]College of Medical and Health Sciences, Bahir Dar University, Ethiopia; [2]Department of Global Health, Research School of Population Health, Australian National University, Australia. Email: Tilahun.Alamnia@anu.edu.au, Phone number +61474173615)

[3]Wubshet Tesfaye (PhD, Health Research Institute, University of Canberra, Australia. Email: Wubshet.Tesfaye@canberra.edu.au)

[3,4]Solomon Abrha (MSc, [3]University of Canberra, Australia. [4]College of Medicine and Health Sciences, Mekelle University, Ethiopia. Email: Solomon.Bezabh@canberra.edu.au)

[2]Matthew Kelly (PhD, Department of Global Health, Research School of Population Health, Australian National University, Australia. Email: Matthew.Kelly@anu.edu.au)

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Abstract

Objectives: Non-communicable diseases (NCDs) are causing a new and devastating health burden in low-income countries. In Ethiopia 39% of deaths are NCD related, but the health system is under-prepared. Evidence of risk factor distributions is needed to inform response. The research question for this review investigates prevalence distributions and sex and age variations of metabolic risk factors among Ethiopian adults.

Research design and methods: This systematic review used the Preferred Reporting Items for Systematic Review and Meta-Analysis guidelines. Studies published until January 06, 2021 were searched from PubMed, Scopus, ProQuest, and Web of Science databases, reference lists and grey literature. Studies reporting prevalence of metabolic risk factors: overweight/obesity, hypertension, impaired glucose homeostasis, and metabolic syndrome among adults were eligible for this systematic review and meta-analysis. Two authors independently extracted data and used the Joanna Briggs Institute (JBI) tool for appraisal. The random effects model was used to conduct meta-analysis using Stata version 16. Subgroup analyses examined differences by region, study year, sample size, and settings.

Results: From 6087 records, 74 studies including 104,382 participants were included. Most showed high prevalence of metabolic risk factors. Meta-analysis revealed pooled prevalence of metabolic risk factors from 12%-24% with the highest observed for overweight/obesity (23.9%, 95% Confidence Interval (CI): 19.9-28.0) and hypertension (21.1%, 95% CI: 18.7-23.5) followed by metabolic syndrome (14.7%, 95% CI: 9.8-19.6) and impaired glucose tolerance (12.4%, 95% CI: 8.7-16.1). The prevalence of overweight/obesity was higher in women. All metabolic risk factors were higher among people aged above 45 years.

Conclusions: Nearly 1 in 5 Ethiopian adults has at least one metabolic risk factor for NCDs, despite heterogeneity of studies limiting the certainty of evidence, the result suggests the need for coordinated effort among policymakers, health care providers, non-governmental stakeholders, and the community to implement appropriate preventive measures to reduce these factors.

Strengths and limitations of the study

- This is the first systematic review and meta-analysis combining studies with similar characteristics in the context of metabolic risk factors for NCDs in Ethiopia.
- The study provided comprehensive evidence of prevalence of metabolic risk factors of non-communicable diseases among adults in Ethiopia which can be easily used for planning intervention to decrease non-communicable disease incidence in the population.

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- The certainty of the evidence of this review is limited by heterogeneity of the included studies which hampered the ability to calculate combined pooled prevalence of multiple risk factors.
- We did not find studies conducted in some regions of the country like Afar, Benishangul-Gumuz, Gambella regions, so the prevalence estimates might not reflect the realities in these regions.

Introduction

Non-communicable diseases (NCDs) are the leading cause of death worldwide, but they disproportionately affect people living in Low and Middle-Income Countries (LMICs).[1] In 2017, more than 73% of global deaths (41 million) were attributed to NCDs.[1] Of these, more than 80% were due to cardiovascular diseases, cancer, chronic respiratory diseases, and diabetes mellitus (DM).[1,2] About 75% (28 million) of all NCDs deaths and 82% (16 million) of the premature deaths occurred in LMICs. In the WHO African Region, the burden of NCDs is gradually increasing and is anticipated to overtake the burden of mortality and morbidity from communicable diseases by the year 2030.[3] The same holds true in Ethiopia where, NCDs were estimated to cause around 43% of deaths in 2019.[4]

Most NCD related deaths are attributed to major risk factors which can be broadly categorized into behavioral (tobacco smoking, alcohol drinking, sedentary lifestyle, and high dietary salt intakes) and metabolic (overweight/obesity, high fasting plasma glucose, high blood pressure, and high level of blood triglycerides) factors.[3] The continual increase in the prevalence of metabolic risk factors, often reinforced by behavioral factors, has been associated with a direct increase in prevalence of NCDs, especially among adults between the ages of 30 and 70 years.[2]

According to the 2017 Global Burden of Disease Study, metabolic risk factors contributed to a substantial proportion (26%) of the total Disability-Adjusted Life Years (DALYs). High blood pressure, high fasting plasma glucose, high body-mass index, and high cholesterol are responsible for 9%, 7%, 6% , and 4% of global DALYs, respectively.[5] Besides, the 2019 estimates on DM prevalence revealed that 463 million people (9%) have diabetes and this number is projected to rise to 700 million (11%) by 2045.[6] LMICs has been disproportionately affected by these problems and the burden of these diseases is likely to increase in developing countries in Africa.[5]

The 2030 United Nations agenda for sustainable development has set a target to reduce one-third of premature mortality form NCDs[7]. In line with this, the World Health Organization

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(WHO) has set out a milestone plan to prevent and control premature mortality from cardiovascular diseases, cancer, DM, or chronic respiratory diseases by 25% in 2025, which can be achieved by reducing the burden of metabolic risk factors and its contributing behavioral problems among adults using affordable and accessible intervention strategies.[8] However, to implement these interventions and achieve the intended goals, up-to-date, representative, and comprehensive evidence is always needed. The Ethiopian government prepared national guidelines for NCDs prevention and management in line with the global target in 2016,[9] but there is a lack of evidence on the successes of ongoing implementation and program evaluations of the proposed strategies. There is also a lack of comprehensive data on the prevalence of the immediate predictors of NCDs to implement focused interventions and make policy decisions.

There has been no systematic review of the prevalence of metabolic risk factors in Ethiopia. Although some studies have reviewed the prevalence of hypertension, their results are inconclusive due to lack of methodological rigor such as including participants with varying characteristics and/or were performed more than five years ago. There is no available comprehensive evidence for the remaining three risk factors: impaired glucose tolerance, overweight/obesity, and metabolic syndrome. Furthermore, previous individual studies have shown a conflicting association between metabolic risk factors and participants' characteristics, such as sex and age. For instance, some showed that hypertension was higher in women and elders, while others in men and in younger people. To this end, we aimed to conduct this systematic review and meta-analysis to (1) determine the pooled prevalence of metabolic risk factors of non-communicable diseases in Ethiopian adults; (2) examine differences in the distribution of metabolic risk factors across different regions, years, sample sizes, and settings; and (3) test the associations of metabolic risk factors with the sex and age of participants.

We hypothesized that (a) metabolic risk factors for NCDs are higher among adults due to the life-course cumulative increase in risk caused by behavioral factors in this age group, (b) there would be variations in the prevalence of these risk factors across regions and settings of the study, and (c) the prevalence of metabolic risk factors would be higher in women and/or elder adults. Estimating the prevalence of metabolic risk factors for NCDs in Ethiopia is essential to identify the priority areas for policy action, inform practices through various intersectoral collaborations, and decide future research directions. The result of this study can be used by

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the government, health authorities, non-governmental stakeholders, and health care providers to design and deliver health interventions to decrease the burden of NCDs.

Methods

Study design

This systematic review and meta-analysis was conducted and reported in accordance with the Preferred Reporting Items for Systemic Reviews and Meta-analysis.[10] It includes key research questions, information sources, eligibility criteria, study selection, data extraction, risk of bias assessment, and data analysis, results and discussion. The process of screening and selection of studies, the results of the analysis and narrative descriptions were supported with figures and tables for reporting this study.

Patient and public involvement

No patients or public were involved in the design, conduct, interpretation, or dissemination of the results of this study.

Research questions

The key research questions that initiated this review were: (1) What is the level of metabolic risk factors of non-communicable diseases among adults in Ethiopia? (2) Is there a difference in the distribution of metabolic risk factors across regions in Ethiopia, by time period, settings, and sample sizes? (3) What is the age and sex differences in the distribution of metabolic risk factors of non-communicable diseases?

Eligibility criteria

Based on these research questions, we identified Population, Exposure, and Outcome (PEO) for eligibility criteria. Since the review was based on prevalence studies, we did not define controls. The population in this study was any adult living in Ethiopia. The adult was defined as person between the ages of 18-64. However, in some studies, adults were referred over the age of 15. To maximize the generalizability of our study findings, we used both definitions for this systematic review and meta-analysis. Exposures were variables that might contribute to high prevalence of metabolic risk factors, including sociodemographic and economic variables but in this study, we only used the age and sex of participants to compare differences in metabolic risk factors. The outcome variables for this study were metabolic risk factors, such as overweight/obesity, hypertension, impaired glucose homeostasis, and metabolic syndrome. Thus, this review included studies conducted in Ethiopia and published in English language

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which reported a prevalence of at least one of the metabolic risk factors (overweight/obesity, impaired glucose tolerance, hypertension, and metabolic syndrome) in adults with a cross-sectional or cohort designs and published until January 06, 2021. Whereas papers with only abstracts, qualitative studies, articles with a case control design, and research involving participants from vulnerable groups (children and pregnant women) or having pre-existing chronic medical conditions were excluded as these factors were considered potential confounders of the outcomes of interest. Studies conducted in other languages were not considered due to translation issues. However, there was no published article found other than English including in Amharic (Ethiopian official language). Eligible studies were independently screened and evaluated by two authors following the predefined data fields.

Information sources

First, we developed search strategies, identified key terms, and selected databases. These were developed in collaboration with all authors. We also searched PubMed Medical Subject Headings (MeSH) for additional keywords to expand our search scope to include all available evidence. Before we performed the main search, the search method was piloted in the selected databases. We did not use any restrictions, including language, to maximize our search results. Overall our key terms include: “metabolic risk factors”, “cardiometabolic risk factors”, “metabolic syndrome”, “dyslipidemia*”, “dyslipidaemia*”, “hyperlipidemia*”, “hyperlipidaemia*”, “high cholesterol”, “hypercholesterolemia”, “hypercholesterolaemia”, “triglycerides”, “hypertriglyceridemia”, “hypertriglyceridaemia”, “overnutrition”, “malnutrition”, “overweight”, “obesity”, “obese”, “blood glucose”, “impaired glucose tolerance”, “glucose intolerance”, “high blood glucose”, “hyperglycemia”, “high fasting plasma glucose”, “insulin resistance”, “diabetes”, “diabetes mellitus”, “blood pressure”, “high blood pressure”, “hypertension”, “systolic blood pressure”, “diastolic blood pressure” and “Ethiopia”. Then two reviewers (TA and WT) independently searched PubMed, Embase via Scopus, ProQuest, and Web of Science databases for relevant studies from their inception to January 06, 2021. The reference lists of the included articles in this systematic review and meta-analysis and grey literatures were reviewed and hand searched to identify additional relevant articles. The search results were then exported to Endnote X9 reference manager software and to Covidence[11] for removing duplicates, screening, and study selection.

PubMed search strategy

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(((“metabolic risk factors” OR “cardiometabolic risk factors” OR “metabolic syndrome” OR dyslipidemia* OR dyslipidaemia* OR Hyperlipidemia* OR hyperlipidaemia* OR “high blood cholesterol” OR hypercholesterolemia OR hypercholesterolaemia OR triglycerides OR hypertriglyceridemia OR hypertriglyceridaemia) OR (overnutrition OR malnutrition OR overweight OR obesity OR obese) OR (“blood glucose” OR “impaired glucose tolerance” OR “glucose intolerance” OR “high blood glucose” OR hyperglycemia OR “High fasting plasma glucose” OR “insulin resistance” OR diabetes OR “diabetes mellitus”) OR (“blood pressure” OR “high blood pressure” OR hypertension OR “systolic blood pressure” OR “diastolic blood pressure”)) AND (Ethiopia)). Full data search techniques and results are in the appendix (supplementary table 1).

Study selection

To perform the screening, the records identified from the search results were exported to Endnote X9 reference manager software (Thomson Reuters, Philadelphia, PA, USA) and then exported to Covidence virtual library.[11] After duplicate records were removed, two reviewers (TA and WT) independently screened the titles and abstracts of the retrieved articles guided by the predefined PEO and eligibility criteria. Following the selection of articles based on their titles and abstracts, a full text review was conducted for the selected studies for further relevance and inclusion. Any disagreements during the screening process were resolved via discussion.

Data collection process

Two authors (TA and WT) independently extracted key data from the included studies. The data extraction was conducted based on the modified version of the Joanna Briggs Institute (JBI) Meta-Analysis of Statistics and Review Instruments (JBI-MAStARI).[12] Data extracted included study characteristics (authors name, type of metabolic risk factor studied, year of study, study area, study setting, study design, sample size, response rate, number of participants with the case), participant characteristics (age, gender, disease condition) and study outcome measures (prevalence of metabolic risk factors). Any disagreements encountered during information extraction were resolved via discussion.

Risk of bias assessment

Two reviewers (TA and WT) independently assessed the quality of the included studies using Joanna Briggs Institute (JBI) checklist[13] developed for methodological quality assessment of prevalence studies that includes criteria of inclusion, description of subjects and the setting,

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reliability and validity of exposure measurement, criteria of measurement, identification and strategies to deal confounding factors, validity and reliability of outcomes measurement, and statistical analysis used. The studies were scored on a scale of 0 to 1 for 8 items and studies with a total score of <4, 4-6 and 7-8 were considered to have low, moderate, and high methodological quality, respectively. Any disagreements in ratings the studies were resolved through discussion between the two reviewers.

Outcome measures

The main outcome of interest for this study were the prevalence of metabolic risk factors among adults in Ethiopia reported in the original articles in percentage (number of cases/ total number of participants). We examined four metabolic risk factors: overweight/obesity, impaired glucose tolerance, hypertension, and metabolic syndrome. The outcome variables were categorized based on the following definitions: Overweight, defined as a body mass index (BMI) of 25-29.9 kg/m^2 and obesity as BMI ≥ 30 kg/m^2); [2,14] Impaired glucose tolerance (a fasting plasma glucose of 100 to 125 mg/dL or oral glucose tolerance test greater between 140 and 199 mg/dL); [15] High blood pressure (a blood pressure cut-off value of ≥140/90 mmHg on the average of blood pressure measurements using a sphygmomanometer); [2,16] and metabolic syndrome (three or more of the following based on adult treatment panel (ATP III) criteria: elevated triglyceride (≥150mg/dL), reduced high density lipoprotein(<40mg/dl for male, and <50 for female), high blood pressure (≥130/85 mg/dL, high fasting plasma glucose (≥110 mg/dL), or increased waist circumference (≥102 cm for men and ≥88 cm for women); [17] or waist circumference of ≥94 cm for men and ≥80 cm for women, and two of elevated triglycerides (≥150 mg/dl) or treatment for this lipid abnormality; reduced HDL-C (<40 mg/dL in men and <50 mg/dL in women) or treatment for lipid abnormality; high blood pressure BP (≥130/85 mm Hg) or treatment for hypertension; impaired fasting serum glucose (≥100 mg/dL) or previously diagnosed with type 2 diabetes based on international diabetes federation criteria).[18] Additional outcomes include the variation of the metabolic risk factors by age and gender of participants.

Data analysis

Relevant information on the study area, study setting, study design, study sample, study response rate, and study outcome were extracted using Microsoft Excel and exported to Stata version 16 (StataCorp LLC, College Station, TX, USA) for the meta-analysis. The characteristics of included studies was summarized and presented using tables and figures

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(forest plots). The random effects restricted maximum likelihood (REML) meta-analysis model was used to pool studies with homogenous outcomes, given it is more robust in terms of variance estimation in small sample studies.[19] Forest plots along with the respective 95% CIs were generated for individual studies and the pooled data to show graphic summaries of the analyses. Heterogeneity across studies was assessed using the I^2 index, values greater than 75% indicated substantial heterogeneity.[20] The Z-statistic was used for testing associations with $\alpha \leq 0.05$ as cut-off to determine statistical significance level. Then we identified studies that reported statistically significant associations between sex or age of participants and metabolic risk factors to extract odds ratios, confidence intervals, and p-values. This is because, often, studies provided odds ratio, confidence intervals, and p-values only when significant associations were observed. The log of odds ratios was then used to determine the pooled association between each metabolic risk factor and the sex and age of the participants.

Publication bias

The presence of significant publication bias was assessed using funnel plot and the Begg and Egger's test. The symmetry of the funnel plot was visually inspected to assess the normal distribution studies around the centre of funnel plot.[21] Then, the Begg's funnel plot and Egger test for small-study effects[22] were used to objectively assess the risk of bias across studies.

Subgroup analysis

Subgroup analyses were performed to examine the difference in prevalence of metabolic risk factors across the following variations:

1) Study regions: this was based on the nine ethno-linguistically based regions of the country (Afar, Amhara, Benishangul-Gumuz, Gambela, Harari, Oromia, Somali, 'Southern Nations, Nationalities, and Peoples Region' (SNNPR), and Tigray) and two city administrations of the country (Addis Ababa city administration and Dire Dawa city council);
2) Years of the study: we divided year of study in to three categories studies: conducted before 2000, studies conducted between 2000 and 2010, and studies performed between 2011 and 2020. This classification was chosen by the authors to show differences in the prevalence of metabolic risk factors in adults across ten-year age ranges, which we thought would allow sufficient variation for comparisons to be made;
3) Study settings: community-based versus institution-based. This classification was intended to compare prevalence of metabolic risk factors among adults selected from

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institutions (schools, banks, and health facilities) and from the general community where there would be a difference in participant characteristics. For instance, the former is more likely to represent educated and health literate adults whereas the latter likely comprises less wealthy and uneducated community members; and

4) Sample size: less than 400 versus 400 and above. The sample size classification was based on the authors' objective to compare the prevalence of metabolic risk factors between studies with small sample size and large sample size.

The results of subgroup analysis were presented in tables with prevalence's, 95% confidence intervals, and I^2 statistics test results. The statistical significance for subgroup analysis results was also determined at a p-value <0.05.

Trim-and- fill analysis

Trim-and-fill analysis were used to assess the effect of remaining studies that might have been included and fill imputed studies based on bias corrected pooled prevalence's. Statistical significance was declared if the p-value is <0.05.

Grading the certainty of evidence

The certainty of the evidence was assessed using the GRADE (Grading of Recommendations Assessment, Development and Evaluation) approach. This approach identifies systematic reviews and meta-analysis as a magnifying glass thorough which analysis of evidence is viewed and applied. Evidence from observational studies generally start with initial low certainty. The certainty of evidence of the outcomes (metabolic risks) was downrated for risk of bias, inconsistency (heterogeneity), indirectness of evidence, imprecision, and publication bias.[23]

Ethics approval

This study is a systematic review and meta-analysis based entirely on the review of existing literature. No human participants or patients were involved in the design and conduct this review. Therefore, ethics approval and consent to participate would not be applicable for this study.

Results

Study selection

A total of 6087 records were identified from database and hand searching of reference lists of the included studies and grey literatures. After removal of duplicates (n=2900), ineligible

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articles (n=3045) were removed articles based on their titles and abstracts. One hundred forty-two publications were then eligible for full-text retrieval and screening. Of these, 14 articles were excluded due to their participants including children and adolescents, 27 studies were excluded due to participants with confounding chronic diseases like diabetes, 2 articles were excluded due to previous similar reports, 18 studies were excluded because they did not explicitly record, measure, or report the outcomes, and 7 articles were excluded due to their study design (case control). No unpublished relevant study was found, and no records were excluded due to a low-quality assessment. Finally, 74 articles involving a total of 104,382 participants were included in this systematic review and meta-analysis. The study selection flow chart is shown in figure 1.

Characteristics and risk of bias of included studies

The characteristics of the included studies in this systematic review and meta-analysis are presented in the appendix (supplementary table 2). Out of 74 studies, 36 of them reported a point prevalence on hypertension,[24-59] 14 on impaired glucose tolerance,[60-73] 18 on overweight or obesity, [74-89] and the remaining eight studies on metabolic syndrome.[90-97] The studies were conducted in — Amhara (n=22), [25,28,29,32,34,38,41,45,50,53,61-63,65,70,72,75,78-81,89] Oromia (n=9), [35,36,46,52,69,71,87,91,97] Southern Ethiopia (n=15), [31,37,40,44,51,55,57-59,64,67,74,85,86,94] Tigray (n=5), [26,42,43,73,83] Somali (n=1),[30] Addis Ababa city administration (n=14), [24,27,33,47,48,54,56,60,66,68,77,88,90,92,93] Dire Dawa city council (n=2) [39,96] regions, and five [49,76,82,84,95] of them were nationwide studies. All studies were cross-sectional in nature and most of them (n=52) were community-based studies. The sample sizes for the studies ranged from 68 to 10,938 participants.

The risk of bias of included studies in this systematic review and meta-analysis were assessed and presented in supplementary table 2. The overall risk of bias in the included studies was considered to be low: sixty-six (89%) had low risk of bias, while the remaining eight studies (11%) had moderate level of risk of bias. None of the included studies had high risk of bias. Overall, the GRADE assessment of quality of evidence provided a low certainty of evidence to support prevalence estimates and associations with the age and sex participants (supplementary table 3).

Prevalence of hypertension

A total of 36 studies involving 47,204 adult participants reported on prevalence of hypertension were included in this analysis. Of these studies, most were conducted in Amhara region

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(n=11),[25,26,28,29,32,34,38,41,45,50,53] Addis Ababa (n=7),[24,27,33,47,48,54,56] Southern Ethiopia (n=9)[31,37,40,44,51,55,57-59] and there was one nationwide study. [49] The prevalence of hypertension among individual studies ranged between 3% in Amhara region [53] to 34.7% in Addis Ababa.[33] Meta-analysis result showed the pooled prevalence of hypertension was 21.1% (95% CI: 18.7-23.4, certainty of evidence: moderate), with substantial heterogeneity among the included studies (I^2= 97.6%; p≤ 0.001) (figure 2 and supplementary table 3).

Prevalence of overweight/obesity

With regard to overweight/obesity, 16 studies involving 28,822 adult participants reported on prevalence of overweight/obesity were included in this analysis [74-89]. Of these, six studies were from the Amhara region.[75,78-81,89] Looking at the distribution of overweight or obesity, the prevalence span from 9.7% from a nationwide study[82] to 56.5% in a study conducted in Addis Ababa. [87] The pooled prevalence of overweight/ obesity was estimated at 23.9% (95% CI: 19.9-28.0, certainty of evidence: low). The presence of substantial heterogeneity in the included studies (I^2= 98.8%; p≤ 0.001) decreased the certainty of the evidence (figure 3 and supplementary table 2).

Prevalence of impaired glucose tolerance

Fourteen studies covering 12,604 adult participants and reporting prevalence of impaired glucose tolerance were included in this meta-analysis. [60-73] Six of these studies were conducted in Amhara region[61-63,65,70,72] and three in Addis Ababa.[60,66,68] The prevalence of impaired glucose tolerance ranged from 2.6% in Southern Ethiopia[64] to 21.7% in Addis Ababa.[68] Overall, the pooled prevalence of impaired glucose tolerance was 12.4% (95% CI: 8.7-16.1, certainty of evidence: low), but there was heterogeneity among the included studies in this meta-analysis (I^2= 98.3%; p≤ 0.001) **(**figure 4 and supplementary table 3).

Prevalence of metabolic syndrome

Besides metabolic syndrome, eight studies targeting 15,932 adult participants which reported on prevalence of metabolic syndrome were included in this analysis.[90-97] Three of the these studies [90,92,93] were from Addis Ababa, while the remaining were from Southern Ethiopia,[94] Oromia,[91,97] Dire Dawa, [96] and nationwide.[95] Metabolic syndrome prevalence varied from 4.8% (95% CI: 1.73%,7.87%) based on national sample [95] to 26% (95% CI: 19.61%,32.39%) in Oromia region.[91] In this meta-analysis, the pooled prevalence of metabolic syndrome among Ethiopian adults was 14.7% (95% CI: 9.8%, 19.6%, certainty of evidence: low). The high level

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of heterogeneity among the included studies (I^2= 98.6%; p≤ 0.001) and publication bias downrated the certainty of evidence (figure 5 supplementary table 3).

Subgroup analysis

Subgroup analysis was performed to investigate sources of heterogeneity among the studies included in the pooled prevalence of metabolic risk factors using regions of study, years of study, settings of the study, and sample size. The subgroup analysis showed that the pooled prevalence of hypertension was 20.9% in studies done after 2010, 21.4% in research conducted with study participants above 400, 20.9% in studies performed at community level, 27.0% in studies done in Addis Ababa. The result of subgroup analysis also revealed the pooled prevalence of impaired glucose tolerance was 11.5% in studies done later 2010, and 12.9% in research done with sample size less than or equal to 400, 15.6% in studies conducted in institution, and 18.6% in studies done in Addis Ababa city administration. With regard to the pooled prevalence of overweight/obesity, it was 23.9% in research performed since 2010, 24.6% in research done in Amhara region, 22.3% in studies done at community level, and 38.6% in original studies conducted with study participants up to 400. Similarly, the pooled prevalence of metabolic syndrome was 23.0% in research performed with study participants less than or equal to 400, 14.7% in studies done in the last ten years, and 18.9% in studies performed in institutions (table 1).

Table 1. Prevalence of metabolic risk factors among adults in Ethiopia grouped by characteristics of included studies.

Type of metabolic risk factor	Subgroup		No of studies	Prevalence with 95% CL	I^2 and p value
Hypertension	Year of study	Before 2000	1	3.00 (0.84, 5.15)	-
		2000 to 2010	3	25.12 (20.10, 30.15)	47.4%, p= 0.14
		After 2010	32	20.90 (17.78, 24.01)	86.2%, p≤ 0.001
	Region of the study	Addis Ababa city administration	7	27.05 (22.63 to 33.47)	86.1%, p= 0.00
		Amhara	10	19.89 (12.54 to 27.25)	95.7%, p≤ 0.001
		Oromia	4	19.18 (11.08 to 27.30)	88.1%, p=0.61
		SNNP	9	19.25 (15.22, 23.18)	77.5%, p≤ 0.001
		Tigray	3	16.83 (13.64, 20.02)	0.0%, p=_0.86
		Somali	1	28.30 (21.74, 34.85)	-
		Dire Dawa city administration	1	24.40 (18.13, 30.66)	-
		Ethiopia	1	15.60 (10.21, 20.98)	-
	Setting	Community based	26	20.97 (17.23, 24.71)	92.0%, p≤ 0.001
		Institution based	10	21.12 (14.80, 27.44)	92.3%, p≤ 0.001
	Sample size	≤ 400	5	17.98 (6.90, 29.07)	85.9%, p≤ 0.001
		> 400	31	21.43 (18.67, 24.19)	96.4%, p≤ 0.001
Impaired glucose	Year of study	Before 2010	1	21.60 (15.578, 27.622)	-

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tolerance		After 2010	13	11.49 (8.18, 14.81)	88.6%, p≤ 0.001
	Region of the study	Amhara	6	9.70 (5.73, 13.68)	82.4%, p≤ 0.001
		Addis Ababa city administration	3	18.56(12.59, 24.52)	69.8%, p= 0.037
		SNNP	2	8.99 (-4.028, 22.02)	95.2%, p≤ 0.001
		Oromia	2	13.80 (10.16, 17.43)	0.0%, p=0.609
		Tigray	1	12.50 (7.55, 17.45)	-
	Setting	Community based	10	10.87(7.21, 14.53)	88.6%, p≤ 0.001
		Institution based	4	15.58 (8.20, 22.96)	87.9%, p≤ 0.001
	Sample size	≤ 400	5	12.91 (8.49, 17.32)	75.7%, p=0.02
		> 400	9	11.76 (7.36, 16.17)	91.4%, p≤ 0.001
Overweight/obesity	Year of study	Before 2010	1	27.20 (20.72, 33.67)	-
		After 2010	15	23.90 (18.14, 29.67)	93.3%, p≤ 0.001
	Region of the study	Amhara	6	24.66 (15.97, 33.35)	92.1%, p≤ 0.001
		Ethiopia	3	12.90(9.33, 16.47)	34.8%, p=0.216
		Addis Ababa city administration	2	25.56 (21.07, 30.05)	0.0%, p=0.601
		SNNP	3	20.94 (9.79, 32.09)	91.0%, p≤ 0.001
		Tigray	1	30.10 (23.42, 36.77)	-
	Setting	Community based	13	22.31 (17.27, 27.36)	90.0%, p≤ 0.001
		Institution based	3	31.97 (8.37, 55.56)	97.6%, p≤ 0.001
	Sample size	≤ 400	3	38.57 (21.83, 55.29)	94.1%, p≤ 0.001
		> 400	13	20.78 (16.04, 25.51)	89.1%, p≤ 0.001
Metabolic syndrome	Year of study	Before 2010	2	14.56 (10.17, 18.96)	18.9%, p=0.236
		After 2010	6	14.72 (8.23, 21.21)	91.1%, p≤ 0.001
	Region of the study	Addis Ababa	3	16.37 (11.87, 20.87)	51.0%, p=0.13
		Oromia	2	22.91 (17.13, 28.68)	43.6%, p=0.18
		SNNP	1	9.60 (5.167, 14.03)	-
		Ethiopia	1	4.80 (1.72, 7.87)	-
		Dire Dawa city administration	1	9.50 (5.08, 13.91)	
	Setting	Community based	3	7.63 (4.22, 11.04)	55.6%, p= 0.105
		Institution based	5	18.93 (14.53, 23.32)	66.3%, p= 0.018
	Sample size	≤400	2	23.01 (17.43, 28.59)	39.4%, p=0.19
		>400	6	11.91 (7.39, 16.42)	83.0%, p≤0.001

Publication bias

The presence of publication bias was first assessed subjectively by observing the symmetrical distribution of the included studies in the funnel plot. The funnel plot test result showed that all the included studies in the pooled prevalence of metabolic risk factors were symmetrically distributed around the center within the 95% confidence limits (figure 6-9). Then, Egger's test for funnel plot was conducted. The result of Egger's test for the funnel plot revealed that there was no publication bias in the included studies for the pooled prevalence of hypertension (p = 0.411). But there was publication bias for the pooled prevalence of overweight/obesity, impaired glucose tolerance, and metabolic syndrome in the included studies (p ≤ 0.001 for overweight/obesity, p ≤ 0.001 for impaired glucose tolerance, and p = 0.005 for metabolic syndrome) (supplementary figure 1- 4)

Trim-and-fill analysis for the prevalence of metabolic risk factors

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Trim-and -fill analysis was conducted to adjust publication bias for pooled prevalence of metabolic risk factors with the estimation of number of missing studies that might exist. The result showed that there were no studies imputed for missing studies for the pooled prevalence of overweight/obesity. Whereas the trim and fill analysis revealed three studies were imputed for the pooled prevalence of impaired glucose tolerance and after adjustment the estimated pooled prevalence of impaired glucose tolerance was 10.5 (7.2,13.7). While for the pooled prevalence of metabolic syndrome one studies was imputed for missing analysis and the estimated prevalence after adjustment was 13.3 (8.7, 17.9). Note that trim-and -fill analysis was not performed for hypertension since there was no publication bias among the included studies (supplementary figure 5 - 7).

Sensitivity analysis

Besides, the subgroup analysis, publication bias, and trim-and-fill analysis we examined the effect of individuals studies on the overall pooled prevalence estimates of metabolic risk factors. However, the level of heterogeneity did not significantly reduce, when we compare the level heterogeneity by removing any one of single study or outliers, mainly studies having larger or smaller sample sizes.

The association of the sex of participants and prevalence of metabolic risk factors

To determine the association between metabolic risk factors and sex of participants log odds ratio was used after identifying studies reported significant association between sex of participants and the prevalence metabolic risk factors. Overall, 18 hypertension studies,[28-33,35,40,43-45,48,49,51,55,59,98] 3 impaired glucose tolerance studies,[64,68,70] 7 overweight/obesity studies,[74,78,83,85,86,88,89] and 7 metabolic syndrome studies [90-92,94-97] reported a significant link between prevalence and the sex of participants. The result of log odds ratio's analysis revealed that the prevalence of hypertension, impaired glucose tolerance, and metabolic syndrome did not show significant difference between men and women. However, a significant difference was observed in the prevalence of overweight/obesity between men and women, with women having 60% higher chance of being overweight/obesity than men (AOR = 0.41 [95% CI: 0.26-0.65], certainty of evidence: low) (figure 10 and supplementary table 3).

The association of the age of participants and prevalence of metabolic risk factors

Similarly, to determine the association of age of participants and the prevalence of metabolic risk factors log odds ratio were calculated after identifying studies reporting significant associations between the age of participants and prevalence metabolic risk factors. Out of 36

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studies on hypertension, 26 studies [25,27,29-35,37,38,40-48,50,51,54,57-59] reported significant association between the age of the participants and the prevalence of hypertension. Similarly, nine studies [74,76-78,80,82,84,85,87] on overweight/obesity, six studies [60,61,63,64,66,73] on impaired glucose tolerance, and six studies [90,91,94-97] on metabolic syndrome reported significant associations between the age of participants and the respective prevalence. The result of log odds ratio showed that the prevalence of metabolic risk factors was significantly associated with the age of participants. The prevalence of hypertension was 3.4 times higher in people aged 45 years and older than people below 45 years old (AOR =3.4 [95% CI: 2.6-4.2], certainty of evidence: moderate). The prevalence of overweight/obesity was 1.6 times higher among people above 45 years old than participants below 45 years old (AOR = 1.6 [95% CI: 1.1-2.4], certainty of evidence: low). Similarly, impaired glucose tolerance (AOR =2.6 [95% CI: 1.4-4.8], certainty of evidence: low) and metabolic syndrome (AOR =3.3 [95% CI: 2.2-4.7], certainty of evidence: low) were significantly higher among people above 45 years old *(*figure 11 supplementary table 3).

Discussion

To the best of our knowledge, this is the first systematic review and meta-analysis to comprehensively examine the prevalence estimates of metabolic risk factors for NCDs in Ethiopia. There was one review[98] reporting the pooled prevalence of hypertension, but the evidence had low certainty due to inconsistency and publication bias and was conducted more than five years ago and considered all adults irrespective of co-existing conditions. Given the lack of comprehensive evidence on risk factors of NCDs at national level, this review targeted metabolic risk factors for NCDs in Ethiopia. The findings of this review are likely to inform future policy decisions.

The World Health Organization has estimated that more than a third of the burden of disease in Ethiopia can be attributed to NCDs, and an estimated 43% of deaths in 2019 were caused by NCDs.[4] Metabolic risk factors are major contributors to global burden of diseases and are immediate indicators for the incidence of NCDs,[5] especially for cardiovascular diseases and diabetes, which warrants particular public measures as the burden varies by region, gender, and age.[5,99] This study revealed a high prevalence of metabolic risk factors for NCDs in Ethiopia — nearly one in five participants suffer from at least one of the studied metabolic risk factors (hypertension, overweight/obesity, impaired glucose tolerance, and metabolic syndrome). They are associated with, and reinforced by, behavioral risk factors such as alcohol consumption, tobacco smoking, poor dietary habits, and physical inactivity.[2] The high level of poverty, and unemployment could also contribute to the metabolic risk factors.[100-102] Therefore,

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the high prevalence of the metabolic predictors for NCDs in this study potentially indicate the presence of various behavioral and environmental stressors underlying these conditions that need to be examined in future studies.

In this study, we found a 21% pooled prevalence of hypertension (certainty of evidence: moderate), which is slightly higher than the 19% reported in a previous relatively smaller review in Ethiopia,[98] but slightly lower than the 24% prevalence reported by the WHO.[103] Also, this pooled prevalence is lower than the prevalence (57%) in older adults found in a multiple country African review with a moderate quality of evidence,[104] or the prevalence (32.3%) of hypertension reported in LMICs collectively despite the quality of evidence being low due a high inconsistency in the included studies.[105] The difference with the report in LMICs is explained in their subgroup analysis of pooled prevalence of hypertension by income levels of countries that the pooled prevalence was 38% for upper middle-income countries and 23% for low-income countries. Regarding the study in African countries,[104] the higher prevalence could be due to the inclusion of older cohort, a population group already at risk of high blood pressure. Also, the prevalence of smoking[106] and alcohol consumption[107] is lower in Ethiopia compared to the rest of Africa. Although the exact causes of hypertension in Ethiopia are not entirely understood, several factors such as smoking, overweight or obesity, physical inactivity, high salt intake, and alcohol consumption habits would have undeniable role for the reported high prevalence. Also, other factors like stress, increased age, urbanization, and genetics could contribute to the reported prevalence of high blood pressure.

Our findings also indicated, although the certainty of evidence is low, the pooled prevalence of overweight and obesity was 23.9%, which is lower than the global prevalence (39%)[108] but higher than the prevalence of overweight (15.9%) reported in demographic health survey analysis of 32 sub-Saharan African countries having a moderate quality of evidence[109] and from the 2011 Ethiopian demographic health survey reported prevalence (12%) in urban women.[76] Poor diet intake including high levels of simple carbohydrates, increased caloric consumption, use of medications, reduced physical activity, and other psychosocial factors are among the predictors of overweight/obesity that could also be applicable in the Ethiopian context. In addition, although not considered in this study, early age malnutrition (including undernutrition) causes metabolic abnormalities like hypothyroidism and insulin resistance later in life and thus could contribute to the increasing overweight and obesity in the country.[110,111] Therefore, due attention should be given to the potentially modifiable determinants of overweight or obesity to mitigate the subsequent risk of NCDs earlier in life.

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This study also showed that the pooled prevalence of impaired glucose tolerance in Ethiopia was 12.4%. This finding is higher than the prevalence (9.1%) reported in 2015 national NCDs survey,[95] the pooled prevalence (6.0%) among childbearing age women in sub-Saharan Africa;[112] comparable to the prevalence (12.0%) of impaired glucose homeostasis in sub-Saharan Africa having a moderate quality of evidence,[113] and lower than the prevalence (22.1%) reported in patients with aldosteronism with a low quality of evidence.[114] The differences may have occurred due to methodological variations, participants characteristics and the available evidence in respective settings. Nevertheless, our finding is a stark reminder on the burden of impaired glucose tolerance putting significant proportion of the adult population at greater risk of DM and other complications. This underlines the need to adequately and timely manage factors that contribute to impaired glucose tolerance, including less physical inactivity (sedentary lifestyle), overweight or obesity, family history of DM, history of high blood pressure, and high cholesterol.

We found a 15% pooled prevalence for metabolic syndrome in this study with the overall low quality of the evidence. This is much higher than a national NCDs survey that reported 4.8% prevalence[95] and comparable to a 16% pooled prevalence among HIV patients although the certainty of evidence was very low.[115] Higher prevalence were reported, despite the quality of the evidence was low, in people with psychiatric disorders (19%),[116] and arthritis (30.5%).[117] The differences in the prevalence among the studies could be explained by the fact that we only considered healthy adult populations without any underlining health conditions in this study, which could have underestimated the magnitude of the burden in the community.

High levels of triglycerides and metabolic syndromes cause significant proportion of heart diseases thereby contributing to the ever-increasing prevalence of NCDs in LMICs and high-income countries alike.[2] This is because of the thickening and hardening of the walls of blood vessels (arteries), which are most of the time associated with poorly controlled obesity/diabetes and improper feeding of saturated fat, sugar, salty foods and limited intake of fruits and vegetables.[118] Poor diet with high-fat content and highly refined, processed foods like sugars, white bread, and dry cereals are associated with greater risks of developing insulin resistance and all other aspects of the metabolic syndrome.[119] Thus, it is important to educate people on the value of healthy diets (which are low in cholesterol, sodium, and sugar contents), staying physically active, quitting smoking, and decreasing alcohol intake to reduce metabolic syndrome and its long-term consequences.

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Finally, notwithstanding the low-quality evidence to support associations, metabolic risk factors showed variation based on gender — the overall risks appeared to be higher in women. This may seem counterintuitive given some behavioral risk factors like smoking and alcohol consumption, which are often associated with metabolic risk factors like high blood pressure, are more common in men.[120] In contrast, there are studies showing the greater tendency of physical inactivity or sedentary lifestyle among women. [121] Nevertheless, a similar trend was observed in Tanzania, where women had a higher prevalence of obesity, and metabolic syndrome.[122] Consistent with our findings, a previous study also indicated that obesity is more common among women,[120] whereas, another study showed men in some LMICs tended to be physically more active.[121] These findings, collectively, suggest that the metabolic risk factors, particularly overweight/obesity, are more common in women, highlighting their increased risk for NCDs. Irrespective of the causes, the findings highlight the need to address this problem through addressing behavioral risk factors that cause metabolic disorders and promoting community screening for early detection of the conditions. Given the link between socioeconomic status and NCDs,[123,124]strategies aimed at addressing socioeconomic disparities could help mitigate factors contributing to stressful life events thereby reducing high blood pressure and subsequent NCDs.

Implications for practice

Metabolic risk factors are immediate precursor of non-communicable diseases which results in mortalities, disabilities, and decrease in quality of life. Preventing and managing these risk factors is of public health importance. This review highlighted the high prevalence of metabolic risk factors among adults, albeit based on limited certainty of evidence, alerts health care workers, health care institutions and the government to focus on prevention activities to reduce these risk factors in the community.

This study informs primary health care providers and public health workers who are working in the community to consider preventions to reduce metabolic risk factors of NCDs in adults. The current primary health care programs in the country hugely focus on maternal and child health programs. However, over the past three decades, NCDs have been increasing rapidly, warranting further research into the causes and strategies to rectify them. Lifestyle education and health promotion strategies mainly aiming at modifying behavioral risk factors for NCDs can be instrumental in realizing the national and global targets. Therefore, primary health care providers need to be involved in awareness creation programs through health education,

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community-based screening, identification, and referral of cases that need health care interventions like hypertension and diabetes. Working with adult and youth centers, campaigning and awareness creation can also help to reduce these risk factors in the community. This review showed metabolic risk factors were more common in women and older adults. Thus, considering targeted approaches tailored for these segments of the community during prevention activities will potentially result in positive outcomes.

Health care workers in hospitals and health centers should consider screening for metabolic risk factors for adults at least by measuring height, weight, blood pressure, and blood glucose levels for all adults along with screening for behavioral risk factors such as physical inactivity, dietary habits, tobacco usage and alcohol consumption. It is important to include topics related to metabolic risk factors for people who come to the health facilities in the morning health education programs. Follow up and counseling should be also considered for peoples who are found to have metabolic risk factors. Besides, proper training for health care providers increases their ability to diagnose and manage metabolic risk factors and prevent NCDs.

This study informs the regional government to prepare management guidelines to revert the rise of metabolic risk factors to decrease the burden of non-communicable diseases. Regional and national health programs should consider non-communicable diseases preventions as one of their majors by designing appropriate interventions such as by preparing protocols and guidelines, equipping primary health care setups with human power, resource, and making accessible for the community. The government of Ethiopia should also consider strategies working on social determinants of health, such as urban planning, education, employment, and public safety which help to reduce risk behaviors to reduce burden metabolic risk factors. Further, it is important to have proper program evaluation in place to oversee implementation of strategies.

Recommendations for future research

Based on the result of this review, the following recommendation are drawn for future research and researchers.

This review showed that most studies used relatively small sample sizes to represent the community, most of which were conducted by less than a thousand participants. Although they were statistically representative of the community, we recommend large scale surveys to guide public health interventions and inform decision making. Nearly all studies were cross sectional

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in nature that showed the problem at a specific point of time. Thus, long term studies, such as prospective cohorts can help to identify the causes of metabolic risk factors in the community.

No intervention has been tested to reduce the metabolic risk factors in Ethiopia. Thus, to identify feasible and effective interventions researchers should consider randomized control trails. Community based interventional studies help to inform the government interventions that are effective to reduce metabolic risk factors. In addition, studies are needed to understand behavioral factors that contribute metabolic risk factors to design and implement appropriate interventions.

Social determinants of health such as gender and age were found to be associated with metabolic risk factors. They are currently the main research and public health agenda in many countries. We also recommend Ethiopian researchers to assess the social gradients of health such as gender, age, education, occupation, income, housing, physical and social environment, public safety, transportation, and access to health care.

Strengths and limitations

This review is the first of its nature combining studies with similar characteristics in the context of metabolic risk factors for NCDs in Ethiopia in a single study. Despite the review has no a priori registered protocol, we strictly adhered to the Preferred Reporting Items for Systemic Reviews and Meta-analysis (PRISMA) guidelines during the design, conducting and reporting of our findings to increase robustness of the methods. The review uses prevalence studies to estimate the level of metabolic risk factors in adults, and most of the included studies had low risk of bias. The number of participants incorporated in this review were more than a hundred thousand. Thus, our prevalence estimate will fairly represent the adult population in Ethiopia. Furthermore, the study tried to show age and sex differences of metabolic risk factors distributions, make our result easy to understand, interpret, put in action.

Although most of the included studies in this review had low risk of bias to represent prevalence's of metabolic risks among adults in Ethiopia, the overall certainty of the evidence is low by heterogeneity of the included studies. Besides, we did not find studies conducted in Afar, Benishangul-Gumuz, Gambella regions, so our prevalence estimates might not precisely reflect the real situations in these regions. Moreover, the heterogeneity among the included studies precluded us from performing further analyses to identify individual study and clinical factors contributing to metabolic risk factors.

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Conclusion

This review showed a high proportion of metabolic risk factors in Ethiopian adults — one in five persons has at least one of the metabolic risk factors. The review summarized pooled prevalence's of overweight/obesity (24%), hypertension (21%), metabolic syndrome (15%), and impaired glucose tolerance (12%). All metabolic risks were higher in older adults, while overweight/obesity appeared to be higher in women. Despite included studies having low risk of bias, the quality of the evidence was low due to high heterogeneity to support these associations, and the small number included studies precluded detailed analysis of the associations. Since metabolic risk factors are immediate precursors for NCDs, they need to be managed and, when possible, reversed to reduce NCDs incidences and long-term consequences on health-related quality of life and other health outcomes. These results are important reminders of the need to effectively implement the multilayered strategies proposed by the Ethiopian government, including strengthening national health policy on NCDs and their risk factors, strengthening national and regional NCDs units, and raising awareness on the burden of these diseases among health workers and the wider community. It also highlights the need to tailor preventative interventions in high-risk populations like older adults and women. Further, it is important to have proper program evaluation in place to oversee the implementation of the strategies. Lifestyle education and health promotion strategies mainly aiming at modifying behavioral risk factors for NCDs can be instrumental in realizing the national and global targets. This study will inform the regional government to prepare management guidelines to significantly reduce the rise of metabolic risk factors to decrease the burden of non-communicable diseases, and for researchers to consider community-based interventional studies.

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Figure legends

Figure 1 Study selection flow diagram.

Figure 2 A forest plot of prevalence (%) of hypertension among adults in Ethiopia.

Figure 3 A forest plot of prevalence (%) of overweight/obesity among adults in Ethiopia.

Figure 4 A forest plot of prevalence (%) of impaired glucose tolerance among adults in Ethiopia.

Figure 5 A forest plot of prevalence (%) of metabolic syndrome among adults in Ethiopia.

Figure 6 Distribution of studies included in hypertension analysis in the funnel graph.

Figure 7 Distribution of studies included in overweight/obesity analysis in the funnel graph.

Figure 8 Distribution of studies included in impaired glucose tolerance in the funnel graph.

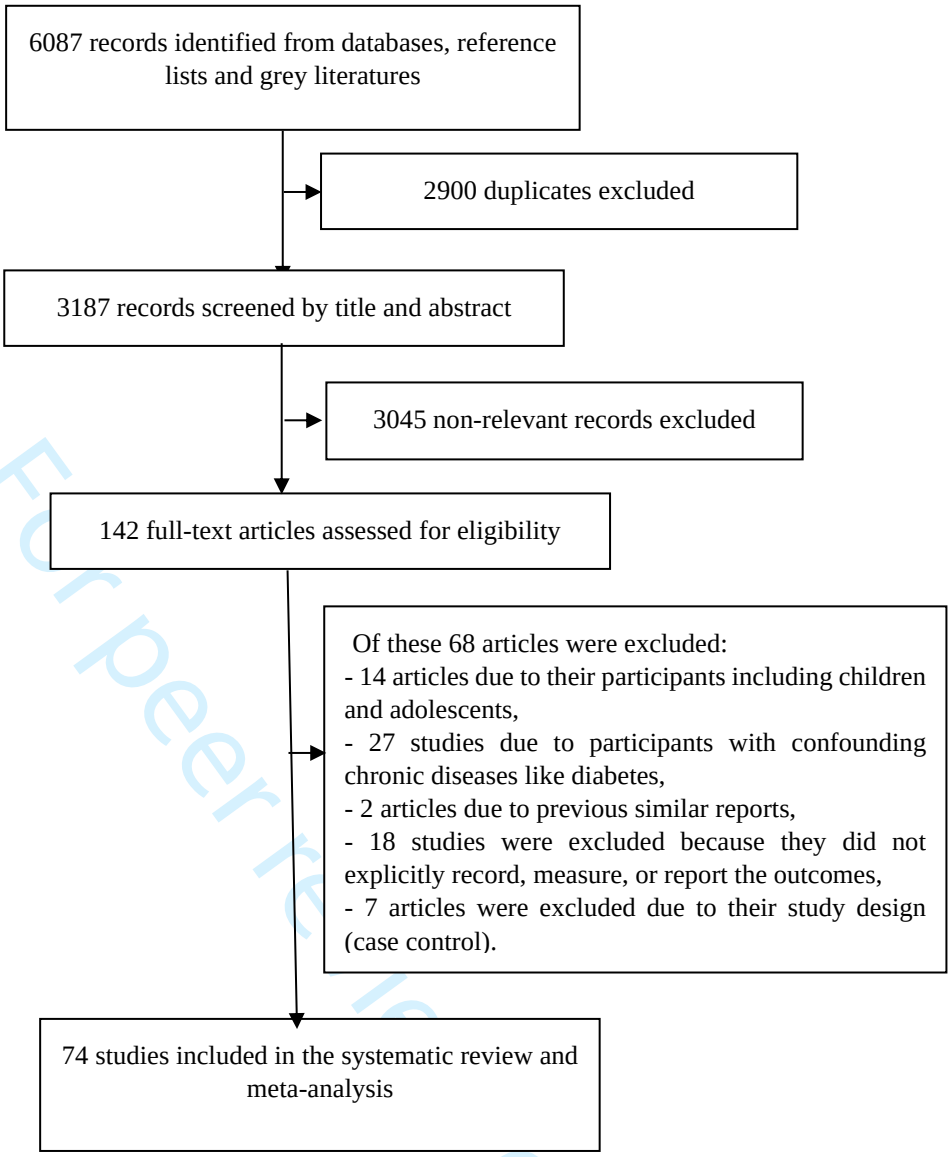
Figure 9 Distribution of studies included in metabolic syndrome analysis in the funnel graph.

Figure 10 A Forest plot of the association gender of participants with the prevalence of metabolic risk factors in Ethiopia (*female as reference*).

Figure 11 A forest plot of the association of age of participant with the prevalence of metabolic risk factors in Ethiopia *(people under the age of 45 years as a reference)***.**

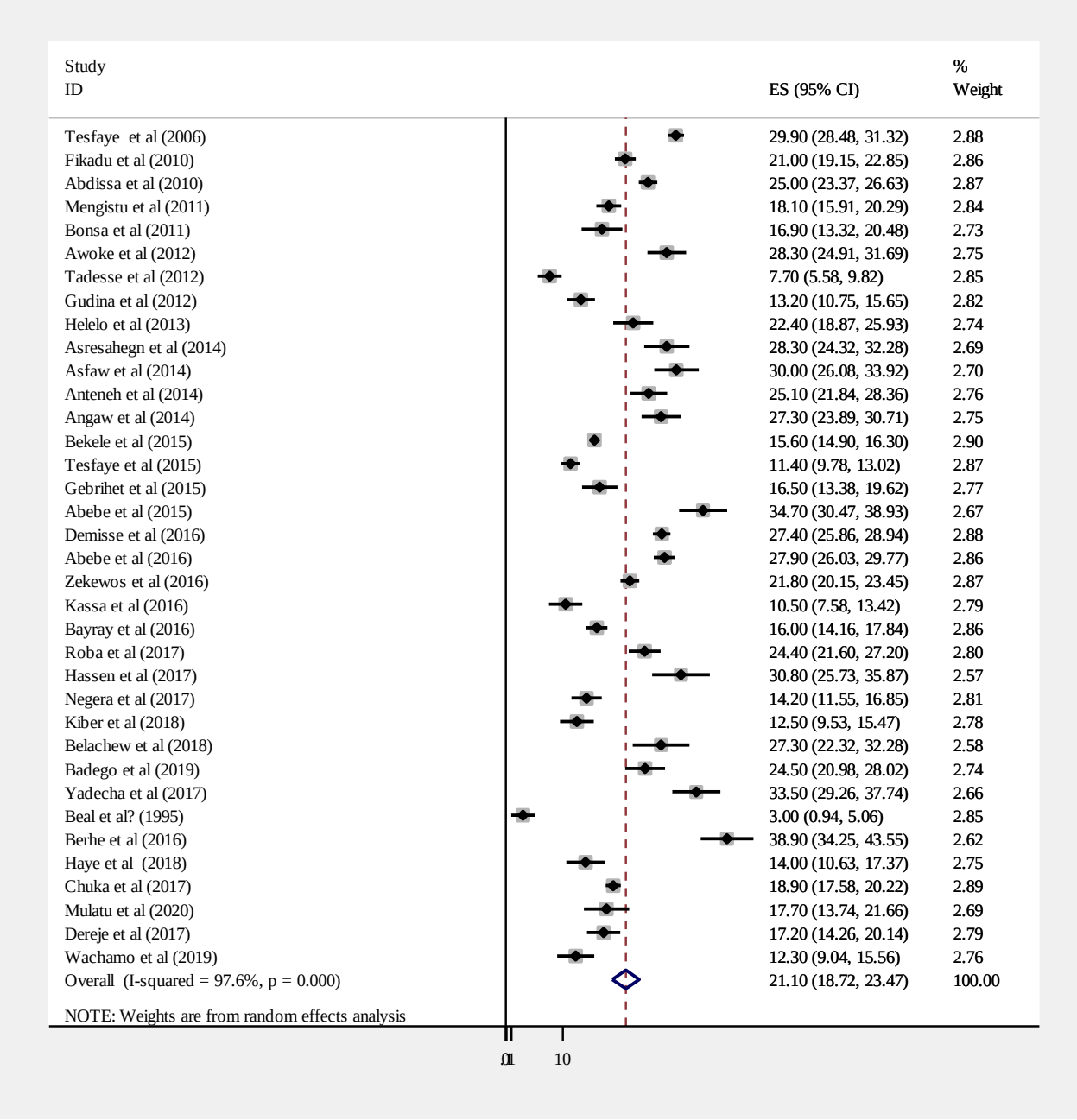
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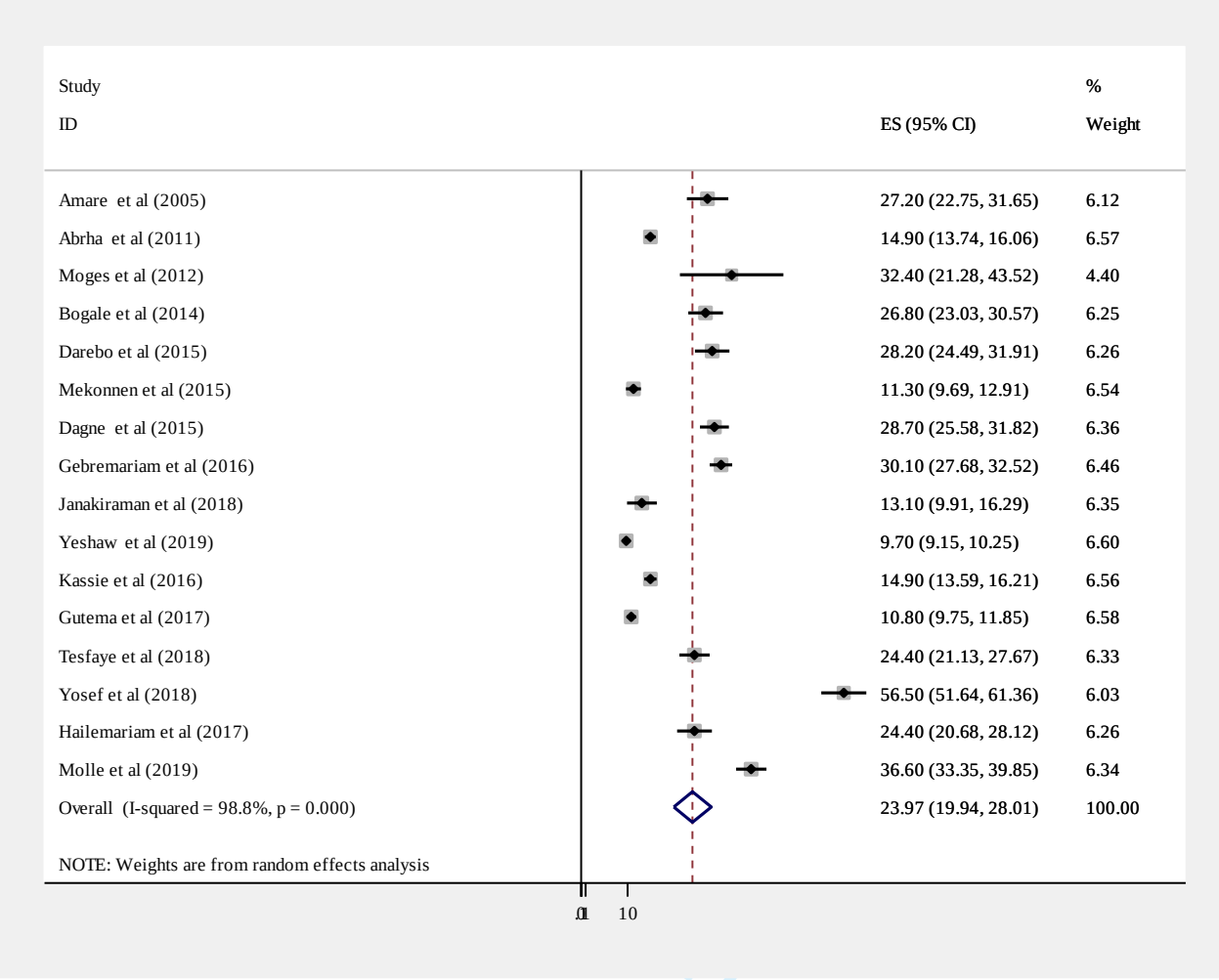
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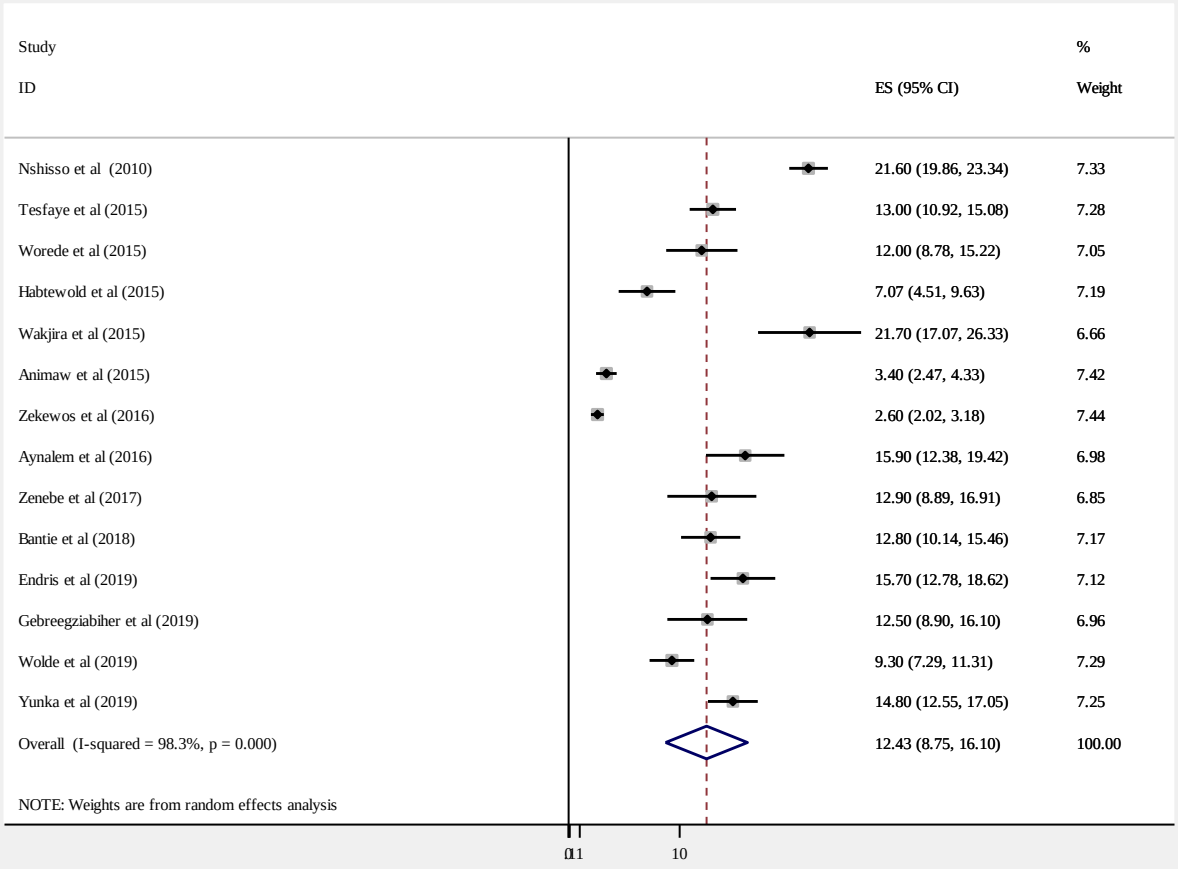
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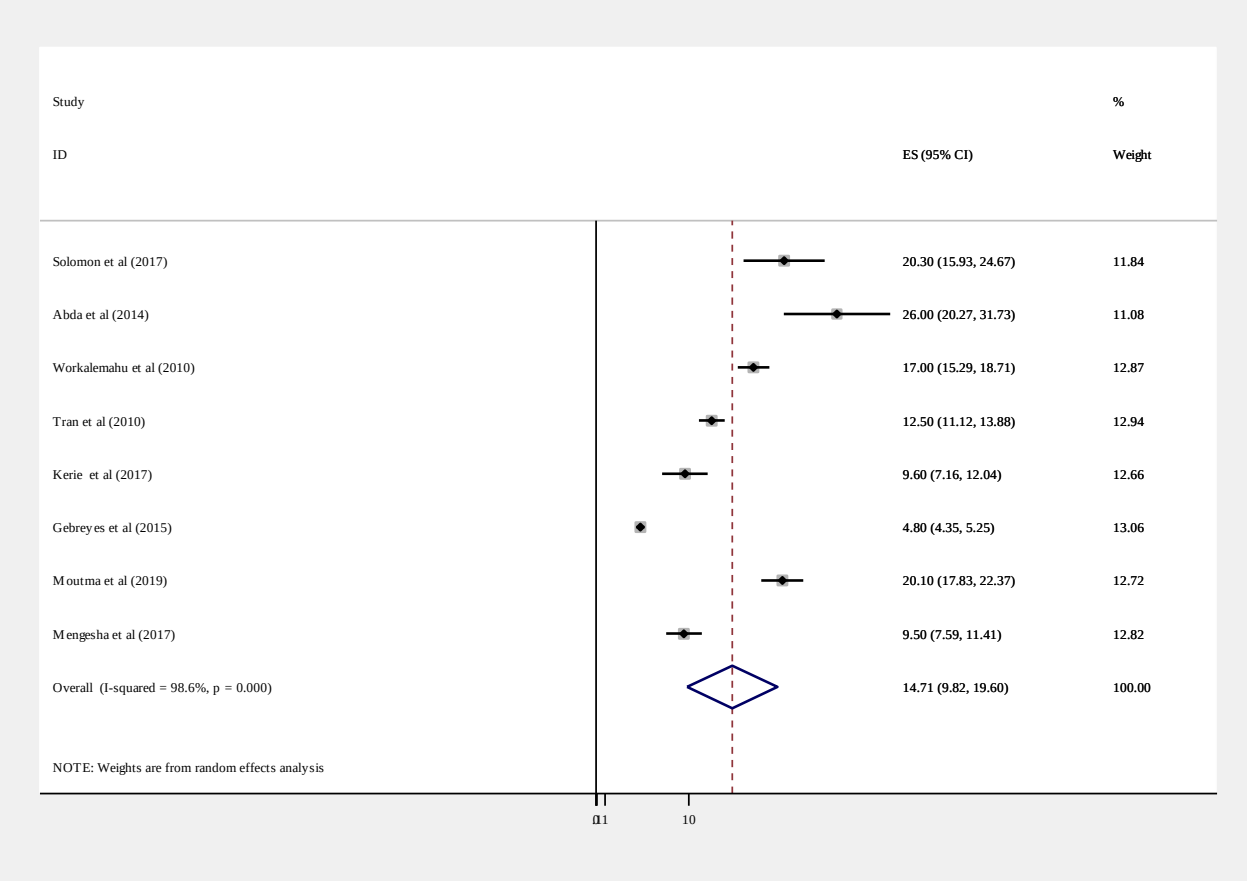

Study
ID
ES (95% CI)
%
Weight
Tesfaye et al (2006) 29.90 (28.48, 31.32) 2.88
Fikadu et al (2010) 21.00 (19.15, 22.85) 2.86
Abdissa et al (2010) 25.00 (23.37, 26.63) 2.87
Mengistu et al (2011) 18.10 (15.91, 20.29) 2.84
Bonsa et al (2011) 16.90 (13.32, 20.48) 2.73
Awoke et al (2012) 28.30 (24.91, 31.69) 2.75
Tadesse et al (2012) 7.70 (5.58, 9.82) 2.85
Gudina et al (2012) 13.20 (10.75, 15.65) 2.82
Helelo et al (2013) 22.40 (18.87, 25.93) 2.74
Asresahegn et al (2014) 28.30 (24.32, 32.28) 2.69
Asfaw et al (2014) 30.00 (26.08, 33.92) 2.70
Anteneh et al (2014) 25.10 (21.84, 28.36) 2.76
Angaw et al (2014) 27.30 (23.89, 30.71) 2.75
Bekele et al (2015) 15.60 (14.90, 16.30) 2.90
Tesfaye et al (2015) 11.40 (9.78, 13.02) 2.87
Gebrihet et al (2015) 16.50 (13.38, 19.62) 2.77
Abebe et al (2015) 34.70 (30.47, 38.93) 2.67
Demisse et al (2016) 27.40 (25.86, 28.94) 2.88
Abebe et al (2016) 27.90 (26.03, 29.77) 2.86
Zekewos et al (2016) 21.80 (20.15, 23.45) 2.87
Kassa et al (2016) 10.50 (7.58, 13.42) 2.79
Bayray et al (2016) 16.00 (14.16, 17.84) 2.86
Roba et al (2017) 24.40 (21.60, 27.20) 2.80
Hassen et al (2017) 30.80 (25.73, 35.87) 2.57
Negera et al (2017) 14.20 (11.55, 16.85) 2.81
Kiber et al (2018) 12.50 (9.53, 15.47) 2.78
Belachew et al (2018) 27.30 (22.32, 32.28) 2.58
Badego et al (2019) 24.50 (20.98, 28.02) 2.74
Yadecha et al (2017) 33.50 (29.26, 37.74) 2.66
Beal et al? (1995) 3.00 (0.94, 5.06) 2.85
Berhe et al (2016) 38.90 (34.25, 43.55) 2.62
Haye et al (2018) 14.00 (10.63, 17.37) 2.75
Chuka et al (2017) 18.90 (17.58, 20.22) 2.89
Mulatu et al (2020) 17.70 (13.74, 21.66) 2.69
Dereje et al (2017) 17.20 (14.26, 20.14) 2.79
Wachamo et al (2019) 12.30 (9.04, 15.56) 2.76
Overall (I-squared = 97.6%, p = 0.000) 21.10 (18.72, 23.47) 100.00
NOTE: Weights are from random effects analysis
10



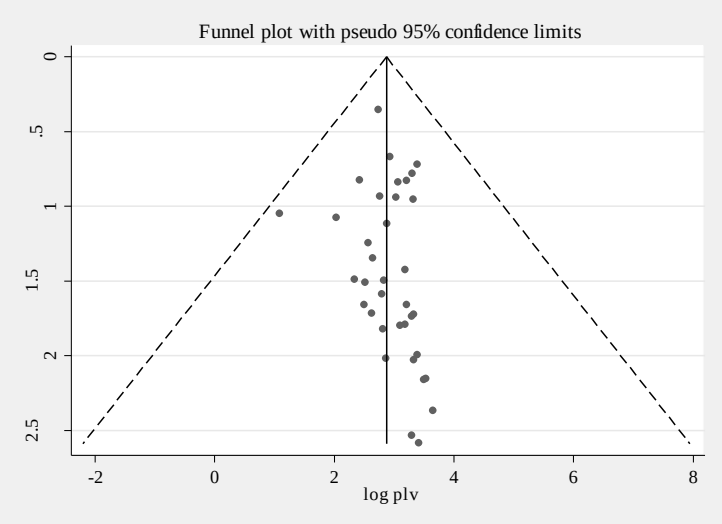
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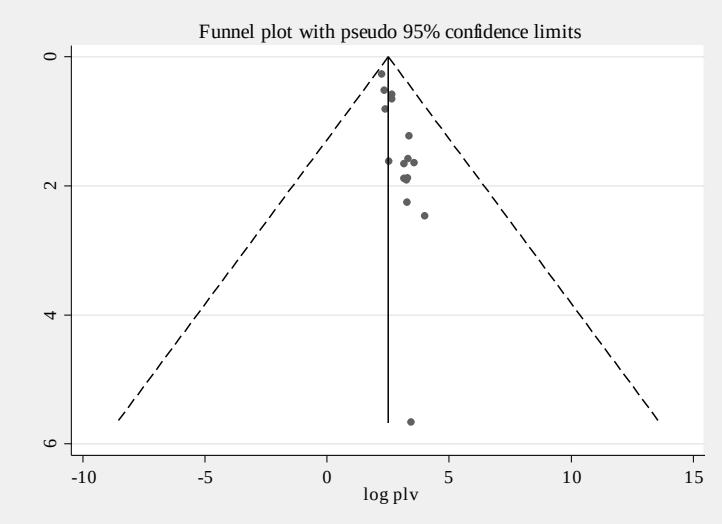
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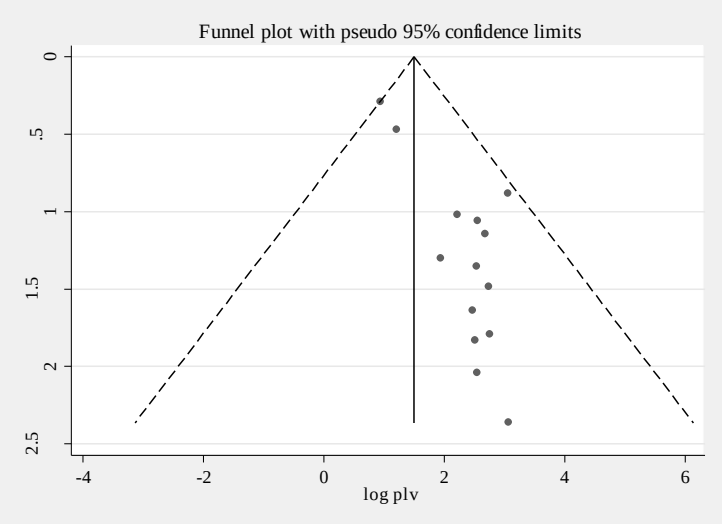
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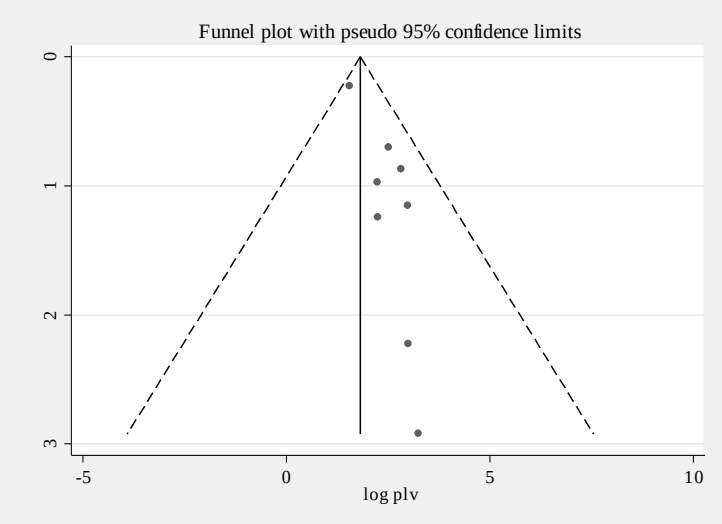
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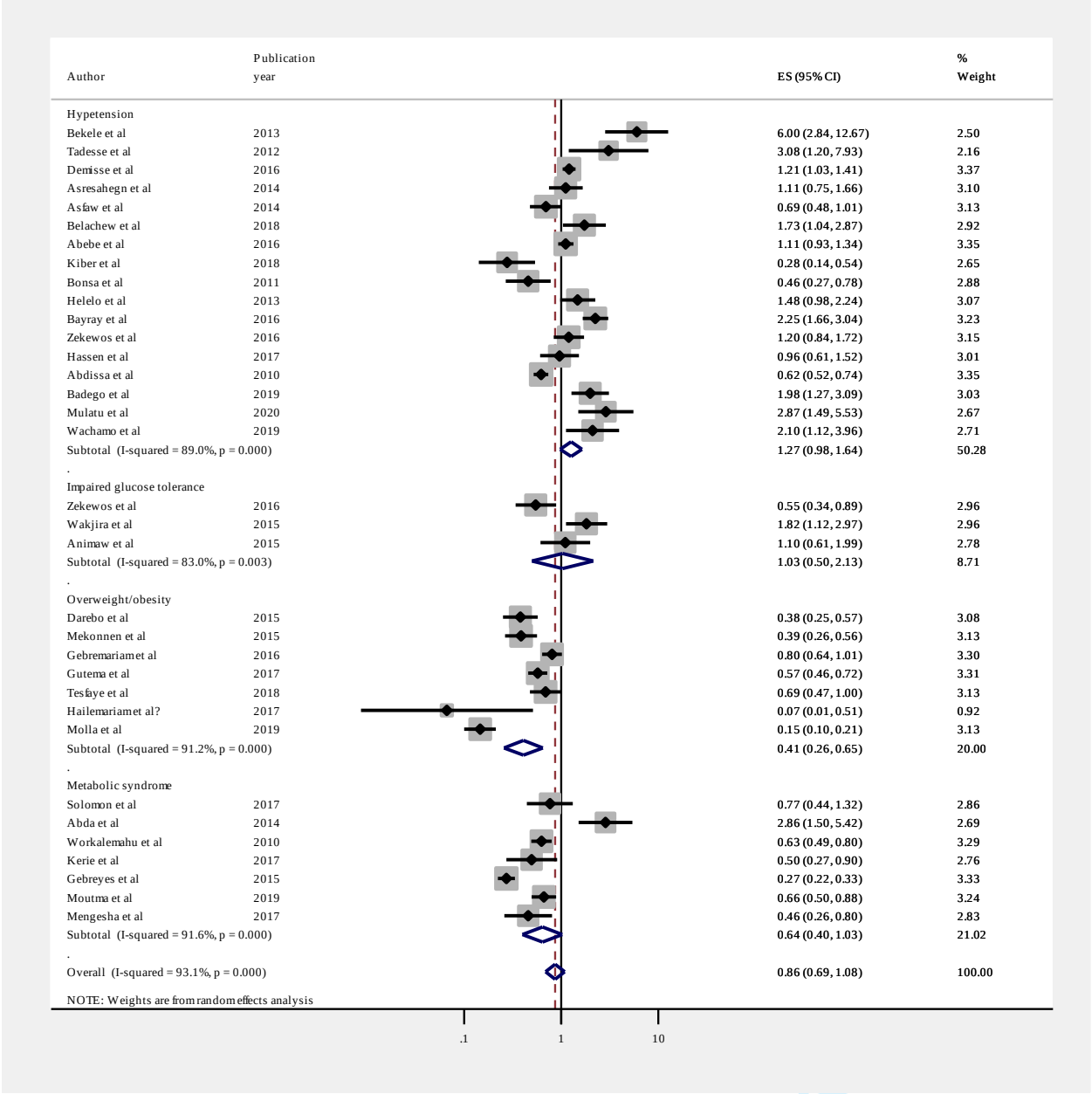
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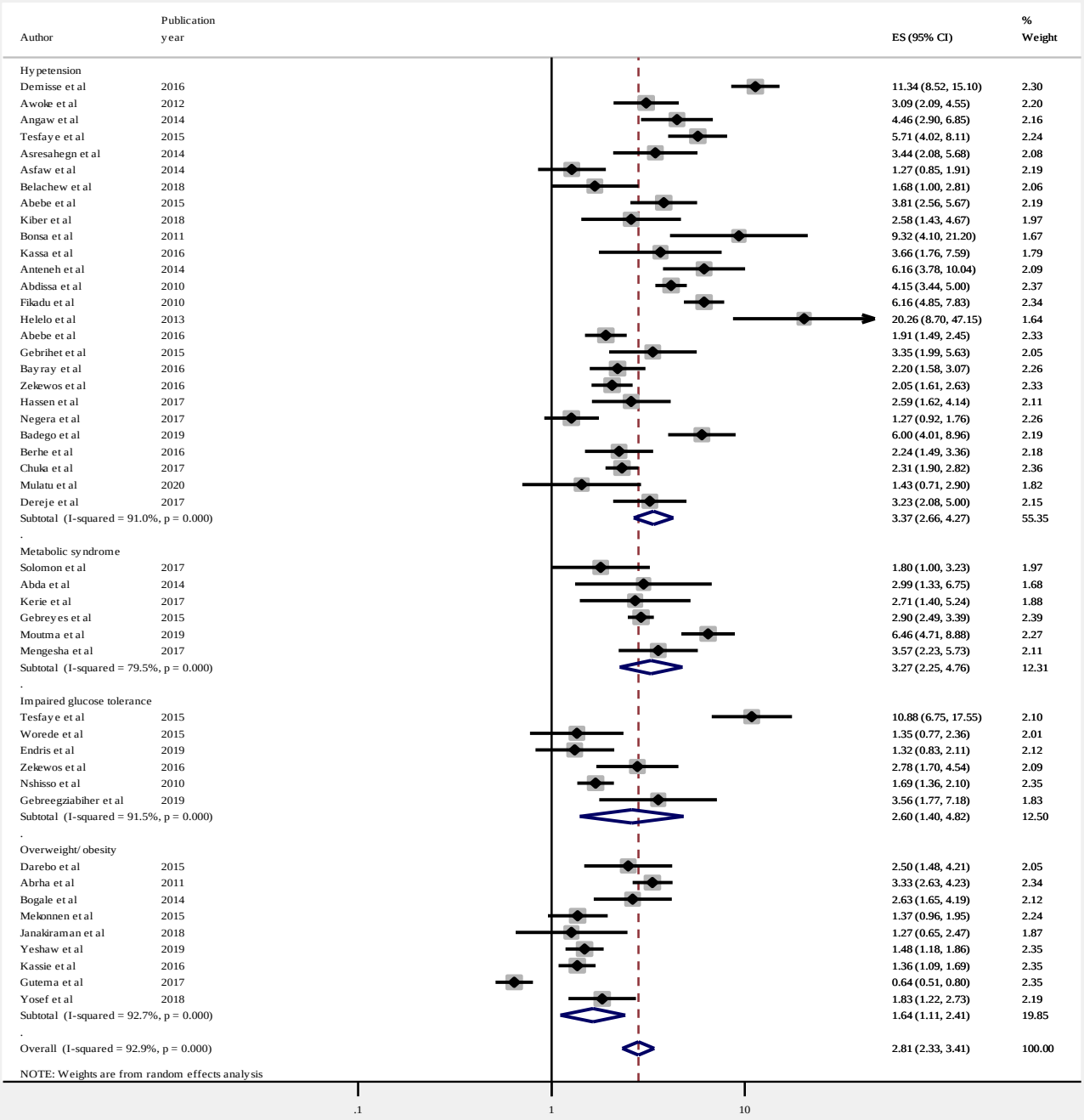
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Author
Publication year
ES (95% CI)
% Weight
Hypetension
Bekele et al 2013 6.00 (2.84, 12.67) 2.50
Tadesse et al 2012 3.08 (1.20, 7.93) 2.16
Demisse et al 2016 1.21 (1.03, 1.41) 3.37
Asresahegn et al 2014 1.11 (0.75, 1.66) 3.10
Asfaw et al 2014 0.69 (0.48, 1.01) 3.13
Belachew et al 2018 1.73 (1.04, 2.87) 2.92
Abebe et al 2016 1.11 (0.93, 1.34) 3.35
Kiber et al 2018 0.28 (0.14, 0.54) 2.65
Bonsa et al 2011 0.46 (0.27, 0.78) 2.88
Helelo et al 2013 1.48 (0.98, 2.24) 3.07
Bayray et al 2016 2.25 (1.66, 3.04) 3.23
Zekewos et al 2016 1.20 (0.84, 1.72) 3.15
Hassen et al 2017 0.96 (0.61, 1.52) 3.01
Abdissa et al 2010 0.62 (0.52, 0.74) 3.35
Badego et al 2019 1.98 (1.27, 3.09) 3.03
Mulatu et al 2020 2.87 (1.49, 5.53) 2.67
Wachamo et al 2019 2.10 (1.12, 3.96) 2.71
Subtotal (I-squared = 89.0%, p = 0.000) 1.27 (0.98, 1.64) 50.28
.
Impaired glucose tolerance
Zekewos et al 2016 0.55 (0.34, 0.89) 2.96
Wakjira et al 2015 1.82 (1.12, 2.97) 2.96
Animaw et al 2015 1.10 (0.61, 1.99) 2.78
Subtotal (I-squared = 83.0%, p = 0.003) 1.03 (0.50, 2.13) 8.71
.
Overweight/obesity
Darebo et al 2015 0.38 (0.25, 0.57) 3.08
Mekonnen et al 2015 0.39 (0.26, 0.56) 3.13
Gebremariamet al 2016 0.80 (0.64, 1.01) 3.30
Gutema et al 2017 0.57 (0.46, 0.72) 3.31
Tesfaye et al 2018 0.69 (0.47, 1.00) 3.13
Hailemariamet al? 2017 0.07 (0.01, 0.51) 0.92
Molla et al 2019 0.15 (0.10, 0.21) 3.13
Subtotal (I-squared = 91.2%, p = 0.000) 0.41 (0.26, 0.65) 20.00
.
Metabolic syndrome
Solomon et al 2017 0.77 (0.44, 1.32) 2.86
Abda et al 2014 2.86 (1.50, 5.42) 2.69
Workalemahu et al 2010 0.63 (0.49, 0.80) 3.29
Kerie et al 2017 0.50 (0.27, 0.90) 2.76
Gebreyes et al 2015 0.27 (0.22, 0.33) 3.33
Moutma et al 2019 0.66 (0.50, 0.88) 3.24
Mengesha et al 2017 0.46 (0.26, 0.80) 2.83
Subtotal (I-squared = 91.6%, p = 0.000) 0.64 (0.40, 1.03) 21.02
.
Overall (I-squared = 93.1%, p = 0.000) 0.86 (0.69, 1.08) 100.00
NOTE: Weights are from random effects analysis
.1
1
10

sex


Author
Publication year
ES (95% CI)
% Weight
Hypetension
Demisse et al 2016 11.34 (8.52, 15.10) 2.30
Awoke et al 2012 3.09 (2.09, 4.55) 2.20
Angaw et al 2014 4.46 (2.90, 6.85) 2.16
Tesfaye et al 2015 5.71 (4.02, 8.11) 2.24
Asresahegn et al 2014 3.44 (2.08, 5.68) 2.08
Asfaw et al 2014 1.27 (0.85, 1.91) 2.19
Belachew et al 2018 1.68 (1.00, 2.81) 2.06
Abebe et al 2015 3.81 (2.56, 5.67) 2.19
Kiber et al 2018 2.58 (1.43, 4.67) 1.97
Bonsa et al 2011 9.32 (4.10, 21.20) 1.67
Kassa et al 2016 3.66 (1.76, 7.59) 1.79
Anteneh et al 2014 6.16 (3.78, 10.04) 2.09
Abdissa et al 2010 4.15 (3.44, 5.00) 2.37
Fikadu et al 2010 6.16 (4.85, 7.83) 2.34
Helelo et al 2013 20.26 (8.70, 47.15) 1.64
Abebe et al 2016 1.91 (1.49, 2.45) 2.33
Gebrihet et al 2015 3.35 (1.99, 5.63) 2.05
Bayray et al 2016 2.20 (1.58, 3.07) 2.26
Zekewos et al 2016 2.05 (1.61, 2.63) 2.33
Hassen et al 2017 2.59 (1.62, 4.14) 2.11
Negera et al 2017 1.27 (0.92, 1.76) 2.26
Badego et al 2019 6.00 (4.01, 8.96) 2.19
Berhe et al 2016 2.24 (1.49, 3.36) 2.18
Chuka et al 2017 2.31 (1.90, 2.82) 2.36
Mulatu et al 2020 1.43 (0.71, 2.90) 1.82
Dereje et al 2017 3.23 (2.08, 5.00) 2.15
Subtotal (I-squared = 91.0%, p = 0.000) 3.37 (2.66, 4.27) 55.35
Metabolic syndrome
Solomon et al 2017 1.80 (1.00, 3.23) 1.97
Abda et al 2014 2.99 (1.33, 6.75) 1.68
Kerie et al 2017 2.71 (1.40, 5.24) 1.88
Gebreyes et al 2015 2.90 (2.49, 3.39) 2.39
Moutma et al 2019 6.46 (4.71, 8.88) 2.27
Mengesha et al 2017 3.57 (2.23, 5.73) 2.11
Subtotal (I-squared = 79.5%, p = 0.000) 3.27 (2.25, 4.76) 12.31
Impaired glucose tolerance
Tesfaye et al 2015 10.88 (6.75, 17.55) 2.10
Worede et al 2015 1.35 (0.77, 2.36) 2.01
Endris et al 2019 1.32 (0.83, 2.11) 2.12
Zekewos et al 2016 2.78 (1.70, 4.54) 2.09
Nshisso et al 2010 1.69 (1.36, 2.10) 2.35
Gebreegziabiher et al 2019 3.56 (1.77, 7.18) 1.83
Subtotal (I-squared = 91.5%, p = 0.000) 2.60 (1.40, 4.82) 12.50
Overweight/ obesity
Darebo et al 2015 2.50 (1.48, 4.21) 2.05
Abrha et al 2011 3.33 (2.63, 4.23) 2.34
Bogale et al 2014 2.63 (1.65, 4.19) 2.12
Mekonnen et al 2015 1.37 (0.96, 1.95) 2.24
Janakiraman et al 2018 1.27 (0.65, 2.47) 1.87
Yeshaw et al 2019 1.48 (1.18, 1.86) 2.35
Kassie et al 2016 1.36 (1.09, 1.69) 2.35
Gutema et al 2017 0.64 (0.51, 0.80) 2.35
Yosef et al 2018 1.83 (1.22, 2.73) 2.19
Subtotal (I-squared = 92.7%, p = 0.000) 1.64 (1.11, 2.41) 19.85
Overall (I-squared = 92.9%, p = 0.000) 2.81 (2.33, 3.41) 100.00
NOTE: Weights are from random effects analysis
.1
1
10

Supplementary table 1: Database search strategies and results

1. PubMed search techniques and results

Serial number	Themes	Search terms and combinations	Results
1	Metabolic related	"Metabolic risk factors" OR "cardiometabolic risk factors" OR "metabolic syndrome" OR dyslipidemia* OR dyslipidaemia* OR Hyperlipidemia* OR hyperlipidaemia* OR "high blood cholesterol" OR hypercholesterolemia OR hypercholesterolaemia OR triglycerides OR hypertriglyceridemia OR hypertriglyceridaemia	288,508
2	Overweight and obesity related	Overnutrition OR malnutrition OR overweight OR obesity OR obese	536,570
3	Blood glucose related	"Blood glucose" OR "impaired glucose tolerance" OR "glucose intolerance" OR "high blood glucose" OR hyperglycemia OR "High fasting plasma glucose" OR "insulin resistance" OR diabetes OR "diabetes mellitus"	925,083
4	Hypertension related	"Blood pressure" OR "high blood pressure" OR hypertension OR "systolic blood pressure" OR "diastolic blood pressure"	83,3024
5	Combination of themes	(#1 OR #2 OR #3 OR #4) AND Ethiopia	2,244

2. Web of science search techniques and results

Serial number	Themes	Search terms and combinations	Results
1	Metabolic related	TS= ("metabolic risk factors" OR "cardiometabolic risk factors" OR "metabolic syndrome" OR dyslipidemia* OR dyslipidaemia* OR Hyperlipidemia* OR hyperlipidaemia* OR "high blood cholesterol" OR hypercholesterolemia OR hypercholesterolaemia OR triglycerides OR hypertriglyceridemia OR hypertriglyceridaemia)	270,247
2	Overweight and obesity related	TS= (Overnutrition OR malnutrition OR overweight OR obesity OR obese)	471,995
3	Blood glucose related	TS= ("blood glucose" OR "impaired glucose tolerance" OR "glucose intolerance" OR "high blood glucose" OR hyperglycemia OR "High fasting plasma glucose" OR "insulin resistance" OR diabetes OR "diabetes mellitus")	721,475
4	Hypertension related	TS= ("blood pressure" OR "high blood pressure" OR hypertension OR "systolic blood pressure" OR "diastolic blood pressure")	625,826
5	Combination of themes	(#1 OR #2 OR #3 OR #4) AND Ethiopia	1,389

3. ProQuest search techniques and results

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Serial number	Themes	Search terms and combinations	Results
1	Metabolic related	ti ("metabolic risk factors" OR "cardiometabolic risk factors" OR "metabolic syndrome" OR dyslipidemia* OR dyslipidaemia* OR Hyperlipidemia* OR hyperlipidaemia* OR "high blood cholesterol" OR hypercholesterolemia OR hypercholesterolaemia OR triglycerides OR hypertriglyceridemia OR hypertriglyceridaemia)	34,463
2	Overweight and obesity related	ti (Overnutrition OR malnutrition OR overweight OR obesity OR obese)	277,110
3	Blood glucose related	ti ("blood glucose" OR "impaired glucose tolerance" OR "glucose intolerance" OR "high blood glucose" OR hyperglycemia OR "High fasting plasma glucose" OR "insulin resistance" OR diabetes OR "diabetes mellitus")	357,066
4	Hypertension related	ti ("blood pressure" OR "high blood pressure" OR hypertension OR "systolic blood pressure" OR "diastolic blood pressure")	145,255
5	Combination of themes	(#1 OR #2 OR #3 OR #4) AND Ethiopia	464

4. Scopus search techniques and results

Serial number	Themes	Search terms and combinations	Results
1	Metabolic related	TITLE-ABS-KEY ("metabolic risk factors" OR "cardiometabolic risk factors" OR "metabolic syndrome" OR dyslipidemia* OR dyslipidaemia* OR Hyperlipidemia* OR hyperlipidaemia* OR "high blood cholesterol" OR hypercholesterolemia OR hypercholesterolaemia OR triglycerides OR hypertriglyceridemia OR hypertriglyceridaemia)	413,733
2	Overweight and obesity related	TITLE-ABS-KEY (Overnutrition OR malnutrition OR overweight OR obesity OR obese)	597,494
3	Blood glucose related	TITLE-ABS-KEY ("blood glucose" OR "impaired glucose tolerance" OR "glucose intolerance" OR "high blood glucose" OR hyperglycemia OR "High fasting plasma glucose" OR "insulin resistance" OR diabetes OR "diabetes mellitus")	1104,668
4	Hypertension related	TITLE-ABS-KEY ("blood pressure" OR "high blood pressure" OR hypertension OR "systolic blood pressure" OR "diastolic blood pressure")	1183526
5	Combination of themes	(#1 OR #2 OR #3 OR #4) AND Ethiopia	1,989

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Supplementary table 2: Characteristics of included studies (n=74) on metabolic risk factors for NCDs in Ethiopia.

Author	Year of study	Study area	Region of the study	Study setting	Age of participant	Types of metabolic risk factor studied	Response rate (%)	Sample size	Quality score of studies	Number of participants with Outcome	Prevalence (%)
Tesfaye et al [1]	2006	Addis Ababa	Addis Ababa City administration	Community	25–64	Hypertension	88.1	4001	7	981	29.9
Awoke et al [2]	2012	Gondar	Amhara	Community	35+	Hypertension	97.6	679	8	192	28.3
Mengistu et al[3]	2011	Humera	Tigray	Community	18+	Hypertension	100	1183	8	214	18.1
Angaw et al[4]	2014	Addis Ababa	Addis Ababa City administration	Institution	18+	Hypertension	96	655	8	172	27.3
Tadesse et al[5]	2012	Gondar	Amhara	Institution	18+	Hypertension	100	610	8	47	7.7
Demisse et a[6]	2016	Gondar	Amhara	Community	18+	Hypertension	94.8	3227	8	837	27.4
Asresahegn et al[7]	2014	Jigjiga	Somali	Community	25–65	Hypertension	98.9	492	8	138	28.3
Asfaw et al[8]	2014	Hossana	South/SNNPR	Community	25–65	Hypertension	99.8	525	7	157	30
Belachew et al[9]	2018	Bahir Dar	Amhara	Institution	18+	Hypertension	100	308	6	84	27.3
Abebe et al[10]	2015	Addis Ababa	Addis Ababa City	Institution	18+	Hypertension	100	487	7	169	34.7

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			administration								
Kiber et al[11]	2018	Debre Markos	Amhara	Community	18+	Hypertension	95.6	477	8	57	12.5
Bonsa et al[12]	2011	Bedele	Oromia	Community	15+	Hypertension	93.8	422	7	67	16.9
Gudina et al[13]	2012	Jimma	Oromia	Institution	15+	Hypertension	100	734	7	97	13.2
Kassa et al[14]	2016	Hawassa	South/SNNPR	Institution	18+	Hypertension	97.25	422	8	43	10.5
Anteneh et al[15]	2014	Bahir Dar	Amhara	Community	30+	Hypertension	99.6	681	8	149	25.1
Roba et al[16]	2017	Dire Dawa	Dire Dawa City council	Community	25-64	Hypertension	96.5	903	8	213	24.4
Helelo et al[17]	2013	Durame	South/SNNPR	Community	31+	Hypertension	96.6	536	7	116	22.4
Abebe et al[18]	2013	Dabat	Amhara	Community	18+	Hypertension	97.3	2200	8	598	27.9
Gebrihet et al[19]	2015	Aksum	Tigray	Community	18+	Hypertension	96	544	8	86	16.5
Bayray et al[20]	2016	Mekelle	Tigray	Institution	18+	Hypertension	99.7	1525	8	243	16
Zekewos et al[21]	2016	Bona District	South/SNNPR	Community	25+	Hypertension	82.1	2401	8	196	21.8
Hassen et al[22]	2017	Kombolcha	Amhara	Community	18+	Hypertension	100	318	8	96	30.8
Negera et al[23]	2017	Illubabor Zone	Oromia	Community	18+	Hypertension	97	668	8	92	14.2
Fikadu et al [24]	2010	Addis Ababa	Addis Ababa City administration	Community	18+	Hypertension	100	1866	8	392	21

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Abdissa et al[25]	2010	Addis Ababa	Addis Ababa City administration	Community	18+	Hypertension	100	2716	8	677	25
Bekele et al[26]	2015	Ethiopia	Ethiopia	Community	15-69	Hypertension	95.5	10,260	6	1527	15.6
Tesfaye et al[27]	2015	Bahir Dar	Amhara	Community	18+	Hypertension	95.5	1472	8	160	11.4
Badego et al[28]	2019	Sidama	South/SNNPR	Institution	18+	Hypertension	94.9	575	8	134	24.5
Yadecha et al[29]	2017	Gimbi	Oromia	Community	18+	Hypertension	98.9	476	7	157	33.5
Beall et al[30]	1995	Ambaras	Amhara	Community	18+	Hypertension	100	263	6	8	3
Berhe et al[31]	2016	Yekatit12	Addis Ababa City administration	Institution	25-64	Hypertension	97.9	423	8	106	38.9
Haye et al[32]	2018	Akaki	Addis Ababa City administration	Institution	18+	Hypertension	98.5	407	7	54	14
Chuka et al[33]	2017	Arba Minch	South/SNNPR	Community	25-64	Hypertension	99.4	3368	8	633	18.9
Mulatu et al[34]	2020	Mizan Aman	South/SNNPR	Community	18+	Hypertension	98.6	356	7	62	17.7
Dereje et al[35]	2017	Hossana	South/SNNPR	Community	18+	Hypertension	98.9	634	8	108	17.2

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Wachamo et al[36]	2019	Hawela	South/SNNPR	Community	18+	Hypertension	98.2	390	7	47	12.3
Tesfaye et al[37]	2015	Addis Ababa	Addis Ababa City administration	Institution	18+	Impaired glucose tolerance	93.3	1003	8	120	13
Worede et al[38]	2015	Koladiba	Amhara	Community	≥20	Impaired glucose tolerance	100	392	8	47	12
Bantie et al[39]	2018	Bahir Dar	Amhara	Community	18+	Impaired glucose tolerance	100	607	7	70	12.8
Endris et al[40]	2019	Dessie	Amhara	Community	18+	Impaired glucose tolerance	98.2	598	8	92	15.7
Zekewos et al[41]	2016	Bona District	South/SNNPR	Community	15+	Impaired glucose tolerance	91.4	2922	8	69	2.6
Habtewold et al[42]	2015	Debre Birhan	Amhara	Institution	15+	Impaired glucose tolerance	95.3	385	7	26	7.1
Nshisso et al[43]	2010	Addis Ababa	Addis Ababa City administration	Institution	18+	Impaired glucose tolerance	100	2153	7	408	21.6
Aynalem et al[44]	2016	Mizan	South/SNNPR	Community	15+	Impaired glucose tolerance	97.1	414	7	64	15.9
Wakjira et al[45]	2015	Addis Ababa	Addis Ababa City administration	Facility	20+	Impaired glucose tolerance	100	304	7	66	21.7
Zenebe et al[46]	2017	Jimma	Oromia	Community	40+	Impaired glucose tolerance	98.1	269	8	34	12.9

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Animaw et al[47]	2015	Bahir Dar	Amhara	Community	18+	Impaired glucose tolerance	95.5	1472	7	48	3.4
Gebreegziabiher et al[48]	2019	Mekelle	Tigray	Community	20+	Impaired glucose tolerance	98.8	325	6	40	12.5
Wolde et al[49]	2019	Gonder	Amhara	Community	18+	Impaired glucose tolerance	96	805	7	72	9.3
Yunka et al[50]	2019	Jimma	Oromia	Community	18+	Impaired glucose tolerance	95.8	955	7	135	14.8
Darebo et al[51]	2015	Hawassa	South/SNNPR	Community	18+	Overweight/ obesity	92.7	565	8	148	28.2
Moges et al[52]	2012	Gondar	Amhara	Community	18+	Overweight/ obesity	100	68	8	13	32.4
Abrha et al[53]	2011	Ethiopia	Ethiopia	Community	15-49	Overweight/ obesity	100	3602	8	534	14.9
Bogale et al[54]	2014	Addis Ababa	Addis Ababa City administration	Institution	18+	Overweight/ obesity	98.87	531	8	141	26.8
Mekonnen et al[55]	2015	Bahir Dar	Amhara	Community	18+	Overweight/ obesity	94.7	1484	8	159	11.3
Dagne et al[56]	2015	Dessie	Amhara	Community	18–64	Overweight/ obesity	93.29	805	8	216	28.7
Janakiraman et al[57]	2018	Gondar	Amhara	Institution	25+	Overweight/ obesity	89	429	8	50	13.1
Amare et al[58]	2005	Gondar	Amhara	Community	18 +	Overweight/ obesity	93	384	6	97	27.2
Yeshaw et al[59]	2019	Ethiopia	Ethiopia	Community	15-49	Overweight/ obesity	100	10 938	8	1058	9.7

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Kassie et al[60]	2016	Ethiopia	Ethiopia	Community	15-49	Overweight/ obesity	100	2848	8	425	14.9
Gutema et al[61]	2017	Arba Minch	South/SNNPR	Community	25-64	Overweight/ obesity	99.4	3368	8	361	10.8
Tesfaye et al[62]	2018	Dilla	South/SNNPR	Community	18-64	Overweight/ obesity	95.6	663	8	155	24.4
Yosef et al[63]	2018	Modjo	Oromia	Institution	18-64	Overweight/ obesity	100	400	6	126	56.5
Hailemariam et al[64]	2017	Addis Ababa	Addis Ababa City administration	Community	18-64	Overweight/ obesity	95	512	7	113	24.4
Molla et al[65]	2019	Gonder	Amhara	Community	18+	Overweight/ obesity	91.5	845	7	174	36.6
Gebremariam et al[66]	2016	Mekelle	Tigray	Community	18+	Overweight/ obesity	90.4	1380	6	415	30.1
Solomon et al[67]	2017	Addis Ababa	Addis Ababa City administration	Institution	18+	Metabolic syndrome	100	325	8	66	20.3
Abda et al[68]	2014	Jimma	Oromia	Institution	20+	Metabolic syndrome	100	225	8	59	26
Workalemahu et al[69]	2010	Addis Ababa	Addis Ababa City administration	Institution	18+	Metabolic syndrome	83.24	1843	7	313	17

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Tran et al[70]	2010	Addis Ababa	Addis Ababa City administration	Institution	18+	Metabolic syndrome	88.78	2205	6	346	12.5
Kerie et al[71]	2017	Mizan	South/SNNPR	Community	18+	Metabolic syndrome	95.7	558	8	51	9.6
Gebreyes et al[72]	2015	Ethiopia	Ethiopia	Community	15-69	Metabolic syndrome	84.5	8673	8	757	4.8
Moutma et al[73]	2019	Eastern Ethiopia	Oromia	Institution	19-64	Metabolic syndrome	97	1200	8	234	20.1
Mengesha et al[74]	2017	Dire Dawa	Dire Dawa city council	Community	25-64	Metabolic syndrome	96.6	903	7	83	9.5

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Supplementary table 3: The completed GRADE checklist for overall quality assessment of evidence for metabolic risk factors review among adults in Ethiopia.

Outcomes	Number of Studies	Risk of bias	Inconsistency	indirectness	imprecision	Publication bias	Certainty of evidence
Hypertension	36	Not serious	Serious	Not serious	Not serious	Not serious	⊕⊕⊕⊕ Moderate
Overweight/ obesity	16	Not serious	Serious	Not serious	Not serious	Serious	⊕⊕⊕⊕ Low
Impaired glucose tolerance	14	Not serious	Serious	Not serious	Not serious	Serious	⊕⊕⊕⊕ Low
Metabolic syndrome	8	Not serious	*	*	*	*	⊕⊕⊕⊕ Low
The association of sex and hypertension	17	Not serious	Serious	Not serious	Serious	Not serious	⊕⊕⊕⊕ Low
The association of sex and overweight/obesity	7	Not serious	*	*	*	*	⊕⊕⊕⊕ Low
The association of sex and impaired glucose tolerance	3	Not serious	*	*	*	*	⊕⊕⊕⊕ Low
The association of sex and metabolic syndrome	7	Not serious	*	*	*	*	⊕⊕⊕⊕ Low
The association of age and Hypertension	26	Not serious	Serious	Not serious	Not serious	Not serious	⊕⊕⊕⊕ Moderate
The association of age and overweight/obesity	9	Not serious	*	*	*	*	⊕⊕⊕⊕ Low
The association of age and impaired glucose tolerance	6	Not serious	*	*	*	*	⊕⊕⊕⊕ Low
The association of age and metabolic syndrome	6	Not serious	*	*	*	*	⊕⊕⊕⊕ Low

*Due to small number of included studies (n <10) could not be formally assessed.

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Reasons to downrate certainty of evidence

1. **Risk of bias** was assessed by the JBIs critical appraisal tool for prevalence studies and categorized as follows:
   Not serious: if the included study in the analysis has low risk of bias score in the quality assessment
   Serious: judged between not serious and very serious or if some of the included study had high risk bias but lower to say very serious.
   Very serious: if more than 50% of the included studies had higher risk of bis score in the quality assessment.
2. **Inconsistency (heterogeneity)** of the included studies was based on the following criteria:
   Not serious: lower heterogeneity (<30%), Overlap of confidence intervals, close variance estimates across studies.
   Serious: if one of the following: high level heterogeneity (30-60 or above) or minimal overlap of confidence intervals or wider variance estimates across studies.
   Very serious: if two or all of the following: higher heterogeneity (30-60%) or above and minimal overlap of confidence intervals and/or wider variance estimates across studies.
3. **Indirectness** of the study was examined by assessing the concordance in population, exposure, and outcome (PEO's) of the study and the evidence
   Not serious: not have indirect comparison, not have different in terms of population, or use the same outcome measure
   Serious: if the studies have any of the following: indirect comparison, or different in terms of population, or different outcome measure
   Very serious: if the studies involved two or more of the following: indirect comparison, different in terms of population, and different outcome measure
4. **Precision** of the evidence was assessed by examining the width of the confidence intervals and associated effect of the exposure
   Not serious: if the confidence intervals of estimates are precise
   Serious: if the estimates involved wide confidence interval
   Very serious: if the estimates involved very wide confidence interval
5. **Publication bias** was assessed by observing the symmetrical distribution of studies in the funnel plot and conducting the Begg's and Egger's test of publication bias.
   Not serious: If study had symmetrical distribution in the final plot and tests did not show significant associations.
   Serious: If study had symmetrical distribution in the final plot but Begg's and Egger's tests showed significant publication bias.
   Very serious: if studies had not symmetrical distribution in the final plot, and also showed publication bias the Begg's and Egger's test.

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Reasons to increase certainty of evidence

1. Large magnitude of effect
   None: if the effect estimate is <2
   Strong: if the effect estimate is >2 or <0.5
   Very strong: effect estimate >4 or <0.25
2. Effect potential residual confounding factors
   Dilute the effect: when the measurement is taken in some selected portion of participants and the actual measurement is expected to result higher estimates.
   Suggest strong effect: when residual confounding is expected to increase the effect but not measured.
3. Dose response gradient
   Yes: if there is dose gradient relationship
   No: if there is no dose gradient relationship

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Egger's test

Std_Eff	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
slope	2.754683	.1588469	17.34	0.000	2.431867	3.077498
bias	.1234977	.1484605	0.83	0.411	-.1782103	.4252056

Supplementary figure 1: Eggers's test for detection of publication bias for studies included in hypertension.

Egger's test

Std_Eff	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
slope	2.136301	.075485	28.30	0.000	1.974402	2.2982
bias	.6232349	.0960689	6.49	0.000	.4171876	.8292822

Supplementary figure 2: Eggers's test for detection of publication bias for studies included in overweight/obesity.

Egger's test

Std_Eff	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
slope	.6438857	.1736983	3.71	0.003	.2654296	1.022342
bias	1.377838	.2219967	6.21	0.000	.8941491	1.861528

Supplementary figure 3: Eggers's test for detection of publication bias for studies included in impaired glucose tolerance.

Egger's test

Std_Eff	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
slope	1.37992	.1365355	10.11	0.000	1.045829	1.71401
bias	1.060526	.2416536	4.39	0.005	.4692211	1.651831

Supplementary figure 4: Eggers's test for detection of publication bias for studies included in metabolic syndrome.

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```
. metatrim pre sep, reffect graph funnel

Note: default data input format (theta, se_theta) assumed.

Meta-analysis

       |  Pooled      95% CI          Asymptotic      No. of
Method |     Est   Lower   Upper  z_value  p_value   studies
-------+-------------------------------------------------------
Fixed  |  13.384  12.999  13.769   68.133    0.000      16
Random |  23.975  19.938  28.012   11.641    0.000

Test for heterogeneity: Q= 1220.870 on 15 degrees of freedom (p= 0.000)
Moment-based estimate of between studies variance = 64.187

Trimming estimator: Linear
Meta-analysis type: Random-effects model

iteration |  estimate     Tn     # to trim     diff
----------+-----------------------------------------
    1     |   23.975      64          0         136
    2     |   23.975      64          0           0

Note: no trimming performed; data unchanged

Filled
Meta-analysis

       |  Pooled      95% CI          Asymptotic      No. of
Method |     Est   Lower   Upper  z_value  p_value   studies
-------+-------------------------------------------------------
Fixed  |  13.384  12.999  13.769   68.133    0.000      16
Random |  23.975  19.938  28.012   11.641    0.000

Test for heterogeneity: Q= 1220.870 on 15 degrees of freedom (p= 0.000)
Moment-based estimate of between studies variance = 64.187
```

Supplementary figure 5: Trim-and-fill analysis for the prevalence of overweight/obesity among adults in Ethiopia.

```
. metatrim pre sep, reffect graph funnel

Note: default data input format (theta, se_theta) assumed.

Meta-analysis

       |  Pooled      95% CI          Asymptotic      No. of
Method |     Est   Lower   Upper  z_value  p_value   studies
-------+-------------------------------------------------------
Fixed  |   6.235   5.825   6.644   29.814    0.000      14
Random |  12.426   8.753  16.100    6.630    0.000

Test for heterogeneity: Q= 764.415 on 13 degrees of freedom (p= 0.000)
Moment-based estimate of between studies variance = 47.131

Trimming estimator: Linear
Meta-analysis type: Random-effects model

iteration |  estimate     Tn     # to trim     diff
----------+-----------------------------------------
    1     |   12.426      60          1         105
    2     |   11.762      66          2          12
    3     |   10.877      73          3          14
    4     |   10.435      74          3           2
    5     |   10.435      74          3           0

Filled
Meta-analysis

       |  Pooled      95% CI          Asymptotic      No. of
Method |     Est   Lower   Upper  z_value  p_value   studies
-------+-------------------------------------------------------
Fixed  |   5.808   5.413   6.203   28.822    0.000      17
Random |  10.467   7.201  13.733    6.281    0.000

Test for heterogeneity: Q= 831.028 on 16 degrees of freedom (p= 0.000)
Moment-based estimate of between studies variance = 44.962
```

Supplementary figure 6: Trim-and-fill analysis for the prevalence of impaired glucose tolerance among adults in Ethiopia.

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```
. metatrim pre sep, reffect graph funnel

Note: default data input format (theta, se_theta) assumed.

Meta-analysis

       |  Pooled      95% CI          Asymptotic      No. of
Method |     Est   Lower   Upper  z_value  p_value   studies
-------+-----------------------------------------------------
Fixed  |   7.055   6.663   7.446   35.317    0.000       8
Random |  14.708   9.820  19.596    5.897    0.000

Test for heterogeneity: Q= 500.276 on 7 degrees of freedom (p= 0.000)
Moment-based estimate of between studies variance = 47.566

Trimming estimator: Linear
Meta-analysis type: Random-effects model

iteration |  estimate    Tn    # to trim     diff
----------+-----------------------------------
    1     |   14.708     21         1          36
    2     |   13.294     22         1           2
    3     |   13.294     22         1           0

Filled
Meta-analysis

       |  Pooled      95% CI          Asymptotic      No. of
Method |     Est   Lower   Upper  z_value  p_value   studies
-------+-----------------------------------------------------
Fixed  |   7.025   6.634   7.415   35.248    0.000       9
Random |  13.298   8.682  17.915    5.646    0.000

Test for heterogeneity: Q= 505.145 on 8 degrees of freedom (p= 0.000)
Moment-based estimate of between studies variance = 47.100
```

Supplementary figure 7: Trim-and-fill analysis for the prevalence of metabolic syndrome among adults in Ethiopia.

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PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) Checklist

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You must report the page number in your manuscript where you consider each of the items listed in this checklist. If you have not included this information, either revise your manuscript accordingly before submitting or note N/A.

Section/Topic	Item No.	Checklist item	Reported on Page No.
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3-4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	N/A
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	7
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	8
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	7-8

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Section/Topic	Item No.	Checklist item	Reported on Page No.
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	8-9
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	8-9
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	9
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	9-10
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	10-11
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	11
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	11
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	11-12
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	11-12
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	11
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	13-15
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	16-20
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	21
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	22
FUNDING			

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Section/Topic	Item No.	Checklist item	Reported on Page No.
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	22

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(6): e1000097. doi:10.1371/journal.pmed1000097

Once you have completed this checklist, please save a copy and upload it as part of your submission. Please DO NOT include this checklist as part of the main manuscript document. It must be uploaded as a separate file.

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MOOSE Checklist for Meta-analyses of Observational Studies

Title: Metabolic risk factors of non-communicable diseases in Ethiopia: a systematic review and meta-analysis.

Item No	Recommendation	Reported on Page No
Reporting of background should include		
1	Problem definition	2-4
2	Hypothesis statement	4
3	Description of study outcome(s)	4,8
4	Type of exposure or intervention used	NA
5	Type of study designs used	5
6	Study population	5
Reporting of search strategy should include		
7	Qualifications of searchers (eg, librarians and investigators)	1
8	Search strategy, including time period included in the synthesis and key words	6
9	Effort to include all available studies, including contact with authors	6
10	Databases and registries searched	6, supplementary table 1
11	Search software used, name and version, including special features used (eg, explosion)	NA
12	Use of hand searching (eg, reference lists of obtained articles)	6
13	List of citations located and those excluded, including justification	10, supplementary Table 2, Figure 1
14	Method of addressing articles published in languages other than English	5, 6
15	Method of handling abstracts and unpublished studies	5, 6
16	Description of any contact with authors	NA
Reporting of methods should include		
17	Description of relevance or appropriateness of studies assembled for assessing the hypothesis to be tested	4, 5
18	Rationale for the selection and coding of data (eg, sound clinical principles or convenience)	7
19	Documentation of how data were classified and coded (eg, multiple raters, blinding and interrater reliability)	7, 8
20	Assessment of confounding (eg, comparability of cases and controls in studies where appropriate)	NA
21	Assessment of study quality, including blinding of quality assessors, stratification, or regression on possible predictors of study results	8
22	Assessment of heterogeneity	9
23	Description of statistical methods (eg, complete description of fixed or random effects models, justification of whether the chosen models account for predictors of study results, dose-response models, or cumulative meta-analysis) in sufficient detail to be replicated	8,9
24	Provision of appropriate tables and graphics	Table 1, Figures 1-11, supplementary table 1-3 supplementary figures 1-7

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Reporting of results should include		
25	Graphic summarizing individual study estimates and overall estimate	Figures 2-5
26	Table giving descriptive information for each study included	Supplementary Table 2
27	Results of sensitivity testing (eg, subgroup analysis)	13,15, Table 1
28	Indication of statistical uncertainty of findings	11-12,15, supplementary table 3

Item No	Recommendation	Reported on Page No
Reporting of discussion should include		
29	Quantitative assessment of bias (eg, publication bias)	14
30	Justification for exclusion (eg, exclusion of non-English language citations)	5
31	Assessment of quality of included studies	11, and supplementary Table 2
Reporting of conclusions should include		
32	Consideration of alternative explanations for observed results	16-20
33	Generalization of the conclusions (ie, appropriate for the data presented and within the domain of the literature review)	22
34	Guidelines for future research	20
35	Disclosure of funding source	22

From: Stroup DF, Berlin JA, Morton SC, et al, for the Meta-analysis Of Observational Studies in Epidemiology (MOOSE) Group. Meta-analysis of Observational Studies in Epidemiology. A Proposal for Reporting. *JAMA*. 2000;283(15):2008-2012. doi: 10.1001/jama.283.15.2008.

NA= Not Applicable

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